# Likelihood approximations distort the ion channel kinetic information in macroscopic currents

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**Abstract** Some ion channel mechanisms leave a mark in the record that does not depend on fitting a model: the flip state of $P2X_2$ showed up as a delay before the macroscopic current rose. The asymmetric activation we reported from the same recordings two decades later leaves no such mark: it rests on evidence ratios under a recursive likelihood, and a published control without the recursion put a non-conformational alternative first. How much likelihood approximations distort what they report, and in which directions, was never measured for macroscopic currents. The diagnostics are classical: for a correctly specified likelihood the score, the log-likelihood's gradient, averages to zero at the true parameters and its covariance equals the reported Fisher information. We tested eight likelihoods, from least squares on the mean current, which leaves the correlated gating fluctuation unmodelled, to a filter conditioning each interval average on both ends. The sweep runs over channel number, noise and interval in a two-state scheme, the smallest that isolates the approximation's own error, ten thousand exact simulations of the process they approximate at every point. Those that leave the correlation unmodelled misreport their information by up to an order of magnitude, threefold on the error bar, wherever the residual retains memory, and no single rescaling repairs it, because the distortion has directions in parameter space. The published control's likelihood is among them. Neither partial correction improves on the likelihood it corrects, and a white residual does not certify calibration. The filter conditioning on both ends is calibrated over almost the whole measured range, with departures growing toward few channels and low noise. The noise at which the least-squares error bar becomes honest lies one to three decades above the noise at which the fluctuations stop carrying the unitary current: no measured region gives both.

## Introduction

Some mechanisms leave a mark in the record that does not depend on fitting a particular model. The flip state we proposed for the $P2X_2$ receptor is one: it shows up as a delay before the current rises (***Moffatt and Hume, 2007***). As inferences get finer, claims are reached that leave no such mark. Two decades later nine kinetic schemes for the same receptor were ranked by Bayesian evidence over the same recordings, in work of our own (***Moffatt and Pierdominici-Sottile, 2025***). Under MacroIR, which averages the current over each acquisition interval and conditions it on the data already observed, a conformational scheme came first; a published control, MacroINR, the same likelihood without the recursion, put the subunit-specific allosteric alternative first, overturning a margin of a factor of six. The direction was reported, the control systematically underestimating the evidence for schemes with conformational intermediates; but the choice between the arms rested on argument, and nothing in those recordings settles which likelihood tells the truth, the way the delay once settled the flip.

What the recordings hold is the reason. Single-channel durations are read off the record (***Neher and Stevens, 1977***; ***Colquhoun and Hawkes, 1981***) and macroscopic currents follow a deterministic relaxation (***Hodgkin and Huxley, 1952***), but most recordings are in neither place: the averaging is incomplete, so a fluctuation around the mean current survives and carries the memory of the gating. That memory is where the remaining information is, since channel number and unitary current enter the mean as a product and no fit to the mean separates them.

Current practice fits the mean and discards the fluctuation. The four approaches surveyed by ***Clerx et al.*** (***2019***) are deterministic fits with a root-mean-square objective, and the IonBench benchmark states that it does not cover optimisation approaches for stochastic models (***Owen and Mirams, 2025***). Discarding the fluctuation usually leaves the point estimate standing. What it damages is the count: samples closer together than the channel's relaxation time are correlated, so a recording supplies fewer independent observations than it has samples while the fit counts them all, and the interval it reports is narrower than the spread it delivers.

Likelihoods that model that correlation have been available for decades, from stationary power spectra and the exact non-stationary two-time covariance (***Katz and Miledi, 1970***, ***1972***; ***Anderson and Stevens, 1973***; ***Conti et al., 1980***; ***Sigworth, 1981***) to whole-record likelihoods (***Celentano and Hawkes, 2004***; ***Milescu et al., 2005***; ***Moffatt, 2007***; ***Stepanyuk et al., 2011***; ***Münch et al., 2022***), differing in what they condition on, MacroIR the most recent (***Moffatt and Pierdominici-Sottile, 2025***). Where they break was stated when they were published and never located: the recursive member that samples instantaneously was expected to degrade as the channel count falls and as the record is averaged, its non-recursive counterpart to have no useful regime at all, the damage falling on the error rates rather than on the estimates (***Moffatt, 2007***).

No statistic computed on a single recording settles it either: one recording is one draw from the sampling distribution, and with the generating model unknown a departure may belong to the approximation or to the model. The question has to be put from outside, over an ensemble drawn from known parameters, and for this class of model that is possible: the forward process simulates exactly where its likelihood cannot be evaluated exactly.

The instruments are classical. For a correctly specified likelihood the score, the gradient of the log-likelihood, has mean zero at the generating parameters and its covariance across recordings equals the reported Fisher information (***Huber, 1967***; ***White, 1982***); how far an approximation departs from either identity is a measurable quantity rather than a test to be passed. Goodness of fit does not answer the question, since an approximation can reproduce the mean current closely and still misreport the information by more than an order of magnitude, and comparing two approximations against each other does not answer it either, because both may be wrong. Near the maximum that same matrix enters the Bayesian evidence through a volume term and a rescaling of the effective sample size, so misreporting it carries the error into every comparison built on it. The correction is derived elsewhere; this paper measures its input and computes no evidence, and it never benchmarks against non-stationary fluctuation analysis (***Stepanyuk et al., 2014***), which answers a different question, returning the channel number and the unitary current with no likelihood and no kinetics, from many exchangeable sweeps.

We measured both departures over a plane whose coordinates are the number of channels, the acquisition interval in units of the relaxation time, and the instrumental noise in units of the gating noise. The rate constants are held fixed. The scheme is two states at an open probability of one half, driven by a single concentration jump, which isolates the error the approximation itself commits; whether it bounds anything richer is a conjecture, not a result of this paper. Ten thousand recordings were simulated at every cell and scored by the eight members of the cost ladder, from least squares to the filter conditioning each interval average on both ends. An implementation error shows up as a departure like any other, so the same test checks the code.

The plane separates two failures usually named as one: what a method gets wrong, and what it says about how wrong it is. On the first the ladder is nearly flat. What separates the members is what they report about those estimates: the ones that leave the correlation unmodelled misreport their information by an order of magnitude wherever the residual retains enough memory, while the one conditioning on both ends of the interval is calibrated over almost the whole plane. The map includes that member in what it audits, since a method whose limits are unmapped has been endorsed rather than characterised.

Making the measurement takes an ensemble. Using it does not: the integrated autocorrelation of the standardized least-squares residual reads the effective interval count off a fit already being made. It also picks up the analogue filter on a record sampled faster than its own bandwidth, and that is a number the reader chose as they chose the sampling rate. What is new here is narrow, and the likelihood's construction is not part of it: that was published with the $P2X_2$ analysis. Its derivatives are. The gradient of the log-likelihood is analytic and the Fisher information follows from the same pass (Methods). Standard errors from fluctuation data go back to ***Anderson and Stevens*** (***1973***) and analytic gradients to ***Milescu et al.*** (***2005***). What this work adds, for a non-stationary protocol read from a single record, is an interval whose calibration has been verified rather than assumed.

## The likelihood family and its diagnostics

A macroscopic current is the summed signal of a population of ion channels, each an independent continuous-time Markov chain over a few conformational states. $N_{\mathrm{ch}}$ is the number of channels and $K$ the number of kinetic states of one channel ($K = 2$ throughout this paper, carried symbolically). Each channel has a generator (rate) matrix $\mathbf{Q} \in \mathbb{R}^{K\times K}$, hence the row-stochastic transition matrix $\mathbf{P}(t) = e^{\mathbf{Q}t}$ over an elapsed time $t$, with entries $P_{i\to j}(t)$ from state $i$ to state $j$, and a conductance $\gamma_k$ in state $k$, collected in $\boldsymbol{\gamma} \in \mathbb{R}^K$.

Writing that likelihood exactly calls for two objects, and neither is available here: a microscopic description keeping each channel discrete, which is explicit and ruled out by cost, and the law of the conductance averaged over the acquisition window, which has no closed form at any channel count although its low-order moments do (Appendix 1). Every member replaces both with Gaussians, and they differ in which object the second Gaussian describes, what it is conditioned on and how its variance is accounted.

### The observable is an interval average

A recorded sample is the current averaged over the acquisition window rather than sampled at an instant, because the current is low-pass filtered in analogue form before it reaches the converter. Writing the sampling period as $\Delta$ and taking the weighting to be uniform across it, the quantity a likelihood must describe is the windowed average

$$\bar{y}_t \;=\; \frac{1}{\Delta}\int_{t-\Delta}^{t} y(\tau)\,\mathrm{d}\tau \;+\; \eta_t, \tag{1}$$

where $y(\tau)$ is the instantaneous population current and $\eta_t$ is the instrumental noise accumulated over the window. This interval average is the physical observable. When $\Delta$ is much shorter than the fastest relaxation of $\mathbf{Q}$ the average sits close to the instantaneous current and the distinction is negligible; when $\Delta$ is comparable to or longer than that relaxation the two diverge, and a likelihood built on instantaneous sampling describes the wrong random variable.

Uniform weighting is a first approximation to what the electronics do, the real kernel being the impulse response of the amplifier's analogue low-pass. The simulator and every member share the uniform window (Methods), so data and likelihood describe the same observable and the window does not enter the measurement here; the closures below are the approximation the diagnostic sees. What the window misses lands in the instrumental part of the predictive variance and not in the gating part the closures approximate. Appendix 1 bounds it and gives the remedy, which is to block-average a record down to that scale before analysis.

### The two Gaussian closures fail in different regimes

To close a distribution is to replace it by the Gaussian with the same mean and covariance. Two closures are available, and every member makes the first; whether it pays for the second is set by what it predicts. The current at an instant is $\boldsymbol{\gamma}$ contracted against the occupancies at one time, and the linear image of a Gaussian is Gaussian; the interval average is a functional of the path inside the window and needs a closure of its own. An instantaneous member drops that evolution instead, which misspecifies the random variable rather than closing it, and in the unhelpful direction, since averaging is what brings the law closer to Gaussian.

> The two Gaussian approximations
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> **Occupancy closure.** One probability per occupancy vector, a count that grows with $N_{\mathrm{ch}}$, is replaced by a Gaussian in the per-channel pair $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the same size at every channel count and living on the simplex. *Validity:* two conditions, not one. The channel count must be high enough for the counts to behave as a continuum, which fails as $N_{\mathrm{ch}}$ falls; and the recorded sample must not resolve the step in predicted current made by moving one channel between states, which fails as the instrumental noise falls. The second is why the boundary of the failing region in the Results runs across the plane of channel count against noise rather than down a line of constant $N_{\mathrm{ch}}$.
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> **Interval-signal closure.** The law of the conductance averaged over one window is replaced by a Gaussian carrying its first two moments. The exact law is telegraph-like, a mixed law closed only at $K = 2$ and proliferating with $K$, which neither the few distinct conductances channel models carry nor the independence of the channels rescues (Appendix 1). *Validity:* the window must span many transitions, or the instrumental noise must be large enough to make the departure immaterial. The second half is exact rather than heuristic: noise independent of the gating current adds to the variance of the recorded sample and leaves every cumulant above the second unchanged, so the standardized departures shrink as the noise grows.
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> The two are controlled by different quantities, the occupancy closure by the channel count and the interval-signal closure by the window against the gating timescale, while instrumental noise relaxes both. That makes them separable by design, which is why the Results sweep a plane of channel count against noise at one window before moving the window along an axis of its own.

The moments the closure keeps are closed-form single-channel objects, independent of $N_{\mathrm{ch}}$ and of the topology (Appendix 1), which is what justifies the closure beyond the two-state case run here.

What they discard does not vanish. It reappears as temporal correlation the likelihood does not predict, which is what the correlation term of the distortion measures (below). Three regimes follow, named here so the maps in the Results have vocabulary: the *microscopic* regime, few channels, where the occupancy closure fails and only the microscopic description is exact; the *telegraphic* regime, windows far shorter than the relaxation time, where the interval-signal closure fails; and the *Gaussian* regime, where both hold.

Two design quantities are reported throughout in natural units, marked by a tilde. Time is measured in the channel time constant $\tau = 1/k_{\mathrm{off}}$, the closing time constant, and current in the unitary current $i$, giving the acquisition interval $\widetilde{\Delta} = \Delta\, k_{\mathrm{off}} = \Delta/\tau$ and the instrumental noise $\widetilde{S} = S\, k_{\mathrm{off}}/i^2$, with $S$ the instrumental noise power spectral density, so $\widetilde{S}$ is the noise power accumulated over one channel time constant measured against the single-channel signal power (Methods). The channel count and the instrumental noise enter the comparison through their ratio, which is the variable the design plane collapses in (Results).

### Least squares models no gating, and the family above it varies in window and recursion

Before any of the choices above comes a prior question: does the method model the gating fluctuations at all? Classical nonlinear least squares answers no. It fits the deterministic mean current and assigns one constant variance to every residual. Least squares is therefore not a member of the family. It is the bottom rung of the ladder of computational cost this paper walks, and it is the anchor against which every member above it is measured, because it is the method most readers are using.

The methods differ along two axes. One is the window: how much of the acquisition interval the predicted current is conditioned on. The other is recursion: a recursive (prequential) member (***Dawid, 1984***) factorizes the likelihood as $\prod_t q_t(\bar{y}_t \mid \bar{y}_{1:t-1})$, so the occupancy distribution carried into each window is the posterior left by the previous ones, while a non-recursive (open-loop) member factorizes as $\prod_t m_t(\bar{y}_t)$, each window propagated from the initial condition and never conditioned on the observed trace.

The window decision is not the family's property alone: any method that predicts a recorded value has to make it, at every level of the gating description and least squares included, which is why least squares has two arms here.

The names are read off the two axes. The suffix is the recursion axis, `NR` for a non-recursive member and `R` for a recursive one; the prefix is the window axis, none for the current at an instant, `M` for the interval mean given the state at the window's start, `I` for the interval mean given the states at both of its ends. Five members follow rather than six, because without an update the state at the end of a window is never observed and the two interval prefixes describe the same method on the non-recursive side. The `M` is the mean conductance and not an abbreviation of macroscopic; every member of the family is macroscopic. `VR` is `MR` with the residual interval variance in place of the total one, the one member whose variance form is a choice rather than a consequence of what it conditions on (Appendix 2). The two least-squares arms sit off this lattice, `LSE` fitting the instantaneous deterministic mean and `ILSE` the same mean averaged over the interval, where the `I` records the window setting and nothing else. Each is the open-loop member at its own window setting with the gating variance replaced by one constant fitted over the whole record, and what the least-squares arm discards is the variance (Methods). Table 1 lists the name each member goes by in the Results.

Two things follow. `IR` occupies the top rung below the intractable exact likelihood, so its standing is structural and set before any measurement; and the lower rungs are established likelihoods rather than constructed foils, the non-recursive instantaneous member being the independent-interval likelihood of ***Milescu et al.*** (***2005***) and the recursive instantaneous member the macroscopic filter of ***Moffatt*** (***2007***) and the generalized filter of ***Münch et al.*** (***2022***). Above all of them sits the exact likelihood; the same model can nonetheless be simulated forward exactly, so the simulation realizes the top of the ladder and is the ground truth every rung below it is judged against.

### The interval likelihood is the instantaneous one applied to the boundary state

A *boundary state* is the pair $(i \to j)$ of a channel's states at the two ends of an acquisition interval; an arrow marks an object that spans the window, so $P_{i\to j}(\Delta)$ is a transition across it and $\bar{\gamma}_{i\to j}$ an average over it. Chaining intervals leaves a single state variable per junction, since the end state of one interval is the start state of the next, and the moments over the $K^2$ pairs are built from objects of side $K$, with no explicit $K^2$ object formed anywhere (Appendix 1).

The interval likelihood is the instantaneous one applied to the boundary state. The recursive member of ***Moffatt*** (***2007***) carries a Gaussian over the channel occupancies and conditions it on each recorded sample: in the per-channel normalization used throughout, with $\boldsymbol{\gamma}$ the per-state

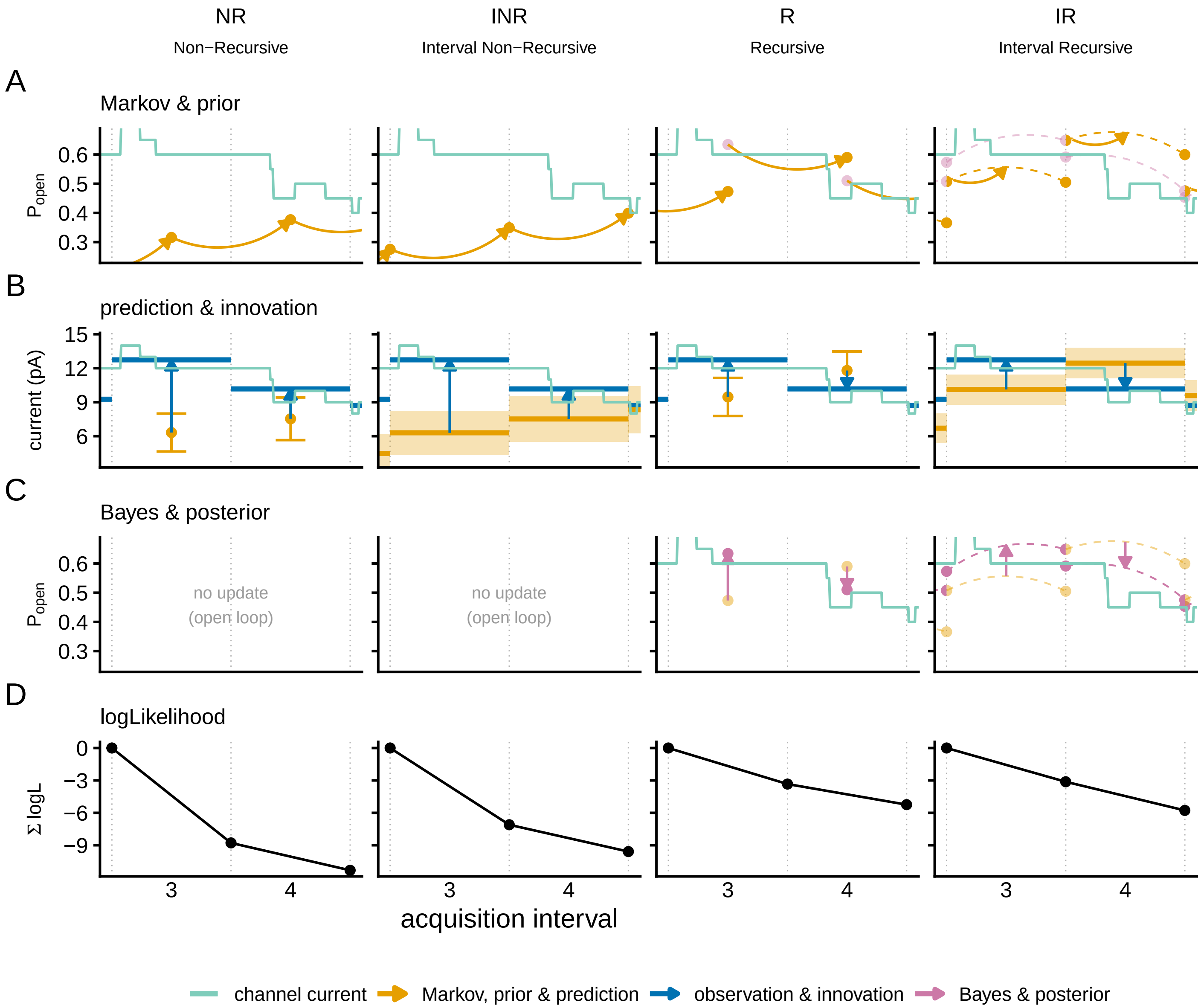


**Figure 1. One filter step, and what each of the two axes changes.** A simulated patch of $N_{\mathrm{ch}} = 20$ two-state channels. The green trace is the current the channels carry, resolved inside the acquisition window, and the recording keeps one number per window, the average over it; dotted lines are the window boundaries and the axis counts windows, each $2$ ms. At the lowest noise of the design plane, $\widetilde{S} = 0.01$, the recorded value sits on the average of that trace. Four representative members score it (columns), the two axes crossed: the open-loop gating likelihood (`NR`), its interval-averaged partner (`INR`), the recursive filter (`R`) and the interval-conditioned filter (`IR`). Read `LSE` and `ILSE` off the open-loop columns with a band of constant height, `MR` and `VR` off the recursive column carrying `INR`'s prediction (Table 1). Rows follow one turn of the cycle: (A) the prior open probability, (B) the predicted current with its spread and the innovation, (C) the Bayesian posterior, absent from the open-loop columns, and (D) the log-likelihood from zero. The window axis is read across row (B): `NR` and `R` predict an instant, a point at the sample with an error bar, `INR` and `IR` the interval average, a segment across the interval with a band. In (A) and (C), `IR` alone carries a pair of occupancies, one at each end of the interval, tied by a dashed arc that is their covariance and implies no path through it. Colour is the filter phase (orange predict, blue observe, purple update, black the score). No number is quoted from this simulated recording.

conductance and $\epsilon^2$ the instrumental variance,

$$y^{\text{pred}} = N_{\text{ch}}\,\boldsymbol{\mu}^{\top}\boldsymbol{\gamma}, \qquad \mathbf{g} = \boldsymbol{\Sigma}\boldsymbol{\gamma}, \qquad \sigma^2 = \epsilon^2 + N_{\text{ch}}\,\boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}\boldsymbol{\gamma},$$
$$\boldsymbol{\mu}^{\text{post}} = \boldsymbol{\mu} + \frac{\mathbf{g}}{\sigma^2}\,\delta, \qquad \boldsymbol{\Sigma}^{\text{post}} = \boldsymbol{\Sigma} - \frac{N_{\text{ch}}\,\mathbf{g}\mathbf{g}^{\top}}{\sigma^2}, \qquad \delta = y^{\text{obs}} - y^{\text{pred}}. \tag{2}$$

This is the Gaussian conditioning of the pair (state, observation) on the value recorded. The gain $\mathbf{g}$ is the covariance between the state a channel is in and the current it carries, and it is the whole of what brings the observation to the state: the predicted variance divides it, and nothing else in the update touches the data. Every moment here belongs to a single channel, and the factors of $N_{\text{ch}}$ are where the counts over the ensemble enter (Appendix 1).

What it gives up is not exactness but *sufficiency*: the three moments it consumes are exact functions of $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, but $\mathrm{Var}[\bar{y} \mid \mathbf{N}]$ depends on $\mathbf{N}$, so $(\mathbf{N}, \bar{y})$ is not jointly Gaussian and Eq. 2 is not the exact Bayesian update, a heteroscedastic emission treated as if it were homoscedastic. Nor can it be used on the interval average, which is not a function of the occupancy at any one time, so no conductance vector on the $K$ states delivers its mean. On the $K^2$ pairs of states a channel occupies at the two ends of the window one does, and every interval member is that substitution closed by a Gaussian (Appendix 1).

In a recursive member the posterior pair becomes the prior for the next window; in a non-recursive member it is not carried forward. Carrying it forward is itself a closure, exact only while the end marginal is a sufficient summary of the boundary posterior, so the recursive members' Gaussian information is model-based rather than the exact information of the process, and how far it departs is what the diagnostics below measure.

## The distortion is a matrix, and only the residual check runs on a real recording

Everything below is read off ensembles of exactly simulated recordings at two parameter vectors, and every figure states which one it shows: the simulation truth $\theta_{\text{sim}}$, never known in an experiment, and the pooled optimum $\theta_{\text{pool}}$, the fit of the member under test to all the recordings of one cell at once (Methods). Their difference is the misspecification bias itself. At $\theta_{\text{sim}}$ a nonzero mean score is visible at all; at $\theta_{\text{pool}}$ it vanishes by construction, so the information comparison is not contaminated by a displaced gradient.

Three quantities are read off the simulation, in increasing strength. The first is the standardized residual $r_t = (y_t - \mu_t)/\sigma_t$, with $\mu_t$ and $\sigma_t^2$ the likelihood's own predictive mean and variance for interval $t$: mean zero, unit variance and no autocorrelation across intervals if the one-step predictive distribution is correct *at the parameters it is evaluated at*. It is the only one of the three an experimentalist can run on a real recording, needing no ensemble and no known truth, and it does need a fitted point. A fit with the baseline and the noise level free absorbs most of the first two properties; the temporal structure is what no fit absorbs. That structure is summarised by the integrated autocorrelation (***Sokal, 1997***) of the residual, $\tau_{\text{int}} = 1 + 2\sum_{k\geq 1}\rho_k$ with $\rho_k$ its lag-$k$ correlation, summed to the maximum lag stated in Methods: it reads one for a white residual, and above one it counts how many intervals a fit effectively has for every interval it believes it has.

The second is the score. With $\ell(\theta)$ the total log-likelihood and $S(\theta) = \nabla_\theta \ell(\theta)$ its gradient, correct specification requires $\mathbb{E}[S(\theta_{\text{sim}})] = 0$. Projected through the parameter covariance, a nonzero mean score becomes a displacement in parameter units, $b = \mathbf{H}^{-1}\,\mathbb{E}[S]$, with $\mathbf{H}$ the Gaussian Fisher information the likelihood itself reports: how far the estimate is pushed. This one cannot be run on an experiment at all, needing a known truth and an ensemble to take the expectation over.

The third and strongest is the information itself. Correct specification requires $\mathrm{Cov}[S] = \mathbf{H}$, the covariance taken over independent recordings of the same cell, which no experiment supplies: one recording gives one score. An approximation can pass the residual and score checks and still violate this one by an order of magnitude. Because the failure is directional, we record it as a matrix,

$$\mathbf{C} = \mathbf{H}^{-1/2}\,\mathbf{J}\,\mathbf{H}^{-1/2},$$

| Member | The states are conditioned on | Predicted mean | Predicted variance beyond the instrumental term | Called, in the Results |
|---|---|---|---|---|
| *Open loop*: no update; every window is propagated from the initial condition | | | | |
| `LSE` | nothing: there are no states | the deterministic mean current at the middle of the window | none: one constant over the whole record, fitted as the residual sum of squares | the classical fit, un-averaged arm |
| `ILSE` | nothing: there are no states | the same mean current averaged over the window | the same constant | the classical fit, or the least-squares arm |
| `NR` | an instant, the middle of the window | the current at that instant | the spread of the instantaneous conductance across the states the channels occupy | the open-loop member |
| `INR` | the state at the start of the window | the interval-mean current given that state | the spread of that interval mean across the occupied states, and the variance of one channel's interval mean given its start state | the open-loop interval member |
| *Recursive*: the posterior of one window is the prior of the next | | | | |
| `R` | an instant, the middle of the window | the current at that instant | the spread of the instantaneous conductance across the states the channels occupy | the recursive instantaneous filter |
| `MR` | the state at the start of the window | the interval-mean current given that state | the spread of that interval mean across the occupied states, and the variance of one channel's interval mean given its start state | the one-endpoint member |
| `VR` | the start, and both ends for the interval variance alone | the interval-mean current given the start state | the spread of that interval mean across the occupied states, and the variance of one channel's interval mean given both its endpoints | the variance-corrected rung |
| `IR` | the states at both ends of the window | the interval-mean current, the unobserved end state marginalized | the spread of the interval mean across the occupied pairs of endpoint states, and the variance of one channel's interval mean given both its endpoints | the boundary-conditioned member |

**Table 1. The eight members as settings of one construction.** One conditioning choice runs through each row: the states an interval is conditioned on are the states its conductance is computed from and the states its update is taken on, so the second column governs the gain as well as the two columns beside it. Reading down, the second column is the window axis and the two blocks are the recursion axis. Three consequences are worth reading off. `LSE` and `ILSE` differ in the mean alone, which isolates the acquisition average with nothing else moving; `NR` and `R`, and `INR` and `MR`, carry identical Gaussians and differ only in whether the recording is fed back into the states; and `MR` and `IR`, taken from the same state, assign the same variance by two decompositions that rearrange into each other, so what separates them is the gain, which is then what makes the variances they report along a recording differ after all. `VR` drops the between-endpoint spread from the third term without restoring it in the second and predicts less than either from any state. In print, `IR` is MacroIR and `INR` is MacroINR. All eight are implemented, and all eight are run over the design plane; the four that cross the two axes are drawn as one turn of the filter cycle in Figure 1, the equation-level specification is Appendix 2, and the dispatched flags and data keys are in Supplementary File 1.

with $\mathbf{J} = \mathrm{Cov}(S)$ the covariance of the total score, cross-interval terms included, so that $\mathbf{C} = \mathbf{I}$ exactly under correct specification. This is the classical information-matrix-equality comparison ***White*** (***1982***), measured here rather than tested, and read as a property of each approximation rather than as a hypothesis to accept or reject. We fix the direction by convention: $C_{ii} > 1$ means the likelihood under-reports the uncertainty in parameter $i$ and is over-confident, $C_{ii} < 1$ means it over-reports and is conservative. The anchor $\mathbf{H}$ is the likelihood's own Gaussian Fisher information, formed from the predictive moments and their parameter derivatives that the score already requires,

$$\mathbf{H} = \sum_t \mathbb{E}\left[\frac{(\nabla\mu_t)(\nabla\mu_t)^\top}{\sigma_t^2} + \frac{(\nabla\sigma_t^2)(\nabla\sigma_t^2)^\top}{2\sigma_t^4}\right], \tag{3}$$

positive semidefinite by construction and costing no extra likelihood pass (Methods). The comparison is not circular: $\mathbf{H}$ is the information the likelihood claims and $\mathbf{J}$ the information it delivers against data from the true process. All three checks run on the whole cost ladder; the bottom rung needs a qualification, its predictive variance being one constant rather than a gating quantity, and the Results state it where the number is quoted.

The same matrix repairs what it measures. The sandwich covariance ***Huber*** (***1967***); ***Godambe*** (***1960***)

$$\Sigma = \mathbf{H}^{-1}\,\mathbf{J}\,\mathbf{H}^{-1} = \mathbf{H}^{-1/2}\,\mathbf{C}\,\mathbf{H}^{-1/2}$$

substitutes the information the likelihood actually carries for the information it reported. It and the bias vector $b$ are first-order approximations (Appendix 3). We verify it directly, without leaning on the expansion, against the empirical covariance of the estimates over the simulated recordings (Figure 2).

The distortion splits into two physically distinct parts. The sample distortion $\mathbf{C}_s = \mathbf{H}^{-1/2}\,\mathbf{J}_s\,\mathbf{H}^{-1/2}$, built from the within-interval score covariance $\mathbf{J}_s = \sum_t \mathrm{Cov}(s_t)$ with $s_t = \nabla_\theta \ell_t$, measures per-sample departure from Gaussianity, to leading order the third and fourth cumulants of the interval current. The correlation distortion $\mathbf{R} = \mathbf{J}_s^{-1/2}\,\mathbf{J}\,\mathbf{J}_s^{-1/2}$ measures the cross-interval correlation of the score, which $\mathbf{J}_s$ misses and which is the local time correlation of the current that ***Milescu et al.*** (***2005***) named as the dominant error of non-recursive fitting. One parameter at a time, $F_t$ is the Gaussian Fisher information of interval $t$ alone and $J_t = \mathrm{Var}(s_t)$, and accumulated it is $J_T = \mathrm{Var}(\sum_{t\le T} s_t)$, not the sum of the $J_t$, that the reported $F_T$ has to match, so per-interval and accumulated calibration are not equivalent. How the two parts compose is Appendix 3.

The correlation part has a reader-facing translation. Where $\mathbf{R}$ is one number times the identity, that number is the factor by which a likelihood over-counts its intervals. Where $\mathbf{R}$ is anisotropic, or $\mathbf{C}_s \neq \mathbf{I}$, no such correction suffices at any factor and the full matrix is needed.

A matrix is not a number, and two scalars summarise it without pretending to replace it. With $\lambda_i$ the eigenvalues of $\mathbf{C}$ on the retained subspace (Methods), $\bar{e} = d^{-1}\sum_i \log\lambda_i$ and $s^2 = \mathrm{Var}_i(\log\lambda_i)$, the population variance with divisor $d$, the two reported throughout are the exponentials

$$m = e^{\bar{e}} = \left(\prod_i \lambda_i\right)^{1/d}, \qquad a = e^{s}, \tag{4}$$

so that $m$ is the typical factor by which the reported information is wrong, the geometric mean of the per-direction factors, and $a$ is their multiplicative spread from direction to direction, both one at correct specification. In error-bar units take the square root of any of these, since the distortion is a ratio of informations: $1.15$ is a $7\%$ error on a confidence interval, $2$ is $41\%$, and $4$ turns a nominal $95\%$ interval into about $68\%$.

## Results

A recursive likelihood cannot be checked against a record by eye: it predicts each sample from the samples before it, so it follows the record whether or not the uncertainty it reports is the uncertainty it delivers. What follows measures it against least squares, on recordings simulated exactly from the process every member approximates.

Throughout, $\widetilde{\Delta}$ and $\widetilde{S}$ are the acquisition interval and the instrumental noise in the natural units defined above, and the eight members are those of Table 1. The two least-squares arms are one flag apart, `LSE` fitting the conductance at the sample and `ILSE` fitting it averaged over the acquisition window, and both are run over the whole design plane, on the same cells as each other and as every other member.

## Only the boundary-conditioned member is centred and reports its uncertainty correctly in every direction

Figure 2 reads the ladder at one design cell ($N_{\mathrm{ch}} = 100$, $\widetilde{S} = 0.01$, $\widetilde{\Delta} = 0.1$), and each panel shows the two quantities the rest of the paper maps. The cloud away from the truth is the bias. The reported ellipse away from the empirical one is the distortion, and it takes two numbers rather than one: its magnitude, the typical factor by which the reported uncertainty is wrong, and its anisotropy, how much that factor changes from one direction to another ($m$ and $a$ of Eq. (4), the two orange numbers of each panel). Both are one when the member is calibrated, and both are read here from a cloud of a thousand maximum-likelihood fits, where every later figure reads them from the score with no fitting and the two constructions agree only to first order (Appendix 3).

The truth and the cloud mean coincide only for the two least-squares arms and for `INR` and `IR`; the other four macroscopic members are displaced, within the dispersion of their own cloud on the rates and beyond it on the unitary current and the channel number. The noise level is biased in `NR` alone (Figure 2–source data 1). That the mean current pins the product while only the fluctuations can separate its factors is the identifiability structure known for models of this kind (***Hines et al., 2014***). The third marker of each panel is where the score and the information alone put that displacement, with no fitting, and it lands on the same side as the cloud and close to it: on the unitary current $-0.287$ against a measured $-0.228$ for `MR`, and nothing for `IR`, where the cloud resolves none. Figure 3 shows the score that produces it. On the rates no member is off by more than a tenth, the two least-squares arms least of all, and the plane is where their rate bias is established (Figure 4A). The interval average is what removes the displacement, in all three families that carry both arms, and recursion on its own does not: it leaves `R` the least centred member of the eight in the rates and further out in the unitary current than the open-loop member it is built from.

The distortion orders the ladder the other way, and not monotonically: the orange numbers read one for `IR` on every pair, within a third of it for `R`, behind `R` in every pair for the two one-endpoint corrections built on it, and an order of magnitude higher, evenly, for the members that leave the correlation unmodelled, `INR` excepted in the noise pair, the one direction its interval variance repairs. What recovers the calibration is the second endpoint entering the gain (Appendix 2).

The factor is not even one number per member: `INR`'s noise-pair distortion factors into more than ten along the channel-number direction and one along the noise axis, the member that repairs the noise estimate reporting the noise uncertainty exactly right and the channel number with a standard error more than three times too small, so no single rescaling serves.

What does serve is a matrix. The distortion is measured as one, and the same matrix corrects the covariance it measures (the sandwich, Appendix 3), which the rest of the paper reports. Whether that correction is right is what this figure settles, dividing the empirical ellipse of each panel by the corrected one: at one, the correction has recovered the covariance the estimator actually has. It does for every gating-aware member, and at a second cell it lifts the coverage of `IR`'s nominal $95\%$ region from $0.914$ to $0.947$ in all six directions at once (Figure 2–figure supplement 1). It fails alike for the two least-squares arms, at a factor of $1.6$ in both, two members that differ in every first-moment quantity, which puts the failure in what they share; they are also the two arms whose analytic Fisher the differenced check flags (Supplementary File 1), at a size consistent with the residual measured here.

Least squares does estimate the channel number, to within $0.012$ in $\log_{10}$, but with the unitary current pinned at its true value that estimate is the mean current under another name, which is

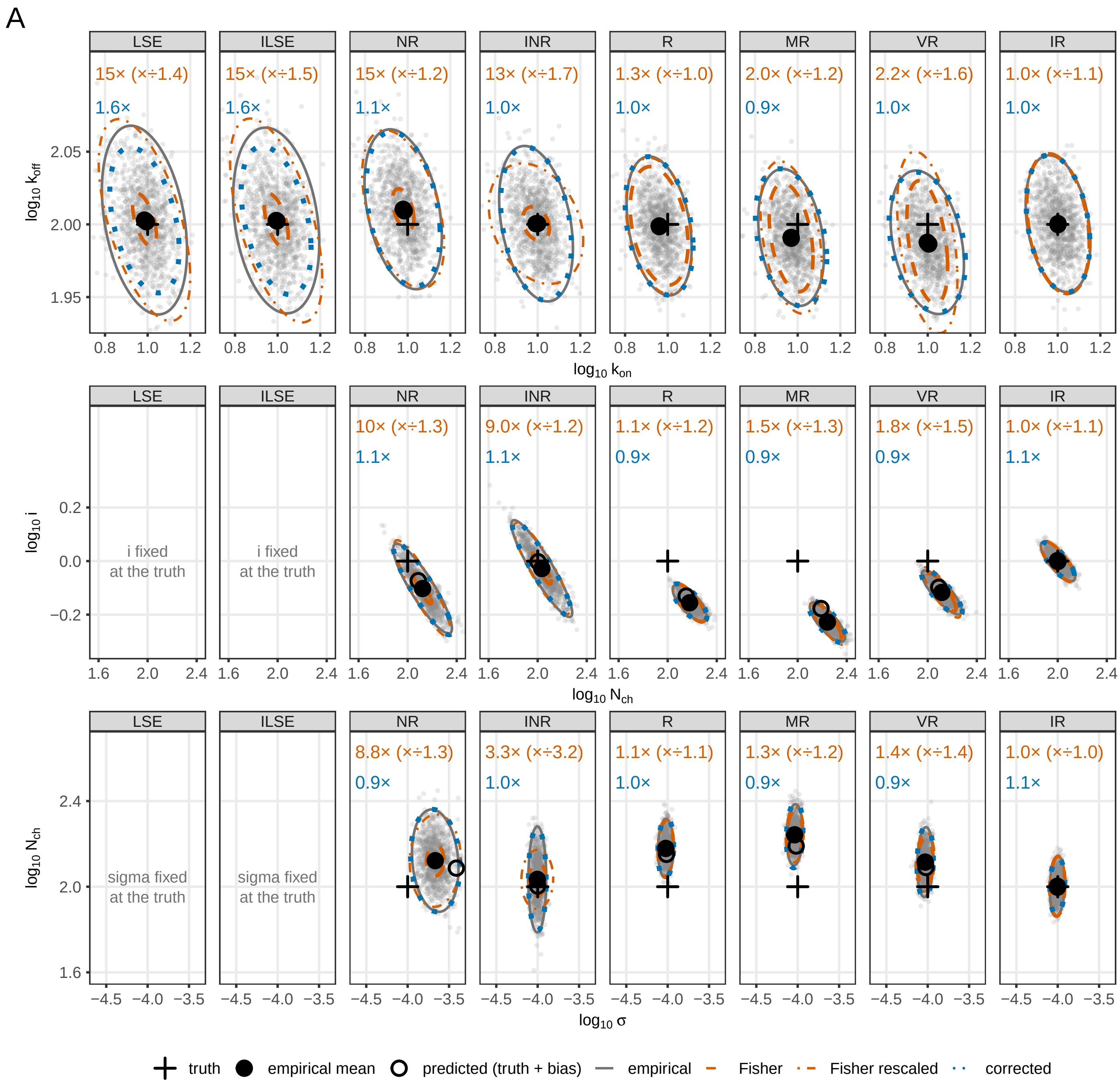


**Figure 2. Parameter recovery and the calibration of the reported interval, for the whole ladder at one design cell.** $N_{\mathrm{ch}} = 100$, $\widetilde{S} = 0.01$, $\widetilde{\Delta} = 0.1$; $10{,}000$ recordings in groups of ten, so one thousand maximum-likelihood estimates per member. Columns are the eight members in ladder order, rows the kinetic, amplitude and noise pairs, with $\sigma$ the instrumental noise level. Each panel carries the cloud of estimates and four $95\%$ ellipses at its mean, named in the key. Above the cloud, the orange numbers are the magnitude of the distortion and, in parentheses, its anisotropy; the blue number is the empirical ellipse over the corrected one. All three read $1.0$ for a calibrated member. Covariances are the Gaussian-Fisher family at the pooled estimate, divided by the group size.

**Figure 2—figure supplement 1.** The sandwich prediction against the empirical multivariate distribution of the estimates, as a Mahalanobis quantile-quantile plot.

**Figure 2—source data 1.** First-moment bias, measured and score-predicted, with $95\%$ bootstrap intervals.

**Figure 2—source data 2.** Distortion magnitude, anisotropy, corrected ratio and eigendirections, per member and parameter pair.

why its other two columns are empty.

## The uncertainty is misreported for two independent reasons, one within a sample and one across them

The miscalibration of Figure 2 is a property of the likelihood and not of the optimiser, and Figure 3 shows what produces it, scoring one common set of one thousand recordings at the same cell, interval by interval, with no fitting. Row (A) already separates the families: the recursive members follow the record, their predictive band tracking the current they score, while the open-loop members and both least-squares arms predict the ensemble and do not. The log-likelihood adds up everything a member gets wrong, and it orders the ladder accordingly: the boundary-conditioned member on top and the two least-squares arms together at the bottom, $119$ nats below it (Figure 3–source data 1). The ranking says nothing about how far from the truth its top sits; every other row is read against a value a correct likelihood must give, and can therefore test. One entry is not the same kind of number, both least-squares arms fitting a noise level to each recording they score.

The figure keeps the three diagnostics apart: the residual carries its variance in row (B) and its memory in the black series of (G); the score its mean in (D), its variance against the information in (E) and (F), and its memory in the coloured series of (G); the information, the precision the member claims, is drawn alone in (C).

The mean of the score is what says whether the estimate will be displaced, and it repeats Figure 2's split: per interval it sits at the chance level for `IR`, near it for `INR`, at four times chance for both least-squares arms, and departs from zero at two fifths to three quarters of the informative steps for the displaced four, so averaging the model over the acquisition window is what removes it. What moves the estimate is the accumulated score divided by the whole information matrix, $b = F^{-1}\,\mathrm{E}[\sum_t s_t]$, and read that way the two interval-averaged members cover zero on all four identified parameters and no other member does. It is the same quantity the third marker of Figure 2 carries at this cell and Figure 4A maps over the plane.

Misreported uncertainty has two independent sources and the figure separates them: the variance of the score estimates the dispersion, and per interval it fails within tens of per cent at the median for every rung of the ladder, while accumulated it fails by an order of magnitude (E against F). The per-sample half the residual gives without any parameter at all: its mean square is one only for the two interval-averaged members, below one for `R`, `MR` and `VR`, and $1.15$ for `NR`, most of that excess sitting in the single interval where the agonist steps on, inside which the occupancy changes and a member that evaluates its variance at a point of the window cannot see it. Both least-squares arms read exactly one over the whole recording by construction, the fitted noise absorbing what the model misses, while running from $0.010$ before the agonist to $1.91$ over the pulse.

The second source is correlation in time, and no per-sample check can see it. `INR` has the calibrated residual variance of `IR` and reports its accumulated information twenty times wrong. The accumulated ratio is the product of the per-sample ratio and a correlation factor, and `IR` carries both at one, $1.07$ and $1.01$, while `R` reaches a comparable $1.07$ as the product of a per-sample variance it under-reports by $28\%$ and a correlation that inflates it by $48\%$, so the calibration of recursion alone is a cancellation. At lag one the score autocorrelation on $k_{\mathrm{off}}$ is $-0.004 \pm 0.004$ for `IR` against $0.191$ for `R` and $0.864$ for `NR`, the local time correlation of the current that ***Milescu et al.*** (***2005***) named as the dominant error of independent-interval fitting, measured at the level of the inference. The same memory shows in the residual, which is the half of this row that transfers to a real recording. Figure 4B carries the second moment over the plane; these are the two mechanisms behind it.

The information on its own says where each parameter is measured. About the channel number and the opening rate it reaches the numerical floor within six intervals of the agonist being removed, while the closing rate survives through the mean and the unitary current through the variance of the channels still open (Figure 3–figure supplements 1 and 2). A member that has conditioned on the past already carries the number of channels still open, so once a parameter reaches the recording only through that number there is nothing further to learn. The two open-

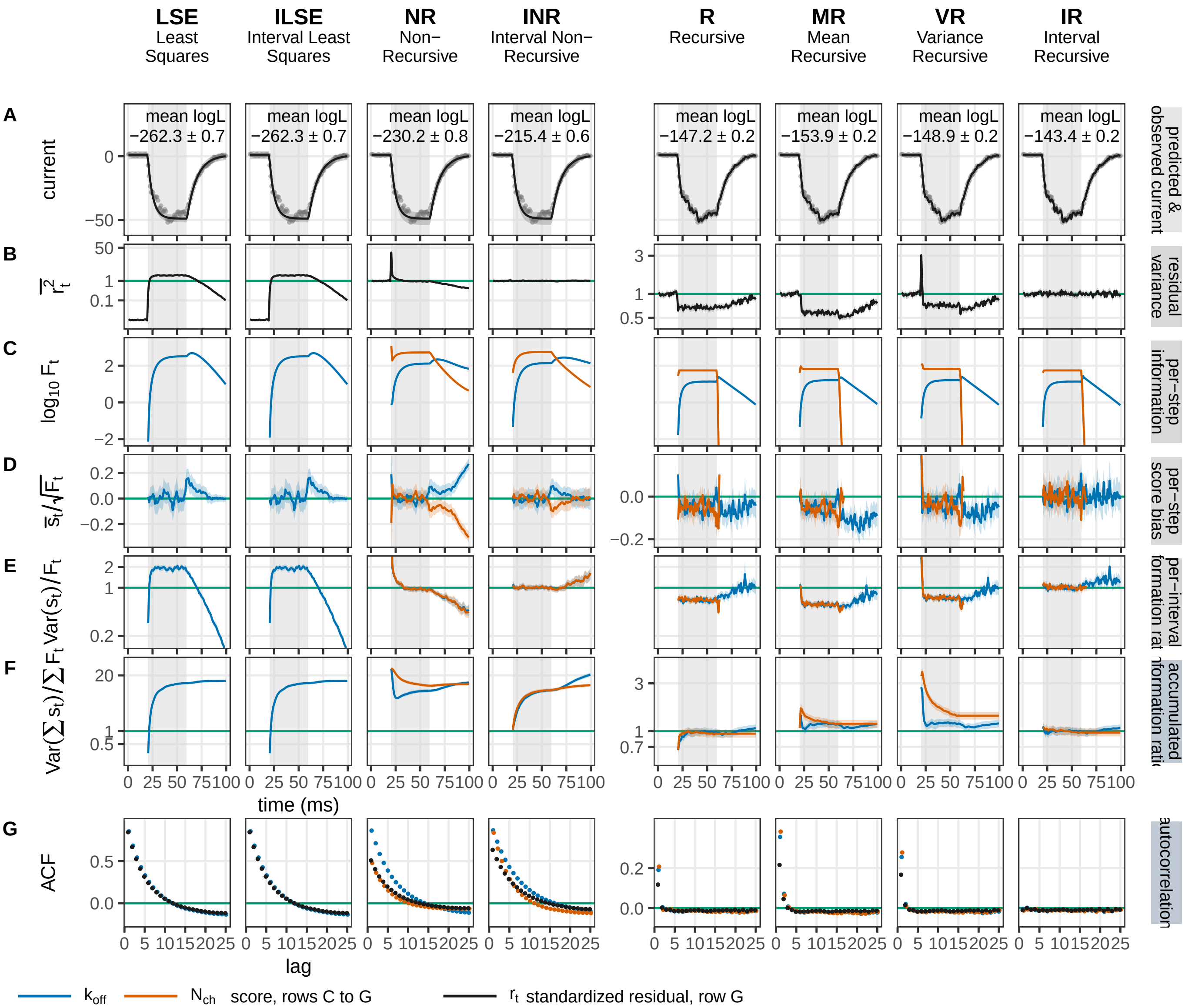


**Figure 3. The calibration cascade in time.** One set of 1,000 recordings ($N_{\mathrm{ch}} = 100$, $\widetilde{S} = 0.01$, $\widetilde{\Delta} = 0.1$) is scored by each of the eight members, in cost-ladder order and with no fitting. Rows A and B are properties of the data, in grey; C to G are per parameter, the closing rate and the channel number in colour. (A) An example recording, its band the member's own predictive $\pm 1$ s.d., annotated with the ensemble-mean log-likelihood. (B) The mean squared standardized residual. (C) The per-interval Fisher information. (D) The mean per-interval score over $\sqrt{F_t}$, standardized so that the two parameters share an axis. (E) The per-interval information ratio $\mathrm{Var}(s_t)/F_t$. (F) The same ratio accumulated, $J_T/F_T$. (G) The autocorrelation against lag, of the residual in black and of the score in colour. Bands are 95% bootstrap intervals over the recordings and curves stop where the information reaches zero. Non-negative quantities are on a logarithmic axis and signed ones on a linear axis, with the calibrated value on the ticks; the four open-loop columns share a y-scale block distinct from the four recursive ones.

**Figure 3—figure supplement 1.** Per-step information against score variance, for the opening rate and the closing rate.

**Figure 3—figure supplement 2.** The same, for the unitary current and the channel number.

**Figure 3—figure supplement 3.** The residual and the score in the three parameters not coloured above.

**Figure 3—source data 1.** The calibration cascade as numbers: per member and coloured parameter, the fraction of informative intervals whose score departs from zero and whose information ratio departs from one, the accumulated ratio $J_T/F_T$ with its interval, the lag-one score autocorrelation with its interval, and the ensemble-mean log-likelihood.

loop members go on reporting information about the channel count through the washout, 15 to 16% of their pulse total against 0.5 to 0.8%, which is the same information counted twice that their accumulated ratio charges them for. The noise level, measured before the agonist arrives, keeps a white score where the kinetic directions do not (Figure 3–figure supplement 3).

## Averaging in open loop centres the estimate, conditioning on both ends calibrates the uncertainty, and conditioning on one end does neither

Figure 4 maps the bias (A) and the distortion (B) of every member against the acquisition interval, the instrumental noise and the channel count, on two parameters, $k_{\mathrm{off}}$ and $N_{\mathrm{ch}}$. Instrumental noise equalises the members. Where it grows past the gating variance every rung reports honestly, so the differences between them are a property of the regime in which the fluctuations still carry information, not of the algorithms in general. Where that crossing sits is worth having in the plane's own coordinates, because it moves: at the open probability of one half the instrumental variance overtakes the gating variance at $\widetilde{S}/N_{\mathrm{ch}}$ equal to 0.0025 at the shortest window and 0.14 at the longest. The cell the two preceding figures use sits a factor of 234 in variance below it, well inside the regime where the gating fluctuations carry the information.

More channels do not repair `R`. Taking the median over the seven intervals at $\widetilde{S} = 0.01$, its $k_{\mathrm{off}}$ distortion (B) sits on a flat floor near 1.4 across four decades of channel count while `IR` converges to one; `NR` moves the other way, from 22 to 77, and the least-squares arm holds a floor near 13. In the channel-number direction the two recursive members separate in sign as well as in size: `IR` stays conservative throughout, while `R` crosses from conservative to over-confident and keeps rising.

The first moment (A) carries a separate cost, and it separates on the other axis. The median absolute bias in the channel number is 0.099 and 0.110 in $\log_{10}$ units for the two members that do not average the model over the acquisition interval, against 0.003 and 0.002 for their interval-averaged counterparts, so averaging is what removes the amplitude bias while recursion alone does not remove it and slightly worsens it: `R`'s channel-number bias of about 0.14 in $\log_{10}$, close to a 35 to 40% error, does not shrink as channels are added, and `NR` carries half of it in the same direction. `IR` is at or near zero everywhere. Least squares has a rate bias of essentially zero over the same plane, and that concession is worth making plainly: the classical fit recovers the rates, and what it gets wrong is how well it says it recovered them.

Modelling the correlation across intervals is what removes the distortion, and recursion is how every member of this family that models it does so. A member that takes one axis and not the other repairs one failure and keeps the other, so the family cannot be ordered on a single scale, and nothing in that is specific to ion channels: it holds for any state model whose observations are averages over a window.

Instrumental noise moves the whole picture continuously, and it moves the window-ignoring members furthest. The closing-rate distortion of least squares (B) falls from its floor near 13 to one over about three decades of added noise, at every channel count, and `R` follows the same pattern about three decades lower. The added variance carries no gating memory, so as it grows it dilutes the temporal structure a window-ignoring likelihood fails to model until there is nothing left to fail on, and the channel count at which each member's error bar becomes honest rises with it, and the design map turns that crossover into one of its boundaries.

Over the grid the three members share, counting every parameter at every cell, the reported uncertainty is within ±15% of the delivered one for 97% of `IR`'s points, 78% of `R`'s and 37% of `NR`'s; over its own wider grid `IR` reads 95%, and what it has left is concentrated at few channels and low noise. Least squares carries no entry, because its distortion matrix has four dimensions against the likelihood members' six and the two counts are not commensurable (Methods). The coverage is unequal and has to be quoted with the number: `IR` spans eleven channel counts from 5 to $10^4$ across the twelve noise levels of the ladder (Methods), not on a full rectangle, and the others cover less.

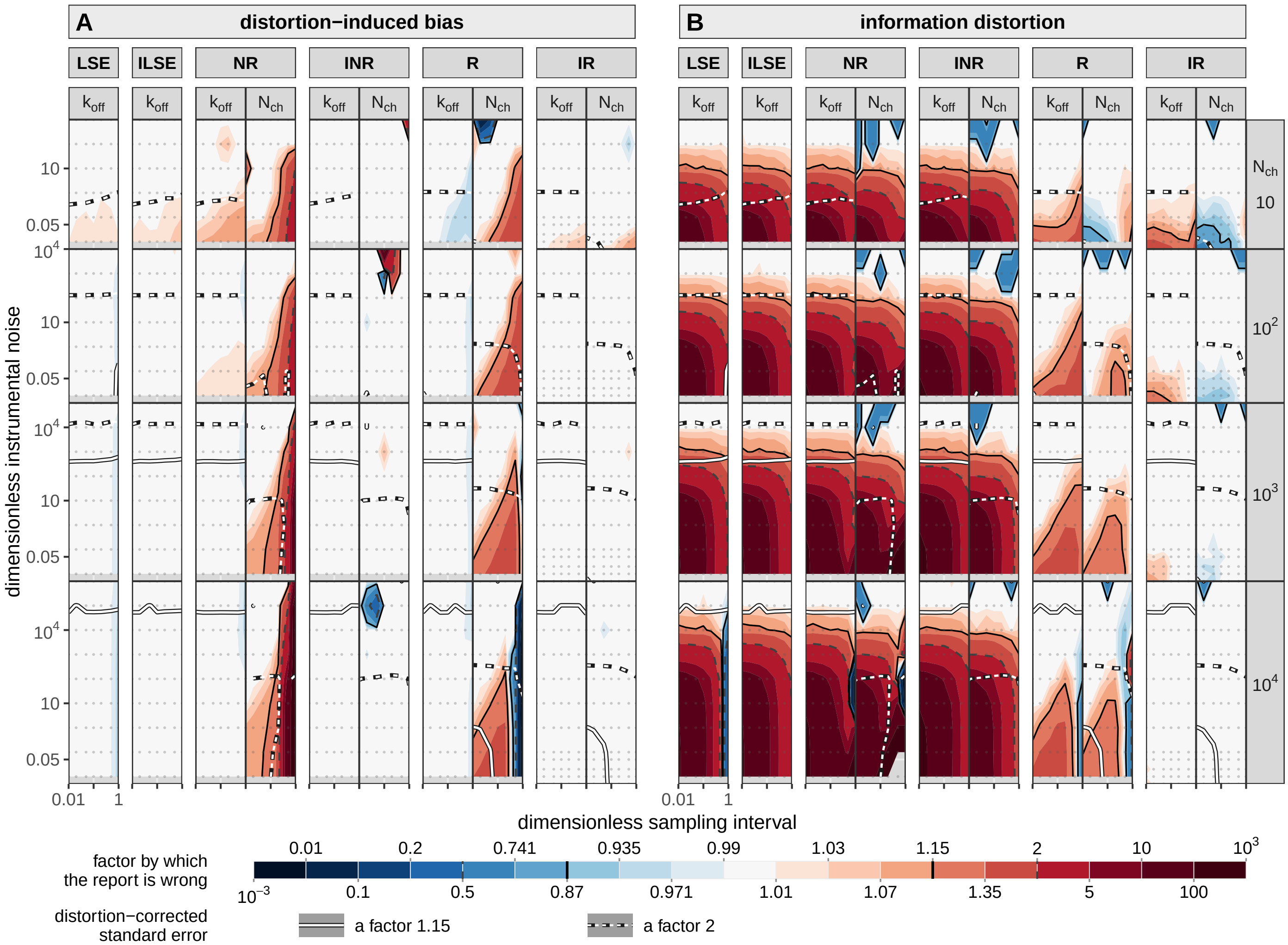


**Figure 4. The design plane: how wrong each member's report is, in both moments, on one scale.** Ten thousand recordings are simulated at every cell of the grid and scored by each member of the cost ladder. (A) The distortion-induced bias, evaluated at the simulation truth $\theta_{\text{sim}}$: how far the estimate is displaced. (B) The information distortion, evaluated at the pooled optimum $\theta_{\text{pool}}$ (Methods): how far the uncertainty a member reports is from the one it delivers. Columns nest twice, member and then parameter, $k_{\text{off}}$ and $N_{\text{ch}}$. Both least-squares arms are on the grid and each holds one column per half rather than two, their four-parameter configuration fixing the unitary current and the noise level at the simulated values (Methods). Rows are the channel count, each given a height proportional to the range of instrumental noise the sweep reaches there. Inside a panel the axes are the sampling interval in units of $\tau$ and the dimensionless instrumental noise, one decade the same distance on both. Colour is shared by the two halves: the factor by which the report is wrong, centred on one, pale right and saturated wrong in either direction, blue conservative, clipped at $10^{\pm 3}$. What a factor means differs between them: in (A) it is a factor on the parameter, so $2$ means the estimate is out by a factor of two, while in (B) it is a ratio of variances, so $2$ means the reported error bar is $\sqrt{2}$, about $40\%$, too narrow. Lines carry two codes: colour says which quantity, dark hairlines cutting the colour field and white lines on a dark casing cutting the distortion-corrected standard error, which asks whether the parameter can be measured at all rather than whether its error bar is honest; dash says which criterion, solid a factor $1.15$ and dashed a factor $2$, the two thresholds used throughout.

**Figure 4—figure supplement 1.** Figure 4's second moment factored into a per-sample part and a temporal-correlation part, over the same plane.

**Figure 4—figure supplement 2.** Can the distortion be undone by counting fewer intervals? The magnitude of the information distortion against its anisotropy.

**Figure 4—figure supplement 3.** The same plane for the four rungs of the recursive family, in both moments.

**Figure 4—figure supplement 4.** What each member achieves: the distortion-corrected standard error.

The anchor itself was checked over the full roster against a differenced Fisher information (Supplementary File 1). For `INR` and `IR` the analytic Gaussian anchor and the differenced construction coincide within one per cent at every channel count. For `NR` the two part company as channels are added, so its largest factors at $10^4$ channels narrow by about a third when re-anchored on the differenced form, without being overturned; at the coarsest intervals there the differenced Fisher is not positive at all, the same ill-conditioned corner the figure greys. For both least-squares arms re-anchoring would enlarge the reported factors by about $1.3$, so their verdict is conservative under the anchor.

The flat floor comes from the per-sample and correlation split: the two members are equally faithful to a single interval and part company on how much temporal dependence they leave in the score, which conditioning on both endpoints removes (Figure 4–figure supplement 1).

Two features of the plane are easy to misread. Where the acquisition interval is far shorter than the channel's correlation time, successive samples carry redundant information and a faithful likelihood has to register that redundancy, so a large correlation distortion at the shortest intervals of (B) is the correct cost of oversampling rather than a defect. And at $N_{\mathrm{ch}} = 10^4$ with $\widetilde{\Delta} = 1$ the amplitude directions of `R` are not identified: the Gaussian Fisher information is ill-conditioned there, so a displacement that costs almost no likelihood is amplified into a large parameter shift. Those cells are drawn grey in both halves, $23$ of them in (A) and $73$ in (B); their sign and their existence are real, their magnitude is a property of the conditioning.

## Deflating the interval count does not repair the error bar, because the distortion differs by direction

A reader who accepts the distortion will reach for the cheapest repair before changing likelihood: declare that the recording holds fewer independent intervals than it appears to, and deflate the count. That number is already measured, and Figure 5 carries it as an axis. The integrated autocorrelation of the standardized residual reads $1.008$ for `IR`, $1.30$ for `R`, $3.93$ for `NR` and $9.26$ for `ILSE` at $\widetilde{S} = 0.01$. The classical fit therefore behaves as though it had about nine times more independent observations than the recording supplies, and the integrated autocorrelation is a quantity an experimentalist can compute on a real recording without an ensemble and without a known truth.

Read along that axis the family splits on the recursion and on nothing else (Figure 5). From a white classical residual out to one carrying twenty times its own length, `ILSE` and `INR` run from calibrated to factors of $57$ and $33$, while `R` and `IR` stay inside $1.26$ and $1.08$ over the same sweep. The split has a reading in the residuals themselves: across the whole grid `IR` leaves its own record carrying between $0.98$ and $1.07$ of its own length, where the least-squares arms leave up to $17.4$ (Figure 5–figure supplement 1). The channel count and the noise level enter only through their ratio, which is the instrumental variance against the gating variance the channels carry, and that ratio is what the measured statistic reads: at matched $\tau_{\mathrm{int}}$ the least-squares distortion agrees to within $1\%$ across four decades of channel count and the five of instrumental noise that accompany them, and holding the ratio fixed holds $\tau_{\mathrm{int}}$ to within $3.5\%$ and the distortion of every member to a median of $1$ to $2\%$, with the exception of the coarsest sampling, where the largest patch is the ill-conditioned corner and least squares there disagrees between channel counts at the same ratio by a factor of $13.6$. Drawn against each of the two axes in turn, the four channel counts are four separate curves against the noise and one curve against the ratio (Figure 5–figure supplement 2). The acquisition interval does not, which is why the panel carries it in colour, and it is the one other number the reader already has: at $\tau_{\mathrm{int}} \approx 2.9$ the same measured memory carries a distortion of $8.9$ at the shortest interval and $1.6$ at $\widetilde{\Delta} = 0.5$. The relation is a mechanism and not a coincidence of that axis: scored against the memory each member leaves in its own residual rather than in the classical one, the two track each other in every member that has any memory to leave (Figure 5–figure supplement 1). It runs one way only. The residual a member leaves white is no certificate, `NR` misreporting its information by up to a factor of $2217$ in cells where its own residual is indistinguishable from white, which is the same blindness a recursive likelihood has by construction, stated at

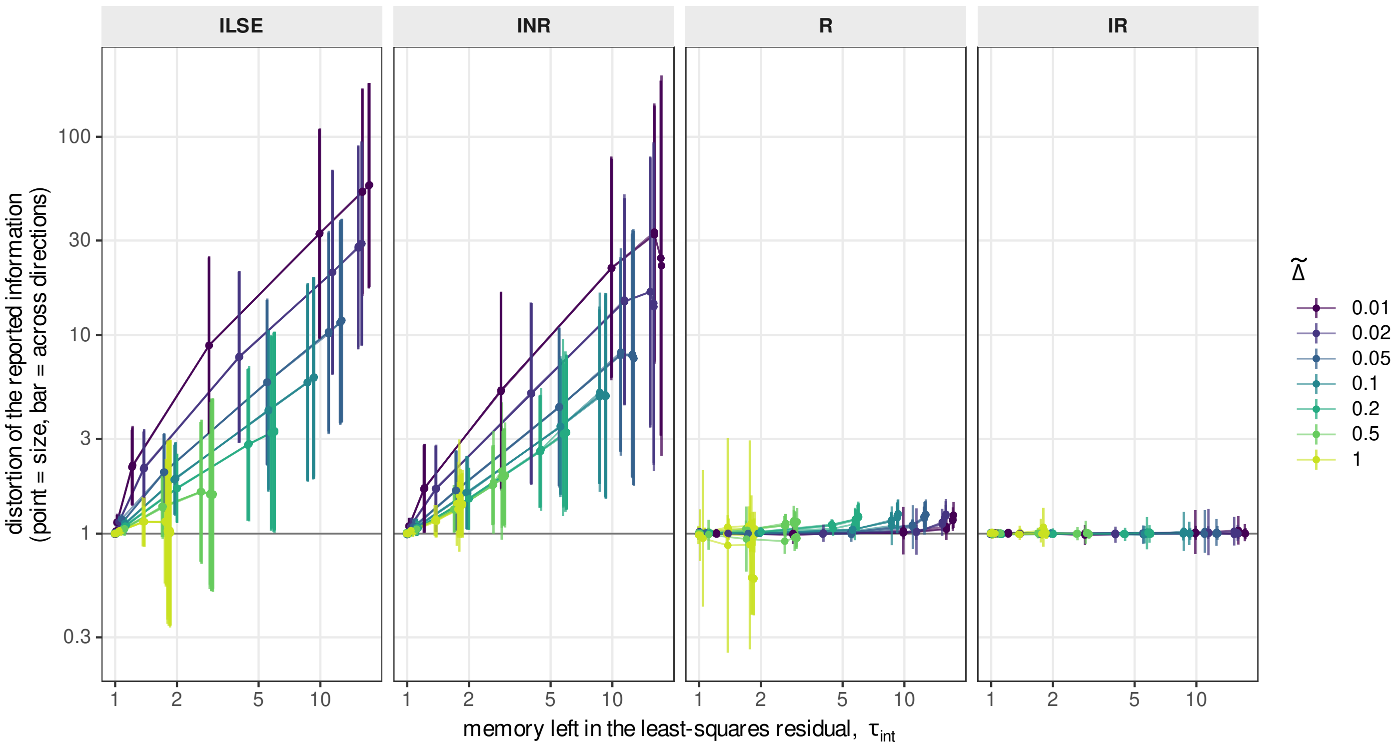


**Figure 5. What each likelihood does to the reported error bar, against a number measurable on a single recording.** Every mark is one cell of the design grid, scored over the same $10{,}000$ recordings as Figure 4, with no binning. The horizontal axis is shared by the four panels and is a property of the least-squares fit alone: $\tau_{\mathrm{int}}$, the integrated autocorrelation of its standardized residual in normalized form, which reads $1$ for a white residual. The vertical axis, also shared, is the distortion of the reported information: the point is its magnitude, the geometric mean of the eigenvalues of the distortion matrix, and the bar runs from that magnitude divided by its anisotropy to the magnitude multiplied by it, so the bar's ends are the reader's best- and worst-determined directions. The error-bar scale is its square root. No threshold is drawn, the criterion belonging to the reader. Colour is the acquisition interval $\widetilde{\Delta}$. Each line joins one channel count at one interval swept over the instrumental noise, so twenty-eight are drawn in each panel and seven can be seen, the four channel counts landing on one another (Figure 5–figure supplement 2). The four members are the two axes of the family crossed (Table 1). Each panel reports what its own fit says about its own parameter set (Methods); nothing is pooled.

**Figure 5—figure supplement 1.** Each member against the memory left in its own residual, rather than in the least-squares residual.

**Figure 5—figure supplement 2.** The instrumental noise and the channel count act only through their ratio.

the start of this section.

That repair works if the distortion is a scalar, every direction wrong by the same factor. The bar on every mark of Figure 5 is that test taken one cell at a time, and Figure 4–figure supplement 2 takes it over the whole plane, through the magnitude and the anisotropy of the distortion (defined above). It is not a scalar. Both least-squares arms and the open-loop interval member over-state their information by a factor of about $5$ to $6$, and the per-direction factor spreads by about $3.2$ around it, flat across four decades of channel count, so directions exist that are wrong by twenty and directions exist that are barely wrong at all. No single effective count serves both, and the claim a reader would otherwise make, that the residual autocorrelation supplies the correction, is refuted by the same measurement that would have supplied the number: the two boundaries of the map below are boundaries on which likelihood to use and not on which correction to apply.

The three recursive members separate on different axes of the same pair. `R`'s magnitude drifts up with the channel count while its spread sits flat at about $1.18$, so its failure is mostly a size it could in principle correct for. `IR` is at one in magnitude at every count and its spread falls toward one; at ten channels that spread is $1.25$, above the strict criterion, which is the same few-channel corner every other measurement in this section points at. `MR` is worse than `R` on both axes and at every channel count.

The same plane over the four rungs of the recursive family adds the second partial correction (Figure 4–figure supplement 3): `VR` sits on `R` along the rates and above it on the channel number. Neither partial correction buys anything on this plane, and the second endpoint is what moves the closing rate to $1.01$.

The interval average is doing two different jobs, and the plane separates them where a single cell cannot. In the second moment the two least-squares arms are the same member at every acquisition interval: their distortion on the closing rate reads $121.0$ against $121.9$ at the finest window and $1.98$ against $2.04$ at the coarsest, and stays within three per cent of itself in all seven. In the first moment they are the same member too until the window reaches half a relaxation time, where the un-averaged arm picks up an offset of $0.013$ in $\log_{10}$, a factor of $1.03$, and the averaged one stays at $0.0009$. So to a fit that does not model the gating variance the interval average does one thing only, which is to keep a window comparable to the kinetics from aliasing the deterministic mean. To the two members that do model it the same step is worth a factor of $9$ to $50$ on that offset at every interval, and a factor of $240$ on the channel number at the coarsest.

The interval axis carries one reading of its own. The distortion of every member that does not recurse scales as the inverse of the window, each halving of the window roughly doubling it, since a shorter window puts more correlated intervals into the same recording and the information-matrix equality fails on the count of them rather than on their length. The two recursive members are flat instead, `R` reading between $1.41$ and $1.50$ at every window, so what it gets wrong is not a property of the window at all. The one departure from the scaling is `NR` at the coarsest window, which reads $70$ against the $6.4$ of the window before it and sits in the same ill-conditioned corner as the greyed cells above.

### Sampling a hundred times faster buys the noise level and nine per cent of a rate

Everything so far asks whether a member's report is honest. The complementary question is what the report is worth, and it is answered by the standard error each member actually delivers, the square root of the diagonal of its distortion-corrected covariance (Figure 4–figure supplement 4). Read along the acquisition interval it says something the design axes do not otherwise make visible: sampling faster is not more data, it is more *independent* data, and only the parameters that live in the uncorrelated part of the signal collect it.

Sampling a hundred times faster, from $\widetilde{\Delta} = 1$ to $\widetilde{\Delta} = 0.01$ at $N_{\mathrm{ch}} = 10^4$, divides the error on the instrumental noise level by $11.8$ to $20.5$ across the noise sweep, close to the factor of ten that a hundredfold increase in independent samples would give and never far from it. It divides the error on the channel number and the unitary current by $6.7$ and $7.7$ at the lowest noise every member reaches, and that gain decays monotonically with instrumental noise, to $1.7$ at $\widetilde{S} = 10$ and to nothing beyond. And it divides the error on either rate by at most $1.09$, anywhere on the plane: a hundredfold increase in samples buys a rate nine per cent.

The three answers are one statement, ordered by the correlation time of whatever each parameter is measured from. The instrumental noise level is the scale of a white process, so every additional sample is independent and its relative error follows the sample count: measured against the interval, the log-log slope is $0.60$ for `IR` and $0.60$, $0.52$ and $0.58$ for the other three that carry it, against the $0.5$ of an inverse square root, while over the whole noise sweep the same quantity varies by a quarter. It is the one parameter whose error the noise level does not set and the sampling rate does. The amplitude pair is measured from the gating fluctuation, which is correlated on the scale of the kinetics, so faster sampling buys the part of its spectrum above that scale, which is real and bounded, and buys nothing once the instrumental noise has covered the fluctuation. The rates are measured from the shape of the deterministic transient, which any interval shorter than the relaxation already resolves, so the extra samples are redundant. Averaging $n$ samples of a correlated process divides the variance by $n/\tau_{\text{int}}$ and not by $n$, and $\tau_{\text{int}}$ is the quantity Figure 5 puts on its horizontal axis.

One consequence is practical and belongs to the reduction rather than to the acquisition. The sampling rate of the rig is not what this axis is: a record acquired fast and uniformly is reduced before it is fitted by grouping consecutive samples and averaging them, and the average of window averages is the average over the window they span, so the reduced record is the same observable at a larger $\Delta$. What the numbers above say is what that grouping costs. For a kinetic question it costs almost nothing, and for an amplitude one it costs the whole gain. The interval members take the window as an argument rather than as a constant, so the groups need not be equal and the schedule can follow the transient; the members written for an instant describe a different random variable once the record is grouped at all.

## In the few-channel corner the boundary-conditioned member misreports two parameters in opposite directions

Figure 6 resolves the corner where `IR` departs, on a scale Figure 4B cannot show. Finding that corner was half the object of the measurement, since a diagnostic that never fires on the member it is applied to says nothing about what it can resolve.

The departure is two-sided within a single cell. At ten channels and $\widetilde{S} = 0.005$ the reported error bar of the closing rate is too small by a factor of $1.62$ at the shortest interval and that of the channel number too large by a factor of $0.65$, so one likelihood is over-confident about a rate and conservative about a count in the same recording. The two parts of the decomposition answer to the acquisition interval in opposite directions: the correlation part of the closing rate falls as the interval coarsens while the per-sample part rises, so sampling more coarsely buys back the memory and pays for it in the per-sample non-Gaussianity. Both parts vanish by $\widetilde{S} = 1$, where the instrumental noise has made the emission Gaussian enough that neither closure is strained.

Two limits travel with the figure. The family applies both of its Gaussian approximations at every step and the score sees only their product, so a departure measured here cannot be assigned to one of them; separating them needs the microscopic recursion, a companion paper's subject. And the grid is truncated where the effect grows: the closing rate's departure is still accelerating at the lowest simulated noise, reading $1.22$, $1.37$ and $1.52$ over the last three noise steps at ten channels, so the figure locates the onset and not the far side.

## No measured regime carries amplitude information and an honest classical error bar at the same time

Figure 7 collects the measurements into one plane in which the measured cells are the input and the four boundaries are the output.

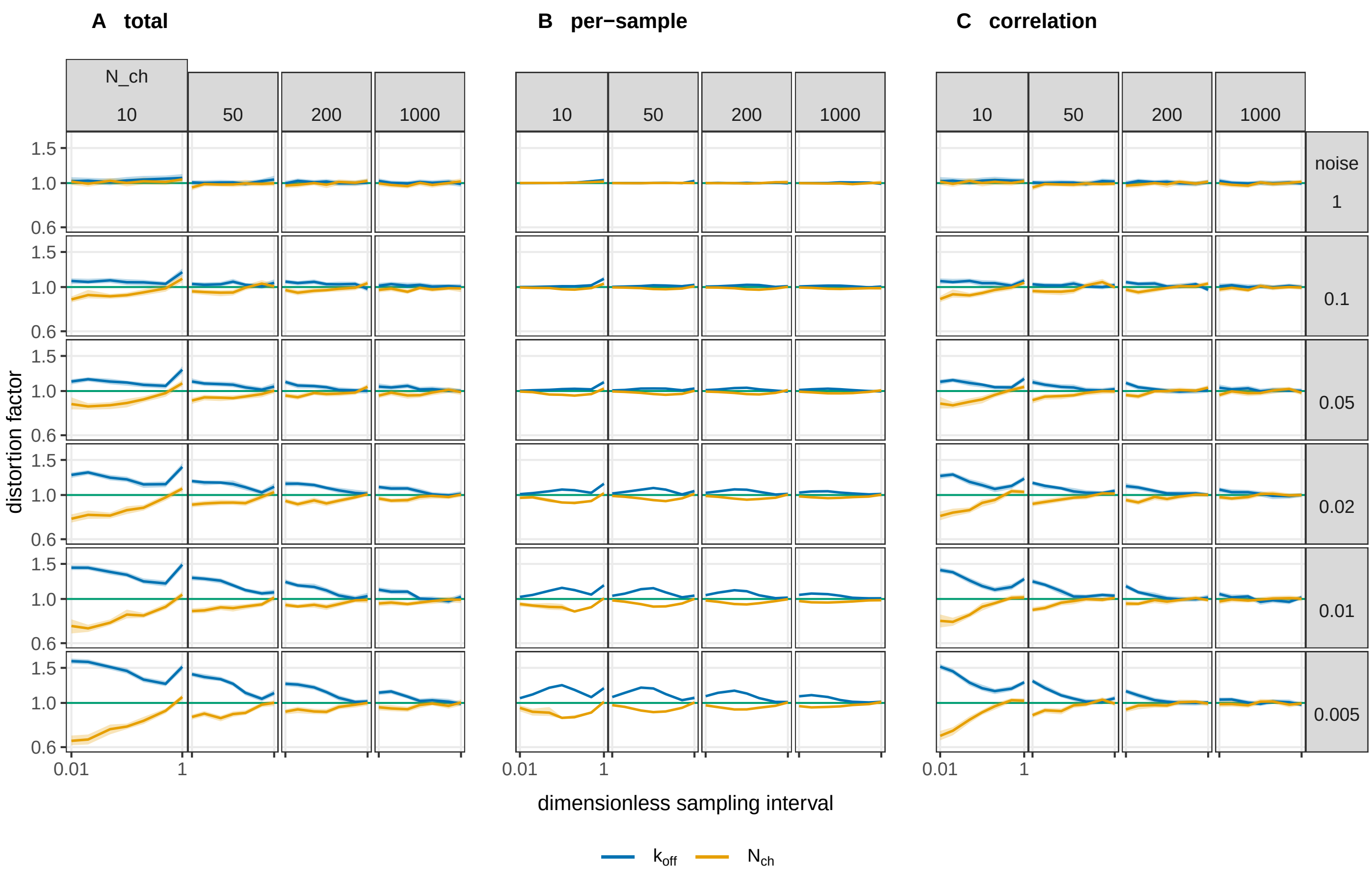


**Figure 6. Where the boundary-conditioned filter stops reporting an honest error bar, and which of the two discarded moments is responsible.** The same measurement as Figure 4, the Gaussian information distortion at the pooled optimum on the same runs, resolved on the corner where $\mathrm{IR}$ departs: on Figure 4's scale a factor of $1.5$ is a faint tint and here it needs an axis. The three blocks are the distortion and the two parts it decomposes into: (A) the total, (B) the part carried by the per-sample non-Gaussianity of the score, and (C) the part carried by its correlation across intervals; their product reproduces the total to within $1\%$ over the whole grid. Each block is a grid of the design plane in the orientation of Figures 4 and 7: the channel count increases to the right, the dimensionless noise $\widetilde{S}$ increases upward, and inside every cell the horizontal axis is $\widetilde{\Delta}$ from $0.01$ to $1$, shared by every panel and labelled once per block. The vertical axis is the distortion factor on a logarithmic scale, shared by all twelve columns so that they can be compared, with the green line at $1$ marking a report that is neither over- nor under-confident. Colour is the parameter. Bands are $95\%$ percentile bootstrap intervals over recordings, of median half-width $0.03$. The two readings the figure supports, and the limits on them, are in the text.

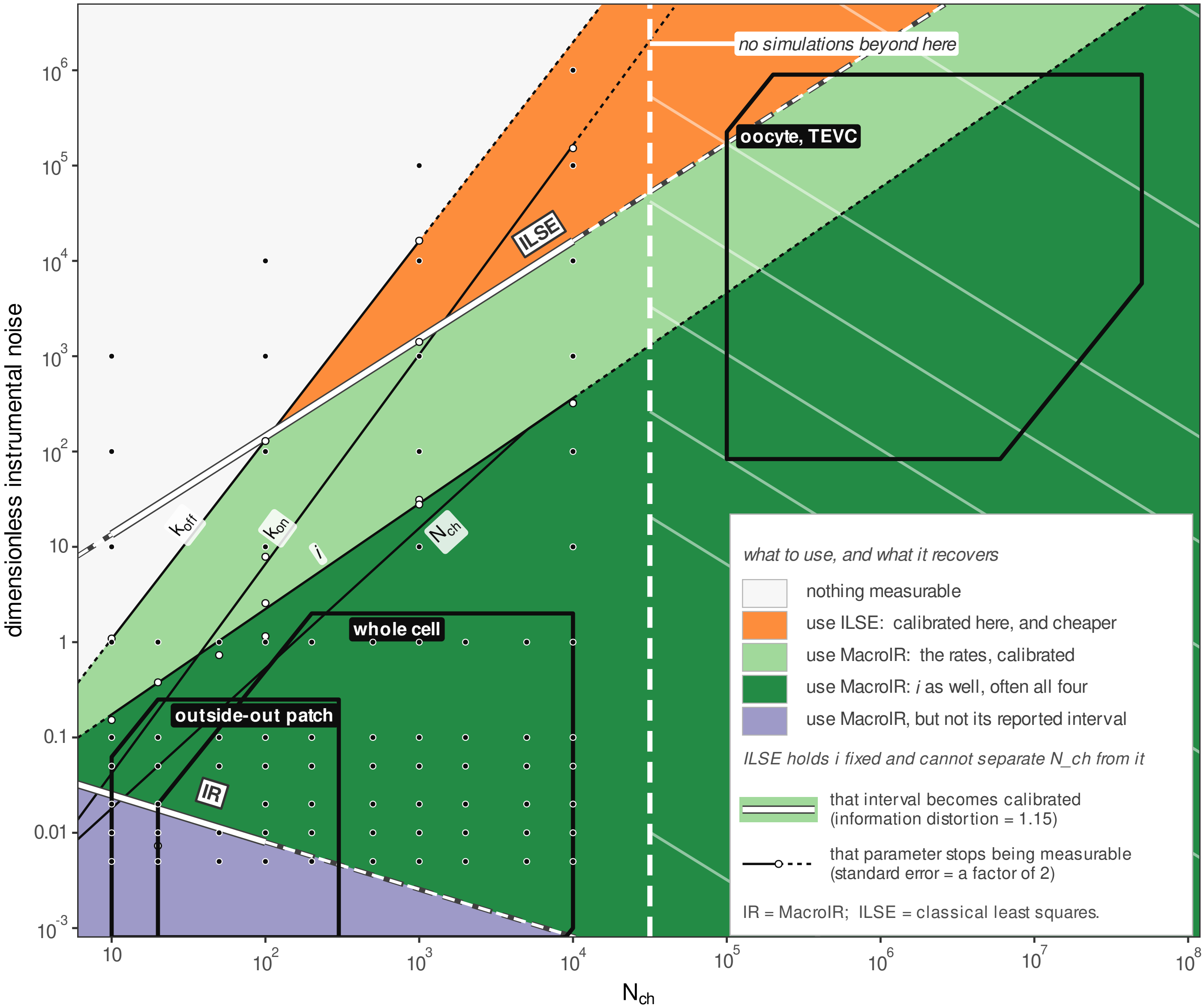


**Figure 7. What a macroscopic recording can be asked for: a usage map over channel number and instrumental noise.** The plane is the design space, the number of channels in the patch against the dimensionless instrumental noise, both on logarithmic scales, at one acquisition interval $\widetilde{\Delta} = 0.1$. Four boundaries divide it. Colour is the parameter the boundary is about. Line type is the question: solid asks whether the parameter can be measured at all, drawn where the distortion-corrected standard error crosses a factor of two; dotted asks whether the reported error bar of least squares is honest, drawn where its information distortion crosses $1.15$; dashed asks the same of the likelihood's own error bar, and is the boundary below which the Gaussian closure is misspecified. Ribbons are the same boundary swept over $\widetilde{\Delta}$ from $0.01$ to $1$, so a ribbon says where a reader's own sampling rate puts the boundary and is not an error bar. Dots are the measured crossings and the line through them is their fitted power law, drawn faint where it leaves the simulated range, which is marked. Vertical strips mark the two vertices, each a range set by the strict and the loose criterion. The boundaries partition the plane into five regions, keyed by what a recording in each can be asked for and named in order in the text, and the three labelled boxes mark where common preparations sit. The map is a concept map, not a phase diagram. Its boundaries are level sets of continuous diagnostics, so a looser criterion moves each of them by up to a decade in noise without changing the layout.

Read upward at a fixed channel count the recording loses capabilities in a fixed order, which gives five regions, numbered from the bottom. Region 0, below the lowest boundary, is where the Gaussian closure is itself misspecified and the recording is not a macroscopic problem, the microscopic regime seen on this plane. In region 1 the likelihood delivers the unitary current, the channel number and an honest interval while least squares reports an interval several times too narrow; in region 2 the amplitude pair stops being separable and the rates alone survive; in region 3 least squares is calibrated too and buys the same answer more cheaply; in region 4 nothing useful is left. The two boundaries that say a parameter has stopped being measurable are level sets of the delivered standard error introduced above, drawn over the whole plane for every member and all five parameters in Figure 4–figure supplement 4, so a reader who wants a criterion other than the factor of two drawn here can take it off that map. The lowest boundary is the one with a negative slope, $-0.56$ in $\mathrm{d}\log_{10}\mathrm{noise}/\mathrm{d}\log_{10}N_{\mathrm{ch}}$, because instrumental noise makes the emission more Gaussian and so relaxes the occupancy closure while hurting everything else. Regions 1 to 4 sit inside the Gaussian regime, and what orders them is how much the data carry rather than whether the closures hold. The telegraphic regime is a statement about $\Delta$, so it is not a region of this plane; it is reached along the interval axis of Figure 4.

The boundary at which the classical error bar becomes honest is the one the mechanism above predicts. Fitted in logarithms over four measured crossings number, with slopes of $1.01$ and $1.03$, a measured slope rather than an assumed scaling.

The ordering of the boundaries is the result the map exists to state: there is no region of the measured plane in which the amplitude still carries information and the classical error bar is simultaneously trustworthy, the two being separated by one to three decades of noise everywhere measured. The scope of that statement has to be stated with it: the classical arm here is `ILSE`, least squares on the deterministic mean current averaged over the acquisition window, with the unitary current held fixed, in its four-parameter configuration. Fixing it is forced and not chosen, since with it free the least-squares information is singular and no maximum-likelihood covariance exists at all, and it is fixed at the value the recordings were simulated from, so the classical arm is helped in this comparison rather than handicapped: two of the six parameters are supplied to it exactly right while every likelihood member estimates all six (Methods). It is not the variance-mean regression: this paper does not benchmark non-stationary fluctuation analysis (***Sigworth, 1980***; ***Heinemann and Conti, 1992***), and nothing here should be read as a measured comparison against it. Which of the two least-squares arms carries the map is not load bearing: recomputing the same boundary on the arm that fits the instantaneous mean moves it by a factor between $0.77$ and $1.39$ in noise across the four channel counts, against the one to three decades that separate it from the boundary below, and the fitted exponent is unchanged at $0.94$ to $1.03$.

Minimising the corrected standard error over the whole noise sweep, the channel number is unreachable below about $17$ channels and the opening rate below about $52$ at this interval, while the unitary current and the closing rate have no floor within the simulated range.

The evidence is thinner than the picture, and the subsection says so rather than leaving it to a caption. The two boundaries at which least squares becomes honest, and the closure floor, are each fitted through four measured crossings. The channel count at which the third region opens spans $120$ to $1368$ depending only on which criterion is used. The lower vertex is extrapolated to $N_{\mathrm{ch}}$ between $3$ and $9$, below ten channels, the floor of the classical arm's grid, and the ten cells that would pin it have not been run.

## Discussion

Over most of the design space the boundary-conditioned member reports an uncertainty close to the one it delivers, and where it does not, the departure is measured rather than assumed. An experimenter can use that member everywhere; where its reported interval is the wrong width is the few-channel corner, and there the repair is the sandwich the diagnostic already supplies (Results).

What buys the calibration is conditioning at both ends of the acquisition interval. Conditioning on the start alone does not buy it, and at the reference cell it costs: the one-endpoint member books the spread of the interval mean across end states as observation noise, while the boundary-conditioned member carries it in the numerator of the gain of Eq. A20 (Appendix 2). The whole distance between the two lies in the update, and a supplement takes that step in its two algebraic halves, with the qualification that the predictive variance divides the gain, so the halves are not a clean separation. The diagnosis is ***Milescu et al.*** (***2005***)'s local time correlation, now measured (Results). Conditioning on both ends leaves a white score, the innovation whiteness that has certified an optimal filter since ***Mehra*** (***1970***).

A method that cannot resolve a mechanism cannot mislead anyone about one, which is why this matters more as the inference improves. The flip state we proposed for $P2X_2$ (***Moffatt and Hume, 2007***), later confirmed (***Jiang et al., 2012***), carried a signature anyone could point to in the record. The asymmetric activation we reported from the same recordings two decades later has no such signature and rests on ratios of evidence between schemes ***Moffatt and Pierdominici-Sottile*** (***2025***), so whatever an approximation gets wrong about the information in a record passes into the mechanism it selects. The full distortion matrix is reported because near the maximum it propagates into the Bayesian evidence, so a likelihood that misreports it carries the error into every comparison built on it, and the covariance repair does not reach it (Appendix 4). The control that reordered the schemes is this study's interval likelihood without the recursion (Methods); at macroscopic channel counts it misreports its information about twentyfold, and the misreported factor is exactly the one the recursion restores. Those recordings are in the regime where this matters: their variance is dominated by the gating fluctuation rather than the instrumental noise, measured in the same paper (***Moffatt and Pierdominici-Sottile, 2025***). Which of the two orderings is right needs the evidences recomputed, which is not done here. That would settle which scheme the recordings prefer, not which mechanism is true: no quantity of recorded data identifies the generating model, which is why this diagnostic needs a simulator whose model is known.

That correlation also tells an experimenter which regime they are in. Over the measured plane the channel count and the instrumental noise act only through their ratio, so the integrated autocorrelation of the standardized least-squares residual reads the regime off a single recording, with no ensemble and no known truth, and Figure 5 says what each member would then do to an error bar at that reading. The reading is necessary and not sufficient: memory implies distortion, whiteness certifies nothing (Results).

Once the agonist is removed, only the closing rate and the unitary current stay informative (Results; Figure 3–figure supplements 1 and 2): an experiment should spend its samples on the jump and its rise.

A second result speaks to how a record should be reduced before it is fitted. Grouping consecutive samples is the same measurement at a larger $\Delta$, with a proportionately smaller instrumental variance and no sample discarded (Results), so the temporal structure of the record becomes something to design: a record can be reduced by orders of magnitude for a kinetic question and hardly at all for an amplitude one; after a concentration change the groups can grow geometrically with the transient they follow, which is legitimate rather than an approximation because the likelihood takes the window as an argument and not as a constant; and the cost of the fit follows the design, since the recursion is linear in the groups it is given.

The ordering of the boundaries is what an experimenter can act on: everywhere on the measured plane, one to three decades of noise separate the point at which the fluctuations stop carrying the unitary current or the channel number from the point at which the classical error bar becomes honest (Figure 7). A reader can run the two identities on their own scheme, at their own open probability, from one call in R or Python (Data and code availability).

As an algorithm, MacroIR is an integrated-measurement augmented Kalman filter, a device long known in control and target tracking ***Zadrozny*** (***1988***), and its predictions coincide with that construction to about $10^{-8}$. The device is conceded; what is specific here is the realization for an exact continuous-time Markov chain of channel occupancies, its differentiation through the recursion, its use from $5$ to $10^4$ channels where the single-molecule integrated filters do not reach, the validity map, and the measurement of what the cheaper members cost. The derivatives are what the diagnostic could not be run without. ***Milescu et al.*** (***2005***) computes them analytically for a likelihood over independent intervals; through a recursion the chain runs the length of the record, which puts the derivative of the spectral reconstruction inside it (Appendix 1) and reaches the filter itself: the occupancy is held on the probability simplex by a smooth damping of the update, because an operation not differentiable where it binds pumps variance into the score (Methods). The lineage is ***Moffatt*** (***2007***) and ***Münch et al.*** (***2022***); the family's first macroscopic application is ***Moffatt and Pierdominici-Sottile*** (***2025***).

***Münch et al.*** (***2022***) asked this paper's question of a filter of this family and answered it by counting coverage over repeated simulated data sets. Three things separate that from what is done here. What their count tests is a posterior, so the prior and the sampler are inside the verdict together with the likelihood; the identities used here are evaluated on the likelihood alone, at parameters that are known, with no prior and no sampler. What their count returns is a probability that a region holds the truth; what is returned here is a matrix, saying by how much and in which parameter the reported uncertainty departs, and separating what each interval commits from what accumulates across them. And a coverage rate costs one posterior fit per replicate at every design point, which keeps such a study to a line of conditions, while a score and an information at known parameters cost one pass each, which is what puts $560$ cells within reach. Their filter maps the instantaneous occupancy to the observation rather than conditioning on the ends of an acquisition interval, which places it with the instantaneous recursive member; what it generalizes is the emission, through state-dependent open-channel noise and a fluorescence observable, and that generalization costs them the analytical update and obliges a two-moment closure of their own, the same obstruction the update of Eq. 2 meets, since the state-dependent term makes the incremental likelihood intractable once "the conjugacy property is lost". Two groups reached one device from different motivations.

***Del Core and Mirams*** (***2025***) argue against filtering on cost, since the moment integration and the correction step run between every pair of samples. The cost is real, and what it buys is now quantified in units of misreported uncertainty. How much cost it is depends on the scheme and the implementation rather than on anything measured here, but how it grows does not: the work per interval is one spectral decomposition of the $k$-state generator per distinct condition and a fixed number of $k \times k$ products, cubic in the state count, in a single pass over the record, so the growth is polynomial rather than combinatorial; one published figure gives its scale, a nine-scheme Bayesian comparison for a ligand-gated receptor taking twelve to fifty-two days per scheme on sixteen to thirty-two cores with a member of this family ***Moffatt and Pierdominici-Sottile*** (***2025***).

Where exchangeable sweeps are plentiful, the classical route is strong and should be said to be. Non-stationary fluctuation analysis (***Sigworth, 1980***; ***Heinemann and Conti, 1992***) returns the channel number and the unitary current with no likelihood at all and a sweep-level bootstrap returns honest intervals with no model, and whether that suffices depends on how many sweeps a preparation tolerates and whether they are exchangeable; how it composes into a complete workflow is not addressed here. The case for a filter is strongest where that route is weakest: few sweeps, rundown, conditions that are not exchangeable, a single precious recording, and an experiment that wants kinetics and amplitudes jointly from the same record. What the alternatives that repair an interval without changing likelihood answer instead is in Appendix 4.

## The perimeter of the measurement

The boundary-conditioned member is an approximation and is not exempt from the failure this paper is about. Over the same grid its distortion runs from $0.645$ on the channel-number direction at $N_{\mathrm{ch}} = 10$, where it over-states its own uncertainty, to $1.701$ on the closing rate at $N_{\mathrm{ch}} = 5$, where it under-states it, so the departure is two-sided and does not become safe by erring conservatively on average. Both excursions sit where the open population is a small integer count, where the occupancy Gaussian is expected to degrade, and non-normality on its own cannot produce a factor-two distortion anywhere on the grid. What produces the departure is not known. The decomposition puts it in the term that accumulates across intervals, and the five candidate criteria measured against the observed boundary were all criteria on the per-sample cumulants of the occupancy, so all five failed to locate it; the measured channel-count exponent of $-0.25$ matches neither the $-0.5$ a skewness mechanism would give nor the $-1.0$ of a kurtosis one, and it is not a clean power law. Stated as a conjecture and in those words: the limits behave like a balance that has to come out just right, several quantities moving together in a way that leaves the departure small over four decades, which is the signature of a conserved quantity or a comparable invariant that has not been identified here.

A second boundary is the measurement model. The instrumental contribution is taken to be Gaussian and white, as the simulator emits it (Methods); where a recording departs from that, through a trapped charge in the headstage or amplifier flicker, the diagnostic would report a distortion belonging to the noise model and not to the gating closure, and we have not run that; unmodelled discrepancy of this kind is documented to produce exactly this overconfidence in whole-cell current fitting (***Lei et al., 2020***).

A boundary of the roster itself: the family carries no open-loop member with the full cross-interval covariance, the generalized-least-squares route of ***Celentano and Hawkes*** (***2004***) and ***Stepanyuk et al.*** (***2011***), so the ladder measures what conditioning buys within this family and does not separate recursion from correlation modelling as such; that rung would test the distinction and is not run here.

A third boundary is the diagnostic's own. The comparison is between two independent implementations, the likelihood and the exact simulator, so everything downstream of the point where they separate is tested: the propagation, the update, the emission arithmetic, the accumulation. Upstream they share one specification, the rate matrix built from the parameters, the conductance vector, the protocol and the sampling times, and a diagnostic built on their agreement cannot see an error there. The general remedy is different in kind, verifying each object against the theorems it must satisfy, and that is a piece of work with its own design, not undertaken here.

The perimeter is a model, and Methods declares it as one: two states at an open probability of one half, a single concentration jump, simulated data, likelihood only. The fixed open probability means a channel count and an opening count differ throughout by a factor of two that no measurement here can settle. What the channel-number axis carries at $p = 0.5$ is the gating variance, so a reader whose receptor sits elsewhere can enter the map at $4Np(1-p)$ in place of $N$; the substitution is exact for the variance and says nothing about the skewness, which vanishes at $p = 0.5$ and which no cell here probes. Richer schemes, the stationary regime, the few-channel boundary, the filtered kernel and experimental data are named companions rather than omissions, and the kernel is one for a stated reason: the condition that makes its residue negligible is hardest to meet at windows short against the relaxation, which is where the boundary-conditioned member's advantage is largest.

What this section adds up to is one hierarchy, stated once: that the tested family misreports its information on this scheme, where, by how much and in which directions, is demonstrated; that the mechanism, temporal correlation the likelihood does not carry, operates in richer schemes is suggested by the mechanism's generality and not shown; that the two-state case bounds anything richer is conjectured; and what a corrected covariance would change in a recomputed evidence is not tested.

What this changes is what the paper hands over. The reader supplies their sweep count, their rundown, their model size and their tolerance, and decides; what they cannot measure for themselves, and what is supplied here, is whether a reported interval is the interval that is delivered. A mechanism once left a mark anyone could point to; choosing between mechanisms now means navigating by likelihood, and the likelihood is the instrument, which has to be calibrated before what it selects can be believed.

# Materials and methods

## Minimal model and simulation

We work with a two-state channel scheme, closed ($C$) and open ($O$), the smallest scheme in which the acquisition-averaging effect we study is present. The closed-to-open transition is agonist dependent with association constant $k_{\mathrm{on}}$ multiplying the agonist concentration; the open-to-closed transition is agonist independent with dissociation rate $k_{\mathrm{off}}$. Only the open state conducts, and the code applies a sign flip so the single-channel current is inward. At the start of every recording all channels are closed, so the occupancy is $(1, 0)$.

The scheme carries six parameters, all estimated in base-ten logarithmic coordinates. The values used to generate the data reported here are $k_{\mathrm{on}} = 10\ \mu\mathrm{M}^{-1}\,\mathrm{s}^{-1}$, $k_{\mathrm{off}} = 100\ \mathrm{s}^{-1}$, a unitary conductance $g$ set to $1$ (the engine parameter `unitary_current`, written $i$ where the noise axis is defined below), and a current baseline $b$ set to $1$ (both in units of $g$), while the white-noise level and the mean channel number were swept (below). Current is expressed throughout in units of the unitary conductance $g$, which corresponds physically to a single-channel current of roughly $1\ \mathrm{pA}$. The dissociation time $\tau = 1/k_{\mathrm{off}} = 10\ \mathrm{ms}$ sets the natural time scale that normalises the acquisition-interval axis. The relaxation at the agonist condition is faster, with rate $k_{\mathrm{on}}[A] + k_{\mathrm{off}} = 200\ \mathrm{s}^{-1}$; the axis uses $\tau = 1/k_{\mathrm{off}}$ by convention. During the agonist step the two rates are equal, $k_{\mathrm{on}}[A] = k_{\mathrm{off}} = 100\ \mathrm{s}^{-1}$, so the open probability the sweep works at is $p = 0.5$. That is where the gating variance of the open population, $N p(1 - p)$, is largest and where its skewness, which goes as $1 - 2p$, vanishes.

The emission model is white instrumental noise only. Both the pink (one-over-$f$) and the proportional noise channels are set to zero, so the term the reader may expect from flicker noise is genuinely absent. For a measurement interval $t$ of duration $\Delta_t$, the predicted current has mean $\bar{y}_t = N_{\mathrm{ch}}\, \mu_t \cdot \gamma_t + b$ and variance

$$\sigma_t^2 \;=\; e_t \;+\; N_{\mathrm{ch}}\, v_t, \qquad e_t \;=\; \mathrm{Current\_Noise}\,\frac{f_s}{n_{\mathrm{samp},t}} \;=\; \frac{\mathrm{Current\_Noise}}{\Delta_t},$$

where $N_{\mathrm{ch}}$ is the channel number, $\mu_t$ the occupancy mean, $\gamma_t$ the interval-averaged conductance, $f_s$ the raw sampling rate, $n_{\mathrm{samp},t}$ the number of raw samples averaged into the interval, $e_t$ the white instrumental variance over the interval, and $N_{\mathrm{ch}}\, v_t$ the channel-gating variance. The white variance $e_t$ falls as the interval lengthens, because more raw samples are averaged into each recorded value. The gating variance $v_t$ is the single-interval conductance variance, whose form differs across the algorithm family and is derived in Appendix 1.

The stimulus is a single concentration jump in three segments, a pre-pulse at $0\ \mu\mathrm{M}$, an agonist step at $10\ \mu\mathrm{M}$, and a post-pulse at $0\ \mu\mathrm{M}$, with raw sampling at $f_s = 50\ \mathrm{kHz}$ throughout. Each recorded measurement is the mean of $n_{\mathrm{samp},t}$ consecutive raw samples, so the averaging window is fixed at acquisition. The averaging is uniform over the window and no analogue filter stage is simulated, so a recorded value is a boxcar average of the trajectory, which is the observable every likelihood in the family is written for (Eq. 1). Recordings from a real amplifier are Bessel filtered instead. That kernel is not simulated here, and how much of it the uniform window misses is bounded in Theory, as a condition on the acquisition window against the filter's response time rather than as an open size.

Ground truth is generated by exact simulation of the continuous-time Markov chain (CTMC) by uniformization, so the interval average of the conductance is realised from an actual sampled trajectory rather than from any moment-closure approximation. Uniformization draws the transition times themselves, so there is no within-interval discretisation to choose and none to justify. The data analysed here are simulated by design: a diagnostic that compares what a likelihood delivers against what it reports needs a known truth, and the inference throughout is likelihood only, with no prior and no evidence computation. The likelihood and the simulator are independent implementations sharing one upstream specification, the rate matrix built from the parameters, the conductance vector, the protocol and the sampling times, written out symbol by symbol in Supplementary File 1; what that sharing costs the diagnostic is stated with the other limitations in the Discussion. Every maximum-likelihood grid cell reported here, the least-squares arm included, used $10^4$ recordings, with one exception: the cell that carries the coverage result ($N_{\mathrm{ch}} = 10$, $\widetilde{S} = 0.005$, `IR`) was run at $10^5$, because coverage of a nominal $95\%$ region in six parameters needs a thousand independent fits and those fits are taken at group size $100$, so $10^5$ recordings are required to produce them. The time-resolved run used $1{,}000$ recordings; the single-recording illustration (Figure 1) was drawn by the substep sampler instead, one $20$-channel trace at $250$ substeps per interval, and does not enter the quantitative comparison.

The simulation seed was set to $0$, which in this engine is the sentinel for a random seed drawn from `std::random_device`, and the resolved seed was never logged. The deposited ensembles are therefore statistically equivalent to, but not bit-for-bit reproducible from, a rerun of the same script. Because each ensemble is large ($1{,}000$ to $10{,}000$ recordings), the reported statistics are stable; the Monte-Carlo standard here is statistical equivalence, not bit identity.

## The likelihood family and its members

The members are the two axes of the preceding section crossed: whether the gating fluctuations are modelled at all, and how much of the acquisition interval the interval-averaged conductance is conditioned on. This subsection gives the flags each setting is dispatched under and nothing else.

Least squares is selected by the family flag *family approximation* $= 2$ and runs non-recursive; every macroscopic member takes $0$. Three flags set the rest: *recursive*, whether the occupancy covariance is propagated between intervals; the endpoint count $\mathrm{av} \in \{0, 1, 2\}$, which counts how many interval endpoints the hidden state is conditioned on and is the program's name for the window axis of the family; and the *variance form*, which separates `MR` from `VR`. Each least-squares arm shares its mean model exactly with the macroscopic member carrying the same averaging flag, `LSE` with `NR` and `ILSE` with `INR`, and differs from it only in the predictive variance: on the recording of Figure 1 those predicted means agree to $9 \times 10^{-16}$ interval by interval. The dispatched flags, the data keys, the conditioning each member applies and the cells each was run over are tabulated in Supplementary File 1. `INR` is the member published as MacroINR in ***Moffatt and Pierdominici-Sottile*** (***2025***), where it served as the control against which MacroIR, the member called `IR` here, was compared.

`IR` is the top implemented rung: it conditions on the pair of channel states at the two ends of the interval, which we call the boundary state, and marginalises the interior analytically, as specified by the construction of Appendix 1. An interval-recursive likelihood of this family was first applied to macroscopic recordings, of the P2X$_2$ receptor (***Moffatt and Pierdominici-Sottile, 2025***), and it coincides with a time-integrated Kalman filter, a correspondence stated and quantified in the Discussion. Every member scores with Eq. 5 and the regularizer of Eq. 7 below; the correspondence between the symbols of Appendix 1 and the names the implementation carries is in Supplementary File 1.

### The likelihood.

Every member of the family delivers, per window, a predicted mean $\bar{y}_t^{\mathrm{pred}}$ and a predicted variance $\sigma^2_{\mathrm{pred},t}$, and the recorded value is scored under the Gaussian they define,

$$\log q_t \;=\; -\tfrac{1}{2}\Big[\log\!\big(2\pi\sigma^2_{\mathrm{pred},t}\big) + \frac{\delta_t^2}{\sigma^2_{\mathrm{pred},t}}\Big], \qquad \delta_t = \bar{y}_t^{\mathrm{obs}} - \bar{y}_t^{\mathrm{pred}}, \tag{5}$$

summed over windows. The recursion enters only through what $(\bar{y}_t^{\mathrm{pred}}, \sigma^2_{\mathrm{pred},t})$ are conditioned on: a recursive member carries $(\boldsymbol{\mu}^{\mathrm{post}}, \boldsymbol{\Sigma}^{\mathrm{post}})$ of Eqs. A20 and A21 into the next window, so Eq. 5 sums to $\log \prod_t q_t(\bar{y}_t \mid \bar{y}_{1:t-1})$; a non-recursive member propagates from the initial condition by Eqs. A14 and A15 alone and the same sum is $\log \prod_t m_t(\bar{y}_t)$. Nothing else in this section is used by the fit.

The initial condition.

The recursion opens with the occupancy the model supplies and with the covariance that occupancy implies for one channel,

$$\boldsymbol{\mu}_0 = \boldsymbol{\pi}, \qquad \boldsymbol{\Sigma}_0 = \mathrm{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top. \tag{6}$$

For the schemes run here $\boldsymbol{\pi}$ is the first basis vector: the record opens at zero agonist with every channel shut, so $\boldsymbol{\Sigma}_0 = \mathbf{0}$ and the spread the filter carries is generated entirely by the propagation of Eq. A15. Eq. 6 is also the one moment at which $\boldsymbol{\Sigma}$ genuinely is a single channel's covariance: from the first update onward, conditioning on a shared recorded value correlates the channels, so the recursion carries $\mathrm{Cov}(\mathbf{N})/N_{\mathrm{ch}}$, and every factor of $N_{\mathrm{ch}}$ here comes from that distinction.

The equations of Appendix 1 are written with the objects the implementation forms, and the correspondence is one to one; it is tabulated symbol by symbol in Supplementary File 1, because a specification whose symbols cannot be found in the code is not a specification.

Two reductions are worth stating because they anchor the family to prior work. Setting the boundary-conditioned conductance variance to zero in the recursive, no-endpoint member (R) recovers the macroscopic-fluctuation update of ***Moffatt*** (***2007***), so R is the recursive ancestor already in print. The single structural difference between the M and I families is that the one-endpoint members drop the boundary cross-covariance term $N_{\mathrm{ch}}\,\widetilde{\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}}$ that IR retains, the tilde being the boundary-pair contraction defined at Eq. A18. Every member is written out on one template in Appendix 2.

The remaining build flags were held fixed in every reported run; their settings, and the naming they disambiguate, are in Supplementary File 1.

The filter carries three numerical safeguards. The first fixes what the boundary-conditioned moments are where a transition cannot occur; without the other two the recursive members return no value at all at the smaller channel counts.

The first sits upstream of the filter cycle, in the boundary-conditioned moments of Eq. A3, built from $\mathbf{F}_1(\Delta)$ and $\mathbf{F}_2(\Delta)$, the two derivatives of the conductance-tilted semigroup carrying the first and second moments of the integrated conductance over paths from $i$ to $j$ (Appendix 1). Both are quotients by the transition probability $P_{i\to j}(\Delta)$, which is zero wherever the transition cannot happen within one window: at zero agonist $k_{\mathrm{on}}[A]$ vanishes identically and $P_{C\to O}$ is zero over the whole pre-pulse segment, so the quotient is $0/0$ in exact arithmetic and, in floating point, a ratio of two quantities no larger than the rounding error that produced them. The implementation does not test for that case; it reads each quotient as a posterior mean and adds a conjugate pseudo-count $\varepsilon$ to numerator and denominator alike,

$$\bar{\gamma}_{i\to j} = \frac{[\mathbf{F}_1(\Delta)]_{i\to j}/\Delta \;+\; \varepsilon\,\frac{1}{2}(\gamma_i+\gamma_j)}{P_{i\to j}(\Delta)\;+\;\varepsilon}, \qquad \overline{m}_{i\to j} = \frac{2[\mathbf{F}_2(\Delta)]_{i\to j}/\Delta^2 \;+\; \varepsilon\,\frac{1}{2}(\gamma_i^2+\gamma_j^2)}{P_{i\to j}(\Delta)\;+\;\varepsilon}, \tag{7}$$

where $\overline{m}_{i\to j}$ is the second moment of the interval-averaged conductance over the same boundary pair, and the variance follows by subtraction, $\bar{v}_{i\to j} = \overline{m}_{i\to j} - \bar{\gamma}^2_{i\to j}$. The two prior means are the endpoint averages of Eq. A8, so the prior the variance inherits is $\left((\gamma_i-\gamma_j)/2\right)^2$ and is never imposed separately. The weight is $\varepsilon = \varepsilon_{\mathrm{mach}}\,\kappa_F(\mathbf{V})$, machine epsilon times the Frobenius condition number of the matrix of right eigenvectors of $\mathbf{Q}$, which bounds the floating-point noise the spectral reconstruction can leave in $\mathbf{P}(\Delta)$, so the pseudo-count is inert wherever the transition probability stands above that noise, takes over where the quotient carries no information, and introduces no fitted constant. Why $\varepsilon$ is keyed to $\kappa_F(\mathbf{V})$, and why the prior is taken from the conductance vector rather than from the marginals of $\mathbf{F}_1$, is in Supplementary File 1.

The second holds the occupancy mean on the probability simplex. Nothing in the Gaussian conditioning of Eq. A20 knows that $\boldsymbol{\mu}$ is a probability vector, and an undamped step can drive its entries negative or above one, so the mean step is scaled by a trust coefficient $\alpha_\mu \in (0,1]$, the largest scaling that keeps the posterior on the simplex. Damping the step is what replaced correcting the state after it. The form of the damping matters here in a way it would not in an ordinary filter, because this paper differentiates the log-likelihood with respect to $\boldsymbol{\theta}$: a hard $\alpha = \min(1,\, c\,\alpha_p)$ leaves a step in $\partial\alpha/\partial\boldsymbol{\theta}$ at the boundary and pumps variance into the score whenever the system sits near it. The coefficient is therefore taken in the smooth form $\alpha = \frac{1}{2}\left(1 + c\,\alpha_p - \sqrt{(1 - c\,\alpha_p)^2 + \epsilon^2}\right)$, a softened minimum of $1$ and $c\,\alpha_p$ that is infinitely differentiable in $\boldsymbol{\theta}$ everywhere, with $\alpha_p$ the largest scaling that keeps the posterior mean on the simplex, $c = 0.9$ and $\epsilon = 10^{-4}$. Reimplementing this with a hard minimum reproduces the mean current and corrupts every quantity this paper measures. The covariance takes its full step, and the truncate-then-renormalise history, the two untied trusts and the $\epsilon/2$ bias arithmetic are in Supplementary File 1.

The third is a pair of constants: a binomial-count floor of $5$, below which the expected count in a state is too small for the Gaussian treatment of that state and the member falls back to its non-recursive form, and a minimum occupancy probability of $10^{-12}$.

## The two least-squares parameter configurations

Least squares runs in two parameter configurations, and they are different statistical models. Both select *family approximation* $= 2$, both fit data from the same simulator, and both minimise the same sum of squares. They differ in which parameters are free, so no quantity computed under one may be compared against the same quantity computed under the other. Which parameters are free is independent of the window setting of the previous section, so the configurations and the arms cross: the six-parameter configuration is run on both arms and supplies the least-squares pair of Figure 3, and the four-parameter configuration is run on `ILSE` alone and supplies every grid cell.

The residual scale is not a free parameter of either and is not given a value. It is integrated out under a scale-invariant prior, so with $\mathrm{SSE} = \sum_i (y_i - \mu_i(\boldsymbol{\theta}))^2$ over the $n$ finite recorded values the objective is the marginal

$$\log L(\boldsymbol{\theta}) \;=\; \log\Gamma\!\left(\tfrac{n}{2}\right) \;-\; \tfrac{n}{2}\log\pi \;-\; \tfrac{n}{2}\log\mathrm{SSE}(\boldsymbol{\theta}), \tag{8}$$

whose maximiser is the ordinary least-squares one, since only the last term depends on $\boldsymbol{\theta}$. What the marginalisation changes is everything downstream of the point estimate. The score is the derivative of Eq. 8 and carries the factor $n/\mathrm{SSE}$ rather than a $1/\sigma^2$ that would have to be supplied, and the information the fit reports is

$$\mathbf{H}_{\mathrm{LSE}} \;=\; \frac{n}{\mathrm{SSE}} \sum_i \frac{\partial\mu_i}{\partial\boldsymbol{\theta}} \frac{\partial\mu_i}{\partial\boldsymbol{\theta}}^{\top}, \tag{9}$$

the Gauss-Newton curvature with $\hat{\sigma}^2 = \mathrm{SSE}/n$ entering as a plug-in prefactor and not as a coordinate. There is therefore no $\partial\sigma^2/\partial\boldsymbol{\theta}$ term to keep or drop: the scale direction is not in the parameter vector at all, which is also why the classical arm carries no score in the noise direction and why its mean squared standardized residual is pinned at one by construction.

The six-parameter configuration feeds the time-resolved figure (Figure 3). Both arms of it are produced by the same script that simulates the ensemble and scores the macroscopic members, so all eight members of that figure score one common set of recordings. All six parameters are free in base-ten logarithmic coordinates: $k_{\mathrm{on}}$, $k_{\mathrm{off}}$, the unitary current, `Current_Noise`, the current baseline and the mean channel number. The time-resolved figures accumulate the per-interval Fisher information and never invert it, so the flat directions of the least-squares fit are harmless there, and two of them are visible rather than hidden. `Current_Noise` has an identically zero score row and an identically zero Fisher row, because the least-squares mean carries no noise term; and the unitary current and the mean channel number form a rank-one degenerate block along the amplitude ridge $N_{\mathrm{ch}}\, i$, so the unitary current is free in this configuration without being separately identified.

The four-parameter configuration produces every grid cell and, where a cell needs one, the per-group estimate cloud. It fixes the unitary current at $1$ and `Current_Noise` at the cell value, leaving four free parameters: $k_{\mathrm{on}}$, $k_{\mathrm{off}}$, the current baseline and the mean channel number. Fixing those two is forced rather than chosen: they are the flat directions of the previous paragraph, and with them free the parameter Fisher information is singular, so no maximum-likelihood covariance exists. Both fixed values are the values the recordings were simulated from. The script builds one parameter vector and passes it both to the simulator and to the fit as its reference point, so the fixed value and the truth are the same number by construction. Least squares is therefore not handicapped in this configuration but helped: two of the six parameters are supplied to it exactly right, while the likelihood members estimate all six. Where it is nonetheless the arm whose reported uncertainty is furthest from the delivered one, that failure is not a lack of information about those two. This is the configuration behind every `ILSE` cell of Figures 4 and 6.

The consequence is carried into the Results. A score, a Fisher information, a distortion matrix or a log-likelihood computed under six free parameters is a different object from the one computed under four, and the two distortion matrices differ in dimension, $6 \times 6$ against $4 \times 4$, so any scalar that accumulates over eigenvalues is not comparable between them. Every statement about least squares in the Results is scoped to one configuration and names it. The statement that least squares cannot deliver the unitary current is a statement about the four-parameter configuration.

### Parameter estimation and the two anchors

Inference in each grid cell is two-stage. First, the ensemble is partitioned into groups of consecutive recordings and a separate maximum-likelihood estimate (MLE) is computed per group, producing a cloud of estimates whose scatter is the empirical parameter covariance. The partition is by position and any remainder is dropped, which costs nothing here because the recordings are independent and identically distributed and $10^4$ divides by both group sizes used. The optimiser is Gauss–Newton with Levenberg damping, warm-started at the simulation truth. Its damping schedule and its three convergence limits are in Supplementary File 1. What stops most fits, however, is a fourth criterion the script cannot reach: the optimiser tests the Newton decrement $\frac{1}{2}\, g^{\mathsf{T}} H^{-1} g$ first, requiring it below $10^{-8}$ for the first ten iterations and below $10^{-4}$ after that, with the three script-set limits acting as fallbacks behind it. Two group sizes were used, $10$ and $100$; with $10{,}000$ recordings these give $1{,}000$ groups of $10$ and $100$ groups of $100$. Second, a single joint MLE is computed over all replicates at once (one group of size $n_{\mathrm{sim}}$), giving the pooled estimate $\theta_{\mathrm{pool}}$; its near-zero gradient certifies that it is a stationary point of the pooled likelihood.

A group whose refit fails outright is dropped from the cloud, so a cloud can hold fewer estimates than the partition has groups, and its size is conditional on convergence. Every fit that did complete carries its convergence status, decided by the criteria above, into the per-group run log. The figures in the body do not filter on that status; the supplements that restrict themselves to converged fits say so.

Warm-starting the per-group fits at the truth is deliberate. It isolates the error of the likelihood from the error of the optimiser, which is what this paper studies; it would be the wrong choice for a study of an estimator, which this is not.

The two anchors, the simulation truth $\theta_{\mathrm{sim}}$ and the pooled fit $\theta_{\mathrm{pool}}$, are distinct points because the Gaussian macroscopic likelihood is misspecified with respect to the exact simulator. Their difference $\theta_{\mathrm{pool}} - \theta_{\mathrm{sim}}$ is the misspecification bias. Every diagnostic below is computed at both anchors, and the two answer different questions: at $\theta_{\mathrm{pool}}$ the score is zero by construction so bias washes out, whereas at $\theta_{\mathrm{sim}}$ the bias is visible but the Fisher information can be indefinite away from the maximum. Each figure states which anchor it uses.

### Diagnostics and the Gaussian-Fisher anchor

Faithfulness is measured with the three diagnostics introduced above: standardized residuals, the score, and the information-distortion matrix comparing the covariance $J$ of the score across replicates with the model's own Fisher information. Their definitions, sign conventions and thresholds are given where they are introduced; this section gives only the computation.

Every diagnostic reported in this paper is anchored on the analytic Gaussian Fisher information, summed over the intervals of a recording and then averaged over the recordings of the ensemble,

$\bar{G} = \mathbb{E}\left[\sum_t \mathrm{GFI}_t\right]$. This is the anchor written $\mathbf{H}$ above, and the expectation is not decoration: for the recursive members the predictive moments of interval $t$ depend on the data already seen, so $\sum_t \mathrm{GFI}_t$ is itself a random matrix and the object the score covariance is compared against is its ensemble mean, taken over the same recordings that supply that covariance. Its per-interval term is the summand of Eq. 3, which follows from the analytic derivative alone and needs no extra likelihood passes. It is positive semidefinite by construction, so no anchor used here can be indefinite.

A second Fisher construction exists in the codebase, a central difference of the analytic score. It is not interchangeable with the analytic anchor per recording, being widely indefinite replicate by replicate, which is one reason the body is anchored on the analytic construction; averaged over recordings it is the check the analytic anchor is measured against, over the full roster, in Supplementary File 1, which also holds its settings and the direction convention of the comparison. An earlier battery anchored on it (data directory `433ed13`) is retained in the deposit; no number reported in this paper is taken from it.

One edge case governs how a cell is summarised, and it is by design rather than by accident. The anchor can be near-singular where a parameter is unidentifiable, which happens once the open population relaxes (Figure 3–figure supplements 1 and 2); there we diagonalize $\mathbf{H}$ and retain the eigenmodes whose eigenvalue exceeds a tolerance set relative to the largest, $\lambda_i > \lambda_{\max} \cdot 10^{-10}$, evaluate every inverse power on that retained subspace alone, and leave the quantity undefined, not finite, when the retained subspace is empty. The dimension $d$ of Eq. 4 is the dimension of that subspace, so a cell where a direction has been dropped is summarised over fewer directions than one where none has. Undefined is not zero, and the maps render those cells in grey to keep the two apart.

The comparison of $J$ against $\bar{G}$ is defined, with its sign convention and its two scalar summaries, in the section that introduces it; on the least-squares branch it carries a further premise, that the residuals are homoscedastic, and the Results state that premise where the ratio is quoted.

Uncertainty on every reported statistic comes from a nonparametric bootstrap over groups, reported as percentile intervals at the $2.5$, $16$, $50$, $84$ and $97.5\%$ quantiles; the intervals are ordinary empirical percentiles. The diagnostic battery used $100$ bootstrap replicates and a maximum lag of $10$, which is the truncation of every autocorrelation it reports, of the score and of the standardized residual alike, and the empirical-versus-theoretical distortion capstone used $200$. The per-group estimate cloud is deposited raw, and every interval quoted from it in this paper is computed downstream in the figure code, from the raw estimates.

## Regime grid and reduction

The acquisition interval was swept over seven levels of $\widetilde{\Delta}$, namely $1$, $0.5$, $0.2$, $0.1$, $0.05$, $0.02$ and $0.01$, realised as $500$, $250$, $100$, $50$, $25$, $10$ and $5$ raw samples averaged per interval, that is step durations of $10$, $5$, $2$, $1$, $0.5$, $0.2$ and $0.1$ ms, giving $10$, $20$, $50$, $100$, $200$, $500$ and $1{,}000$ steps in the record. The total record duration is held fixed at $100$ ms across the sweep, so the interval is varied without varying the amount of data. Within each record the pre-pulse, agonist and post-pulse segments hold a fixed ratio of $20{:}40{:}40$ percent of the steps.

The noise axis needs its conversion stated once, because the quantity the figures plot and the quantity the engine consumes differ by a fixed factor. The engine parameter is `Current_Noise`, a white-noise power spectral density in units of $g^2\,\mathrm{s}$, and it reaches the likelihood only through the variance of one recorded point, $e_t = \mathrm{Current\_Noise}/\Delta_t$, with no correlation time of its own. The sweep holds that density fixed and shortens the window, so the variance of one recorded point rises as $1/\Delta_t$ and the noise per unit bandwidth is unchanged. In the natural units defined above, $\widetilde{\Delta} = \Delta\, k_{\mathrm{off}}$ and $\widetilde{S} = S\, k_{\mathrm{off}}/i^2$; at the run values $k_{\mathrm{off}} = 100\ \mathrm{s}^{-1}$ and $i = 1$ the reference cell used throughout is $\widetilde{S} = 0.01$. Both $\widetilde{S}$ and the gating fluctuation are per-sample variances, which is what makes them comparable. The instrumental variance of one recorded sample is $i^2\widetilde{S}/\widetilde{\Delta}$, so $\widetilde{S} = 1$ is a sample one channel time constant long whose noise carries a standard deviation of exactly one single-channel current, and a decade of that axis is half a decade of current. The gating variance of the same sample rises as the window shortens only until the window is shorter than the channel's own correlation time, after which it saturates, while the instrumental variance keeps rising as $1/\widetilde{\Delta}$; at the open probability of one half the two are equal along

$$\frac{\widetilde{S}}{N_{\mathrm{ch}}} = \frac{2\widetilde{\Delta} - 1 + e^{-2\widetilde{\Delta}}}{8\,\widetilde{\Delta}},$$

which is $\widetilde{\Delta}/4$ for windows short against that correlation time and a quarter for long ones.

The mean channel number $N_{\mathrm{ch}}$ and the noise level are the other two map axes. Every cell entering a body figure is at $n_{\mathrm{sim}} = 10^4$, enforced by the figure code, so a cell run at a smaller ensemble cannot enter a map silently. The noise axis is a fixed ladder of twelve levels, and every cell of every algorithm sits on one of them:

$$\widetilde{S} \in \{0.005,\ 0.01,\ 0.02,\ 0.05,\ 0.1,\ 1,\ 10,\ 10^2,\ 10^3,\ 10^4,\ 10^5,\ 10^6\}, \tag{10}$$

denser below one, where the crossover the map measures sits, and by decades above it. The level $\widetilde{S} = 1$ is where the instrumental noise over one channel time constant equals the single-channel signal power, so the ladder is dense on the side where the gating fluctuations still carry information and coarse on the side where they do not. Coverage is deliberately not a rectangle and it differs by member: `IR` spans eleven channel counts from $5$ to $10^4$ across all twelve noise levels, and every other member covers less. The per-member cell manifest is in Supplementary File 1.

Each grid cell is reduced through five stages, from the per-group estimate cloud to the empirical-against-theoretical distortion capstone; the stages and their outputs are listed in Supplementary File 1.

## Reproducibility and computational environment

The engine is `macro_dr`, built as C++20, and each analysis is a script in a small typed domain-specific language run in a single process (***Moffatt, 2025***). Every analysis output file stamps the build's short git commit hash as its first row, and the data directories are named by that hash, so results from different code versions cannot overwrite one another.

The engine that produced these results is not the build behind the P2X$_2$ analysis of ***Moffatt and Pierdominici-Sottile*** (***2025***). Every score reported here comes from automatic differentiation of the likelihood with respect to $\boldsymbol{\theta}$, propagated through the recursion itself: value and derivative travel through the same code, carried together by one templated C++ type, so the derivative of the spectral reconstruction is taken where the reconstruction is, with its divided differences and its coincidence limits (Appendix 1). The Fisher information does not arrive with it. It is an expectation under the model rather than a derivative of the recorded log-likelihood, so it is wired by hand, as the Gaussian form of Eq. 3 over the predictive moments and the first derivatives the score has already produced. The distortion quantities follow from the two. The earlier study's fits were obtained by Markov chain Monte Carlo and needed no gradient, so the score-based diagnostics are new here rather than carried over from that work, and the dependence is what forces the trust coefficient above to be infinitely differentiable.

The production batteries ran under SLURM with the linear-algebra back end pinned to a single thread so that OpenMP parallelises the per-simulation and bootstrap loops; the resource settings and the one environment variable that is load-bearing are in Supplementary File 1.

All resampling that operates on a fixed dataset, the bootstraps and the group partitioning, is deterministic and reproducible given the data.

## Use of generative artificial intelligence

Generative artificial intelligence was used in the preparation of this manuscript. Anthropic's Claude models, accessed through the Claude Code command line interface, assisted with drafting and editing the manuscript text, with checking numerical claims in the text against the code and the output files they come from, and with writing analysis and plotting scripts. They were not used to generate, alter or select data, to derive the likelihood family or the diagnostics, or to form the interpretation of the results. No figure or figure panel is AI generated. The author reviewed and verified all content, and takes full responsibility for the work, including any part of it that was drafted with such assistance.

## Data and code availability

All code needed to reproduce the study is openly available. The analysis pipeline that generates the figures, together with the simulation configurations that define every run, is deposited in a public repository (`macroir-validity`) and archived at Zenodo, with the DOI issued on acceptance. The simulation and inference engine is the `macro_dr` software ***Moffatt*** (***2025***), referenced at the pinned version that produced the deposited data; each output file records the engine commit hash that generated it. No experimental data were used: every dataset in the paper is simulated, and is regenerable from the deposited configurations up to the simulation seed, which was not recorded for the original freeze (Materials and methods); the later differenced-Fisher battery injects an explicit seed through its deposited configuration (Supplementary File 1).

A separate library, `macroir`, carries the boundary-conditioned likelihood, its score, its Gaussian Fisher information and the distortion diagnostic of this paper out of the engine and into a small self-contained C++ core, with an R package and a Python package over it; the diagnostic returns the distortion matrix, its eigenvalue spectrum and the first-order bias from one call in either language. The two language packages are bindings over that one core, and they were checked against it on a single reference cell, which is the evidence for the statement that the three interfaces return the same numbers.


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## Author contributions

L.M. conceived the study, developed the theory, implemented the algorithms, designed and ran the simulations, analysed the data, and wrote the manuscript.

## Competing interests

The author declares no competing interests.


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## Appendix 1

### The interval likelihood, derived

The body states what each member of the family conditions on and what that buys. This appendix carries the derivation behind those statements: the boundary-state construction and the moments it needs, what separates the start-conditioned members from the boundary-conditioned one, and the correspondence with the linear-filtering literature. It is written to be checked rather than read in sequence, and every equation the body or Methods refers to is numbered here.

**Three marks, and each one says one thing**

**An arrow means the object spans the acquisition window**, of duration $\Delta$: $P_{i\to j}(\Delta)$ is a transition across it, $\bar{y}_{i\to j}$ an average over it, $\bar{y}_{0\to\Delta}$ a recorded value formed from it. **No arrow means one instant**: $\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0$ are the occupancy mean and covariance where the window opens, $\boldsymbol{\mu}_\Delta, \boldsymbol{\Sigma}_\Delta$ where it closes. **A dot marks a collapsed end**: $a_{i\to\cdot} := \sum_j P_{i\to j}(\Delta)\, a_{i\to j} = \mathbb{E}[a \mid X_0 = i]$, as in $\bar{y}_{i\to\cdot}$; for a probability the weight is already inside and the collapse is the plain margin. Across a recording a subscript $t$ indexes the window, as in $\bar{y}_t$; inside a window the two ends are $0$ and $\Delta$, and a collapsed object is defined the moment its pair-resolved version is.

### The acquisition kernel, and what the uniform window misses

The likelihood implements the boxcar of Eq. 1 exactly, while a real acquisition kernel is that boxcar composed with the amplifier's analogue low-pass, so what is unmodelled is governed by one dimensionless group, the cutoff frequency against the acquisition window, $f_c\Delta$. For any component flat on the scale of $\Delta$, the effect on the two objects the likelihood uses, the variance of a recorded value and the covariance between consecutive ones, is

| $f_c\Delta$ | 0.2 | 0.5 | 1 | 2 | 5 | 10 | 20 |
|---|---|---|---|---|---|---|---|
| variance deficit | 61.5% | 29.6% | 14.7% | 7.4% | 2.9% | 1.5% | 0.74% |
| lag-one correlation $\rho_1$ | 64.1% | 21.1% | 8.6% | 4.0% | 1.5% | 0.75% | 0.37% |
| error on a confidence interval | 61% | 19% | 8.3% | 3.9% | 1.5% | 0.74% | 0.37% |

From $f_c\Delta = 1$ upward the deficit is $0.147/(f_c\Delta)$ and $\rho_1$ is $0.075/(f_c\Delta)$, while $\rho_2$ and every longer lag are zero to four decimals: the residue is a nearest-neighbour effect of known size. The total is conserved, $\mathrm{Var} + 2\sum_k \mathrm{Cov}_k = S/\Delta$ to six decimals at every setting, forced by the filter having unit gain at zero frequency, so what a likelihood allowing no such covariance sees is a misallocation of variance between the diagonal and its neighbours rather than a loss of it. The pole count does not enter: four poles, eight poles and a Gaussian of the same $-3$ dB frequency agree to the third decimal from $f_c\Delta = 0.5$ upward, so the width can be quoted as the Gaussian equivalent $\sigma_f = \sqrt{\ln 2}/(2\pi f_c) = 0.1325/f_c$, a ten-to-ninety rise time of $0.34/f_c$, which is the convention electrophysiology already uses for filter width (***Colquhoun and Sigworth, 1995***). And the kinetics are the protected part: the deficit is first order in $1/(f_c\Delta)$ for the instrumental noise and second order for a component carrying a correlation time of its own, because unit gain at zero frequency preserves the integral of an autocovariance and the block-averaged variance of a short-correlated component depends on that integral alone to leading order. At $f_c\Delta = 5$ the white term loses $2.94\%$ and a gating term with correlation time $\Delta$ loses $0.11\%$, a factor of $27$, so the residue lands in the instrumental-noise partition, which is where the Discussion already argues an unmeasured departure of the noise model lives.

What does not scale with the group is the interval holding the concentration jump: group delay plus rise displace and smear the transient by about $0.69/f_c$ at four poles, a fixed time rather than a fraction of $\Delta$, so that one interval carries an error the expansion says nothing about, and the honest recipe discards or flags the first interval after a step.

One property of this family follows from the same algebra and a point-sampling method does not have it. Because the observable *is* the integral, block-averaging a record down to a coarser $\Delta$ produces exactly the observable of Eq. 1 at any decimation factor, with no aliasing question to answer; after decimation the only unmodelled piece left is the residual kernel, which is the table above. That is what makes “average the record first” a scoping statement rather than a workaround.

## The interval average, and how the members are built

A single channel switching among its conductance levels traces a piecewise-constant path, the random telegraph signal when $K = 2$, and the window delivers its integral over one acquisition interval,

$$A_{0\to\Delta} = \int_0^\Delta \gamma\left(X_s\right) \mathrm{d}s, \tag{A1}$$

where $X_s$ is the state of the channel at time $s$. Its law is mixed: an atom at $\Delta\gamma_i$ carrying the probability of never leaving state $i$, and a continuous part from the paths that switched. Its density is a series over the number of transitions in the window, exact at every $K$, closed only at $K = 2$, and proliferating with $K$. Two things that might rescue it do not. Channel models carry few distinct conductances, every shut state at $\gamma = 0$ and the open states often sharing one $\gamma$, so $A_{0\to\Delta}$ is that conductance times the time spent in the open aggregate however large $K$ is; but the aggregate is Markov only if $\mathbf{Q}$ is lumpable, which for the schemes that motivate this work it is not (***Colquhoun and Hawkes, 1981***). And the population multiplies the difficulty rather than adding to it, since the laws of independent channels convolve and a recorded sample constrains only the sum, so the posterior over channels does not factorize (***Moffatt, 2007***).

Every moment in the construction belongs to a single channel. Writing $\mathbf{X}$ for the indicator vector of the state one channel occupies, which is one on that state and zero elsewhere, $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$ is its probability vector and $\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{X})$ its covariance, which is $\mathrm{diag}(\boldsymbol{\mu}) - \boldsymbol{\mu}\boldsymbol{\mu}^\top$ for a channel nothing has been observed about and vanishes when its state is known. Channels are independent of each other, so the counts $\mathbf{N} = \sum_c \mathbf{X}^{(c)}$ over the ensemble have $\mathbb{E}[\mathbf{N}] = N_{\mathrm{ch}}\boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{N}) = N_{\mathrm{ch}}\boldsymbol{\Sigma}$, which is where every factor of $N_{\mathrm{ch}}$ in Eq. 2 comes from.

## The boundary-conditioned moments, and how they are evaluated

The ladder needs two conductance objects, both built from the boundary-conditioned single-channel moments $\bar{\gamma}_{i\to j}$ (the mean interval-averaged conductance of a channel that starts in $i$ and ends in $j$) and $\bar{v}_{i\to j}$ (its variance). Both are moments of the integrated conductance $A_{0\to\Delta}$ of Eq. A1, taken conditional on the chain starting in $i$ and ending in $j$, and both are available in closed form, with no object larger than $K \times K$ formed anywhere. They are derivatives at $\alpha = 0$ of the conductance-tilted semigroup $e^{(\mathbf{Q}+\alpha\boldsymbol{\Gamma})\Delta}$, the generator biased by the conductance. Writing

$$\mathbf{F}_1(\Delta) = \frac{\partial}{\partial\alpha} e^{(\mathbf{Q}+\alpha\boldsymbol{\Gamma})\Delta}\bigg|_{\alpha=0}, \qquad 2\,\mathbf{F}_2(\Delta) = \frac{\partial^2}{\partial\alpha^2} e^{(\mathbf{Q}+\alpha\boldsymbol{\Gamma})\Delta}\bigg|_{\alpha=0}, \tag{A2}$$

with $\boldsymbol{\Gamma} = \mathrm{diag}(\boldsymbol{\gamma})$, the $(i \to j)$ entry of the first is $\mathbb{E}[A_{0\to\Delta}\,\mathbf{1}\{X_\Delta = j\} \mid X_0 = i]$ and of the second the same expectation with $A^2_{0\to\Delta}$ in place of $A_{0\to\Delta}$; the factor of two is the two triangular halves of the double integral the second derivative unfolds into. Dividing by the transition probability conditions on the endpoints and dividing by $\Delta$ turns the integral into an average, so

$$\bar{\gamma}_{i\to j} = \frac{[\mathbf{F}_1(\Delta)]_{i\to j}}{\Delta\, P_{i\to j}(\Delta)}, \qquad \bar{v}_{i\to j} = \frac{2[\mathbf{F}_2(\Delta)]_{i\to j}}{\Delta^2\, P_{i\to j}(\Delta)} - \bar{\gamma}_{i\to j}^{\,2}. \tag{A3}$$

Both derivatives are evaluated from an eigendecomposition of the generator, which is the route the implementation takes and the one the P2X2 analysis of ***Moffatt and Pierdominici-Sottile*** (***2025***) was run on. Writing $\mathbf{Q} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{W}$, with $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_K)$ the eigenvalues, $\mathbf{V}$ the matrix of right eigenvectors and $\mathbf{W} = \mathbf{V}^{-1}$, the transition matrix is $\mathbf{P}(\Delta) = \mathbf{V}e^{\boldsymbol{\Lambda}\Delta}\mathbf{W}$ and the two derivatives are the same sandwich with the conductance carried in the eigenbasis. With $\widehat{\boldsymbol{\Gamma}} = \mathbf{W}\boldsymbol{\Gamma}\mathbf{V}$ and $\circ$ the entrywise product,

$$\frac{\mathbf{F}_1(\Delta)}{\Delta} = \mathbf{V}\big(\widehat{\boldsymbol{\Gamma}}\circ\mathbf{E}_2\big)\mathbf{W}, \qquad (\mathbf{E}_2)_{ab} = E_2(\lambda_a\Delta, \lambda_b\Delta), \tag{A4}$$

$$\frac{2\,\mathbf{F}_2(\Delta)}{\Delta^2} = 2\,\mathbf{V}\mathbf{M}\mathbf{W}, \qquad M_{ab} = \sum_c \widehat{\Gamma}_{ac}\,\widehat{\Gamma}_{cb}\,E_3(\lambda_a\Delta, \lambda_c\Delta, \lambda_b\Delta), \tag{A5}$$

where $E_2$ and $E_3$ are the first and second divided differences of the exponential,

$$E_2(x,y) = \frac{e^x - e^y}{x-y}, \qquad E_3(x,y,z) = \frac{e^x}{(x-y)(x-z)} + \frac{e^y}{(y-x)(y-z)} + \frac{e^z}{(z-x)(z-y)}. \tag{A6}$$

Both are symmetric in their arguments and both are smooth where the denominators vanish, so repeated and near-repeated eigenvalues are read off the coincidence limits $E_2(x,x) = e^x$ and $E_3(x,x,x) = e^x/2$, with the one-pair-coincident form between them (Supplementary Information). Eq. A4 is the form in which the first moment is given in the supplement of ***Moffatt and Pierdominici-Sottile*** (***2025***); the second moment was not stated there. The eigendecomposition is computed once per agonist concentration and reused by every window at that concentration, after which each window costs the $O(K^3)$ contraction of Eq. A5.

An equivalent evaluation avoids diagonalizing $\mathbf{Q}$ altogether. $\mathbf{F}_1$ and $\mathbf{F}_2$ are the strictly upper blocks of the exponentials of the augmented generators

$$\mathcal{Q}_2 = \begin{pmatrix} \mathbf{Q} & \boldsymbol{\Gamma} \\ \mathbf{0} & \mathbf{Q} \end{pmatrix}, \qquad \mathcal{Q}_3 = \begin{pmatrix} \mathbf{Q} & \boldsymbol{\Gamma} & \mathbf{0} \\ \mathbf{0} & \mathbf{Q} & \boldsymbol{\Gamma} \\ \mathbf{0} & \mathbf{0} & \mathbf{Q} \end{pmatrix}, \tag{A7}$$

of sizes $2K\times 2K$ and $3K\times 3K$, whose exponentials are block triangular with $e^{\mathbf{Q}\Delta}$ on the diagonal (***Van Loan, 1978***), so that $e^{\mathcal{Q}_3\Delta}$ on its own returns the transition matrix and both moments with no spectral machinery, and where no reliable eigensolver is at hand it is the construction to use. The route taken here is the spectral one because this paper propagates the derivative of the log-likelihood with respect to the kinetic parameters through every object the filter builds, and the eigenvalues and eigenvectors of $\mathbf{Q}$ move with those parameters in closed form, by a first-order perturbation of the same decomposition. Every number reported in this paper comes from the spectral route.

Both expressions are $0/0$ where a transition is unreachable, which is not a numerical edge case here: at zero agonist $k_{\mathrm{on}}[A]$ vanishes identically, so $P_{C\to O}(\Delta) = 0$ exactly over the whole pre-pulse segment. Conditioning on a single transit whose time, in the short-window limit, is uniform over the window makes the average $\gamma_j + (\gamma_i - \gamma_j)\,U$ with $U$ uniform on $(0,1)$, so the limit is the midpoint in the mean and the uniform-split variance,

$$\bar{\gamma}_{i\to j} \to \tfrac{1}{2}(\gamma_i + \gamma_j), \qquad \bar{v}_{i\to j} \to \tfrac{(\gamma_i-\gamma_j)^2}{12}, \tag{A8}$$

which depends on the conductance vector alone. The implementation does not branch on the singular case: it reads the two quotients of Eq. A3 as posterior means and shrinks them towards an endpoint prior with the same midpoint mean and the second moment

$\frac{1}{2}(\gamma_i^2 + \gamma_j^2)$ (Methods), whose implied variance $((\gamma_i - \gamma_j)/2)^2$ is three times the limit above, a deliberately conservative floor, with a pseudo-count whose weight is the floating-point noise of the spectral reconstruction, so the limit is approached smoothly and no fitted constant enters (Methods). The start-conditioned mean conductance is the $K$-vector

$$\bar{\gamma}_{i\to\cdot} \;=\; \sum_j P_{i\to j}(\Delta)\,\bar{\gamma}_{i\to j}, \tag{A9}$$

which conditions on the state at the interval's start alone and is the object `INR`, `MR` and `VR` use. The boundary-conditioned mean conductance is the same average retained on both endpoints, so it stays a $K \times K$ object rather than collapsing to a vector, and is the object `IR` uses.

One generalization, and the whole family is inside it.
Eq. 2 takes a channel's conductance to be a number once its state is known. That is true of the current at an instant and of nothing else. One step of generality covers every member below: alongside the conductance, supply a *per-state variance* $\mathbf{w}$, the variance the recorded value still has once the state is fixed. Channels being independent given their states, those variances add over the ensemble, so

$$\mathbb{E}\big[\bar{y}\,\big|\,\mathbf{N}\big] = \boldsymbol{\gamma}^\top\mathbf{N}, \qquad \mathrm{Var}\big[\bar{y}\,\big|\,\mathbf{N}\big] = \epsilon^2_{0\to\Delta} + \textstyle\sum_s N_s\, w_s, \tag{A10}$$

and the extra variance is linear in the counts, exactly as the mean is. Taking the expectation over the state distribution turns the predictive variance of Eq. 2 into

$$\sigma^2 \;=\; \epsilon^2_{0\to\Delta} \;+\; N_{\mathrm{ch}}\,\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma} \;+\; N_{\mathrm{ch}}\,\boldsymbol{\mu}^\top\mathbf{w}, \tag{A11}$$

with every other line of Eq. 2 unchanged. Setting $\mathbf{w} = \mathbf{0}$ returns the published filter. This is the same device ***Münch et al.*** (***2022***) add to the same filter to model open-channel noise, from a different motivation; here the state-dependent term will carry the gating variance the chosen state leaves unexplained rather than a property of the pore.

Four contractions, and the members differ in three of them.
Written this way, Eq. 2 contracts a member's objects against its own state distribution and covariance, and it does so four times:

$$\underbrace{\boldsymbol{\mu}^\top\boldsymbol{\gamma}}_{\text{the mean}} \qquad \underbrace{\boldsymbol{\mu}^\top\mathbf{w}}_{\text{the state-dependent noise}} \qquad \underbrace{\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}}_{\text{what the state contributes}} \qquad \underbrace{\boldsymbol{\Sigma}\boldsymbol{\gamma}}_{\text{state against current, }\mathbf{g}}$$

two against $\boldsymbol{\mu}$ and two against $\boldsymbol{\Sigma}$, and everything else in Eq. 2 is a scalar. Extending the filter to a recorded value that spans the window therefore cannot require a new algorithm. It requires choosing a state variable and supplying $(\boldsymbol{\gamma}, \mathbf{w})$ on it, and the members below are those choices. The per-state variance is not a free third choice beside the other two: Eq. A10 defines it as whatever variance the chosen state leaves unexplained, so naming the state names it.

### The three steps that build the interval update

What stops Eq. 2 from being used on the interval average is that the average is not a function of the occupancy at any one time, so no conductance vector on the $K$ states delivers its mean. On the $K^2$ boundary pairs one does: a channel that starts in $i$ and ends in $j$ contributes $\bar{\gamma}_{i\to j}$ on average, so the interval average is linear in the boundary counts again. That is the whole of the move, and three steps finish it. Run Eq. 2 on the pairs, with $\bar{\gamma}_{i\to j}$ in place of $\boldsymbol{\gamma}$ and the boundary covariance in place of $\boldsymbol{\Sigma}$. Charge what the conductance still varies by once the pair is fixed to $\epsilon^2$, as though the recording carried an extra noise of variance $\bar{v}_{i\to j}$

per channel, of a size that depends on the pair that channel took. Marginalize over the start index, which the next window has no use for. Every equation in this subsection is one of those three steps carried out, and the two objects the interval likelihood is usually written with are those two contractions read on the larger space.

Conditioning on the pair rather than on one end is the same device as static condensation, or as the spatial Markov property, applied on the time axis: the filter holds the states at the interval boundaries and marginalizes the trajectory between them analytically. It is not a transition state in the mechanistic sense, and the word *transition* is not used for it anywhere in this paper.

The boundary state's own Gaussian follows from the prior and the propagator. Let $\mathbf{X}_{0\to\Delta}$ be the indicator of the pair a single channel takes, one on the $(i,j)$ it starts and ends in and zero on the other $K^2-1$, and let $(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ be the channel's state at the window's start. It leaves $i$ and lands in $j$ with probability $P_{i\to j}(\Delta)$, so

$$\mu_{i\to j} = \mu_{0,i}\, P_{i\to j}(\Delta), \tag{A12}$$

$$(\boldsymbol{\Sigma}_{0\to\Delta})_{(i\to j),(k\to l)} = P_{i\to j}(\Delta)\left(\boldsymbol{\Sigma}_0 - \mathrm{diag}(\boldsymbol{\mu}_0)\right)_{ik} P_{k\to l}(\Delta) \;+\; \left[\mathrm{diag}(\boldsymbol{\mu}_{0\to\Delta})\right]_{(i,j),(k,l)}. \tag{A13}$$

The two terms of Eq. A13 are the two sources of spread. The first carries the uncertainty already in the prior across the window, once on each index. The second is the uncertainty a channel known to be leaving $i$ still has about where it lands, the covariance of a single draw from row $i$ of $\mathbf{P}(\Delta)$, which is diagonal on the pairs because a channel takes exactly one of them. Subtracting $\mathrm{diag}(\boldsymbol{\mu}_0)$ in the first term is what keeps the prior's own single-channel part from being counted twice, since the second term regenerates it at the resolution of the pairs.

Summing the start index out of Eqs. A12 and A13 gives the occupancy Gaussian at the end of the window,

$$\boldsymbol{\mu}_\Delta = \mathbf{P}(\Delta)^\top \boldsymbol{\mu}_0, \tag{A14}$$

$$\boldsymbol{\Sigma}_\Delta = \mathbf{P}(\Delta)^\top\left(\boldsymbol{\Sigma}_0 - \mathrm{diag}(\boldsymbol{\mu}_0)\right)\mathbf{P}(\Delta) + \mathrm{diag}(\boldsymbol{\mu}_\Delta), \tag{A15}$$

term by term, the diagonal of the one becoming the diagonal of the other. This is the propagation an unconditioned member performs, and it is Eq. A13 seen from the end of the window alone. The first line of Eq. 2, on the boundary state, is the predictive. Its mean is

$$\bar{y}^{\mathrm{pred}}_{0\to\Delta} \;=\; N_{\mathrm{ch}}\, \boldsymbol{\mu}_0^\top \bar{\boldsymbol{\gamma}}_0, \tag{A16}$$

since $\boldsymbol{\mu}_{0\to\Delta}$ contracted against $\bar{\gamma}$ is Eq. A9 summed against the prior, and its variance has three terms,

$$\sigma^2_{0\to\Delta} \;=\; \epsilon^2_{0\to\Delta} \;+\; N_{\mathrm{ch}}\, \widetilde{\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}} \;+\; N_{\mathrm{ch}}\, \boldsymbol{\mu}_0^\top \mathbf{w}. \tag{A17}$$

The first is the $\epsilon^2$ of Eq. 2, the instrumental noise now taken over the window, which carries the acquisition averaging and decreases with $\Delta$; its parametric form is given in Methods. The third is the noise charged in step two, $N_{\mathrm{ch}} \sum_{i,j} \mu_{i\to j} \bar{v}_{i\to j}$, which is what Eq. A17 writes with $\mathbf{w}$, the start-conditioned interval variance, once the end index is summed. The middle term is $\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}$ of Eq. 2 evaluated on the pairs, and the tilde is what marks that: under it the two $\boldsymbol{\gamma}$ stand for $\bar{\gamma}_{i\to j}$ and $\boldsymbol{\Sigma}$ for the $K^2\times K^2$ boundary covariance, not for the $K$-vector and the $K\times K$ matrix they name everywhere else. Written out with $\boldsymbol{\Sigma}_{0\to\Delta}$ of Eq. A13 and then collapsed by substituting it,

$$\begin{aligned}\widetilde{\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}} \;&=\; \sum_{i,j}\sum_{k,l} \bar{\gamma}_{i\to j}\left(\boldsymbol{\Sigma}_{0\to\Delta}\right)_{(i,j),(k,l)} \bar{\gamma}_{k\to l}\\ &=\; \bar{\boldsymbol{\gamma}}_0^\top\left(\boldsymbol{\Sigma}_0 - \mathrm{diag}(\boldsymbol{\mu}_0)\right)\bar{\boldsymbol{\gamma}}_0 \;+\; \boldsymbol{\mu}_0^\top\left(\mathbf{H}\mathbf{1}\right),\end{aligned} \tag{A18}$$

where $H_{i\to j} = \bar{\gamma}^2_{i\to j}\, P_{i\to j}(\Delta)$ and $\mathbf{1}$ is the vector of ones. The two summands are the two terms of Eq. A13 in order, and only $K$-vectors and $K\times K$ matrices survive, so the $K^2\times K^2$ covariance is never built. Reading the tilde as $\bar{\boldsymbol{\gamma}}_0^\top \boldsymbol{\Sigma}_0 \bar{\boldsymbol{\gamma}}_0$ keeps the first summand and drops the second, which is the single-channel spread of the interval mean across boundary pairs and does not vanish when the occupancy covariance does.

The gain is the same contraction taken once. Built on the pairs it is the $K^2$-vector $\sum_{k,l} (\boldsymbol{\Sigma}_{0\to\Delta})_{(i,j),(k,l)} \bar{\gamma}_{k\to l}$, and step three sums its start index away, leaving the $\mathbf{g}_{0\to\Delta}$ the interval update uses,

$$\mathbf{g}_{0\to\Delta} \;=\; \widetilde{\boldsymbol{\gamma}^\top \boldsymbol{\Sigma}} \;=\; \mathbf{P}(\Delta)^\top \big(\boldsymbol{\Sigma}_0 - \mathrm{diag}(\boldsymbol{\mu}_0)\big)\bar{\boldsymbol{\gamma}}_0 \;+\; \mathbf{G}^\top \boldsymbol{\mu}_0, \qquad G_{i\to j} \;=\; \bar{\gamma}_{i\to j}\, P_{i\to j}(\Delta), \tag{A19}$$

the covariance between the state $\mathbf{X}_\Delta$ a channel ends the window in and the current that same channel contributes to the window average. Over the ensemble it is $\mathrm{Cov}(\mathbf{N}_\Delta, \bar{y}) = N_{\mathrm{ch}}\, \mathbf{g}_{0\to\Delta}$, with $\mathbf{N}_\Delta$ the counts by state at the end of the window, and that is the form the update uses. Again the two summands are the two terms of Eq. A13: $\mathbf{G}$ carries one power of $\bar{\gamma}$ where $\mathbf{H}$ carried two, and the end index is left open instead of being summed against $\mathbf{1}$.

The second line of Eq. 2, marginalized the same way, is the update, with $\delta = \bar{y}^{\mathrm{obs}}_{0\to\Delta} - \bar{y}^{\mathrm{pred}}_{0\to\Delta}$ the innovation,

$$\boldsymbol{\mu}^{\mathrm{post}} = \boldsymbol{\mu}_\Delta + \frac{\mathbf{g}_{0\to\Delta}}{\sigma^2_{0\to\Delta}}\,\delta, \tag{A20}$$

$$\boldsymbol{\Sigma}^{\mathrm{post}} = \boldsymbol{\Sigma}_\Delta - \frac{N_{\mathrm{ch}}\, \mathbf{g}_{0\to\Delta}\mathbf{g}^\top_{0\to\Delta}}{\sigma^2_{0\to\Delta}}. \tag{A21}$$

Conditioning and summing the start index commute here, because the sum is linear and the correction is rank one on the index it acts on, so the downdate crosses it unchanged, with $\mathbf{g}_{0\to\Delta}$ standing where the boundary gain stood. What the sum discards is the posterior correlation between the interval current and the state the window *started* in, which costs nothing inside the window and is what the recursion pays for below.

Two of the three steps are exact and one is the approximation. Eqs. A12 and A13 are the true moments of $\mathbf{X}_{0\to\Delta}$, and so are the three the predictive consumes. Writing $\mathbf{N}_{0\to\Delta} = \sum_c \mathbf{X}^{(c)}_{0\to\Delta}$ for the boundary counts over the ensemble, channels are independent given the pairs each of them took, so given $\mathbf{N}_{0\to\Delta}$ the recorded average has mean $\sum_{i,j} \bar{\gamma}_{i\to j} N_{i\to j}$ and variance $\epsilon^2_{0\to\Delta} + \sum_{i,j} \bar{v}_{i\to j} N_{i\to j}$ exactly, and the state-dependent noise of step two, having no covariance with the counts that carry it, enters $\sigma^2_{0\to\Delta}$ and changes neither $\bar{y}^{\mathrm{pred}}_{0\to\Delta}$ nor $\mathbf{g}_{0\to\Delta}$. That last point is what makes charging it to the noise term legitimate rather than convenient. What is approximate is the Gaussian, in the two places the box above names: the counts are not Gaussian, which is the occupancy closure, and neither is the window average of one channel given its pair, so replacing its law by the two numbers $\bar{\gamma}_{i\to j}$ and $\bar{v}_{i\to j}$ is the interval-signal closure. The two together still do not make the pair $(\mathbf{N}_{0\to\Delta}, \bar{y})$ jointly Gaussian, since its conditional variance depends on the state, so the update conditions the Gaussian that carries the exact moments. The factor $N_{\mathrm{ch}}$ in the downdate, and its absence from the mean correction, come from the same place. All three of $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$ and $\mathbf{g}_{0\to\Delta}$ belong to one channel, while what is observed belongs to the ensemble, so the conditioning is done on $\mathrm{Cov}(\mathbf{N}_\Delta, \bar{y}) = N_{\mathrm{ch}}\mathbf{g}_{0\to\Delta}$ against $\mathrm{Cov}(\mathbf{N}_\Delta) = N_{\mathrm{ch}}\boldsymbol{\Sigma}_\Delta$. The rank-one term picks up the channel count twice and the return to one channel gives one back, leaving a single factor; the mean correction picks it up once and gives it back, leaving none.

### The start-conditioned members, stated forward

The construction above reaches `IR` by choosing the boundary pair as the state. The members that stop at the interval's start are stated here from their own assumptions rather than as that construction with the collapse brought forward.

MR takes the recorded value to depend on the occupancy where the window opens. Formally this asserts

$$\bar{y}_{0\to\Delta} \perp X_\Delta \,\Big|\, X_0, \tag{A22}$$

that once the state a channel started in is known, the recorded average says nothing further about where that channel ended. Everything MR does follows from Eq. A22, and the member is complete without any object larger than the $K$ states. What it must supply is a conductance and a per-state variance on that state, and both are conditional moments of $A_{0\to\Delta}$ given the start alone:

$$(\bar{\gamma}_0)_i = \mathbb{E}\left[\tfrac{1}{\Delta} A_{0\to\Delta} \,\Big|\, X_0 = i\right], \qquad (\sigma^2_{\bar{\gamma}_0})_i = \mathrm{Var}\left[\tfrac{1}{\Delta} A_{0\to\Delta} \,\Big|\, X_0 = i\right], \tag{A23}$$

the first being Eq. A9 and the second the per-state variance Eq. A11 then asks for. Both come from the same tilted semigroup of Eq. A2 that any interval member needs, but from its derivatives contracted against the vector of ones rather than read entry by entry: what MR requires are the row sums, and the row sums are integrals of $K$-vectors, so MR is reached without ever forming a pair.

Eq. A22 is an assumption about the process and can be examined on its own terms. It is false for any window of nonzero length: a channel that starts shut and delivers current during the window is more likely to have ended open than one that delivers none, so the recorded average does carry information about the end state beyond what the start state gives. IR is the member that declines to assume it, and the boundary pair is the state on which the recorded average is linear with no conditional expectation taken at all. MR is therefore a correctly derived filter for a measurement model that is false, rather than an approximate version of IR; how much the falsehood costs is Eq. A28 below.

### What separates the start-conditioned members from the boundary one

Run as the same three steps on the $K$ states instead of the pairs, with $\bar{\boldsymbol{\gamma}}_0$ in place of $\boldsymbol{\gamma}$, two algebraic differences separate them from the one that conditions on both ends. The first is the form of the third term of Eq. A17. MR carries the total per-start-state interval variance, the spread of the interval-averaged conductance of a channel known only to have started in $i$, while VR and IR carry the residual variance left once the end state is also conditioned on, $\sum_j P_{i\to j}(\Delta)\,\bar{v}_{i\to j}$. By the law of total variance the two forms differ by the spread of the interval mean across end states,

$$\underbrace{\sum_j P_{i\to j}(\Delta)\left[\bar{v}_{i\to j} + \left(\bar{\gamma}_{i\to j} - \bar{\gamma}_{i\to\cdot}\right)^2\right]}_{\text{total, MR}} = \underbrace{\sum_j P_{i\to j}(\Delta)\,\bar{v}_{i\to j}}_{\text{residual, VR, IR}} + \mathrm{Var}_j\left[\bar{\gamma}_{i\to j} \mid i\right], \tag{A24}$$

so MR charges to the observation variance a spread that the end state, once conditioned on, would resolve. The second difference is the boundary cross-covariance term

$$N_{\mathrm{ch}}\, \widetilde{\boldsymbol{\gamma}^\top \boldsymbol{\Sigma} \boldsymbol{\gamma}} \tag{A25}$$

which IR retains and both start-conditioned members drop. It enters twice, in the second term of Eq. A17 and in the gain $\mathbf{g}_{0\to\Delta}$ of Eqs. A20 and A21.

The two differences are not independent, and the tempting reading of them, one step that moves the variance followed by one that moves the gain, is wrong in both directions. Taken from a common prior they cancel in the predicted variance: what the boundary term adds to the second term of Eq. A17 is $\boldsymbol{\mu}_0^\top \mathrm{Var}_j[\bar{\gamma}_{i\to j} \mid i]$, which is what the residual form removes from the third, so from the same state MR and IR assign the interval average the same variance by two different decompositions and the whole of what separates them is the gain (Appendix 2). In the other direction, the predicted variance divides the gain, so changing the variance form changes the update as well: `VR` removes that spread from the third term without restoring it in the second, predicts less variance than either from the same state, and its posterior departs from `MR`'s at once. The equality is therefore between two maps from a prior to a predicted variance, and not between two recordings. Once the gains differ the priors differ, and the variances the two members report along one recording differ with them, which is what the Results measure across the regime grid.

### Both members against the exact law

Neither member is measured against the other. Both are approximations to one joint law, that of $(\mathbf{N}_\Delta, \bar{y})$ under the true process, and given $(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ that law has exact first two moments which either member can be held against. Two elementary decompositions supply them, and they say different things about the two blocks.

The variance: both members are exact, and their slots differ.
By the law of total variance, conditioning on the counts where the window opens,

$$\mathrm{Var}\left[\bar{y}\right] = \epsilon^2_{0\to\Delta} + \underbrace{\mathbb{E}\left[\mathrm{Var}(\bar{y} \mid \mathbf{N}_0)\right]}_{N_{\mathrm{ch}}\,\boldsymbol{\mu}_0^\top \mathbf{w}} + \underbrace{\mathrm{Var}\left[\mathbb{E}(\bar{y} \mid \mathbf{N}_0)\right]}_{N_{\mathrm{ch}}\,\bar{\boldsymbol{\gamma}}_0^\top \boldsymbol{\Sigma}_0 \bar{\boldsymbol{\gamma}}_0}, \tag{A26}$$

which is Eq. A17 read with the start-conditioned objects: the middle term is $\bar{\boldsymbol{\gamma}}_0^\top \boldsymbol{\Sigma}_0 \bar{\boldsymbol{\gamma}}_0$, the tilde dropped, and $\mathbf{w}$ is the total slot of Eq. A24. Read with the boundary objects instead, the tilde of Eq. A18 in the middle and the residual slot in $\mathbf{w}$, the same quantity is Eq. A17 again. **Both members report the exact variance of the recorded value** given $(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, so they agree with each other for the reason that each agrees with the truth, and nothing is measured by that agreement. The variable conditioned on in Eq. A26 is the count vector and not one channel's state: conditioning on a single channel's $X_0$ returns the same expressions only while $\boldsymbol{\Sigma}_0$ is still that channel's own covariance, which it stops being at the first update.

The covariance: one member is exact and the other is short by a named amount.
By the law of total covariance, conditioning on the same counts and carried on the per-channel scale,

$$\frac{1}{N_{\mathrm{ch}}}\mathrm{Cov}\left[\mathbf{N}_\Delta,\ \bar{y}\right] = \underbrace{\frac{1}{N_{\mathrm{ch}}}\mathrm{Cov}\left[\mathbb{E}(\mathbf{N}_\Delta \mid \mathbf{N}_0),\ \bar{y}\right]}_{\mathbf{g}^{\mathrm{MR}}_{0\to\Delta}} + \frac{1}{N_{\mathrm{ch}}}\mathbb{E}\left[\mathrm{Cov}(\mathbf{N}_\Delta,\ \bar{y} \mid \mathbf{N}_0)\right]. \tag{A27}$$

The first term is the start-conditioned gain $\mathbf{g}^{\mathrm{MR}}_{0\to\Delta} = \mathbf{P}(\Delta)^\top \boldsymbol{\Sigma}_0 \bar{\boldsymbol{\gamma}}_0$ exactly. The second is what Eq. A22 sets to zero: given the counts the channels are conditionally independent, so it is the per-channel conditional covariance summed over the population, and a conditional independence is precisely the statement that it vanishes. It does not vanish. `IR` carries the whole of Eq. A27, so the difference between the two members in this block is that second summand and nothing else,

$$\mathbf{g}^{\mathrm{IR}}_{0\to\Delta} - \mathbf{g}^{\mathrm{MR}}_{0\to\Delta} = \left(\textstyle\sum_i \mu_{i\to j}\, D_{i\to j}\right)_j, \qquad D_{i\to j} = \bar{\gamma}_{i\to j} - (\bar{\gamma}_0)_i, \tag{A28}$$

which is Eq. A19 minus $\mathbf{P}(\Delta)^\top \boldsymbol{\Sigma}_0 \bar{\boldsymbol{\gamma}}_0$ written out. The deviation $D_{i\to j}$ is a property of the process and not of either algorithm: it is how far the interval average of a channel that ended in $j$ sits from the average over all the places it might have ended. It has zero row sums, $\sum_j P_{i\to j}(\Delta) D_{i\to j} = 0$, since $(\bar{\gamma}_0)_i$ is that very average, and it therefore has zero mean under $\mu_{i\to j}$ as well, so the gap between the two forms of Eq. A24 is exactly its variance under the law of the pairs. Eq. A28 does not depend on $\boldsymbol{\Sigma}_0$ at all. The same deviation is invisible in one block and decisive in the other because it enters the variance squared and the covariance once: a second moment can sit in either of the two variance terms without the total noticing, which is Eq. A24, while a collapsed first moment has nowhere to go, a noise term having no first moment. That asymmetry between $D^2$ and $D$ is the whole of the difference between the two members, and it is a consequence of Eq. A22 rather than a defect of the arithmetic that implements it.

### The same construction in the linear-filtering frame

Everything above was Gaussian conditioning: a joint law over (state, recorded value), its three blocks, and Bayes. Nothing was borrowed from linear filtering, and the derivation does not need it. The correspondence with that literature is nevertheless exact; the relation is conceded in the Discussion, and the equations that make it checkable are here.

For a state $\mathbf{x}$ and a scalar observation $y$ jointly Gaussian, the filter's update is

$$\mathbf{x}^{\text{post}} = \mathbf{x}^{\text{prior}} + \frac{\mathrm{Cov}(\mathbf{x}, y)}{\mathrm{Var}(y)}\big(y - \mathbb{E}[y]\big), \qquad \mathbf{P}^{\text{post}} = \mathbf{P}^{\text{prior}} - \frac{\mathrm{Cov}(\mathbf{x}, y)\,\mathrm{Cov}(\mathbf{x}, y)^\top}{\mathrm{Var}(y)}, \tag{A29}$$

which is Eq. 2 up to the per-channel convention, the source of the factors of $N_{\text{ch}}$ there and not here. The two frames do not differ in the update at all: both are the conditioning of a joint Gaussian, and what is usually called the gain is the ratio of the cross block to the observation variance. The entire content of either derivation is the computation of the three blocks $\mathbb{E}[y]$, $\mathrm{Var}(y)$ and $\mathrm{Cov}(\mathbf{x}, y)$ for a recorded value that is an average over the window.

The standard difficulty is that $\bar{y}$ is not a function of the state at any single time, so no observation matrix $\mathbf{H}$ satisfies $\bar{y} = \mathbf{H}\mathbf{x}_\Delta + v$. The device is state augmentation: carry the running integral $z_t = \int_0^t \boldsymbol{\gamma}^\top \mathbf{N}(s)\,\mathrm{d}s$ alongside the counts, so that $(\mathbf{N}, z)$ propagates jointly and the recorded value is $z_\Delta/\Delta$ plus instrumental noise, which *is* linear in the augmented state. This is the treatment of an observation that is a flow rather than a stock, and it is a known construction rather than a new one (***Zadrozny, 1988***). Augmentation moves the problem rather than solving it: the filter now needs the variance of the accumulator and its covariance with the occupancy at the end of the window, and in general those are not free. For a continuous-time Markov chain they are, and by the two derivatives already written as Eq. A2: the $(i \to j)$ entry of $\mathbf{F}_1$ is $\mathbb{E}[A_{0\to\Delta}\mathbf{1}\{X_\Delta = j\} \mid X_0 = i]$ and of $2\mathbf{F}_2$ the same with $A^2_{0\to\Delta}$, which is exactly what the augmented filter asks for.

Condition on the counts where the window opens; given those the channels are independent, and the three blocks are

$$\mathbb{E}\big[\bar{y}\big] = N_{\text{ch}}\,\boldsymbol{\mu}_0^\top \bar{\boldsymbol{\gamma}}_0, \tag{A30}$$

$$\mathrm{Var}\big[\bar{y}\big] = \epsilon^2_{0\to\Delta} + \frac{\mathrm{Var}(z_\Delta)}{\Delta^2}, \tag{A31}$$

$$\frac{\mathrm{Cov}\big[\mathbf{N}_\Delta, \bar{y}\big]}{N_{\text{ch}}} = \frac{\mathrm{Cov}\big[\mathbf{N}_\Delta, z_\Delta\big]}{N_{\text{ch}}\Delta} = \mathbf{P}(\Delta)^\top \boldsymbol{\Sigma}_0 \bar{\boldsymbol{\gamma}}_0 + \Big(\textstyle\sum_i \mu_{i\to j} D_{i\to j}\Big)_j. \tag{A32}$$

Eq. A30 is Eq. A16, Eq. A31 is Eq. A17, and Eq. A32 is $\mathbf{g}^{\text{MR}}_{0\to\Delta}$ plus Eq. A28, which is to say the $\mathbf{g}_{0\to\Delta}$ of Eq. A19. The second term of Eq. A32 needs nothing from the boundary construction: for one channel, $\mathbf{F}_1$ divided by $\Delta$ and by the transition probability gives $\mathbb{E}[\mathbf{1}\{X_\Delta = j\}\,\frac{1}{\Delta}A_{0\to\Delta} \mid X_0 = i] = P_{i\to j}(\Delta)\bar{\gamma}_{i\to j}$, the product of the two conditional expectations is $P_{i\to j}(\Delta)(\bar{\gamma}_0)_i$, and their difference is $P_{i\to j}(\Delta)D_{i\to j}$. The deviation of Eq. A28 is therefore what the first derivative of the tilted semigroup leaves once the product of the two first moments is taken out, and the agreement between the two routes is a result rather than an input.

**The augmented-state filter and the boundary-state filter compute the same three blocks**, so they perform the same update and return the same posterior over the occupancy.

The augmented state has $K+1$ entries against the boundary state's $K^2$, so *the pair is a device of the derivation and not information the filter must carry*, which is the same conclusion Eqs. A14 and A15 reached from the other side. Given $(\mathbf{N}_\Delta, z_\Delta)$ the recorded value is deterministic, so what Eq. A17 splits between its second and third terms sits entirely inside $\mathrm{Var}(z_\Delta)$, and that is why the cross block needs only $\mathbf{F}_1$ while the variance block needs $\mathbf{F}_2$ as well. The equivalence holds per window, with a reset: the augmented filter must discard the posterior on $z$ and open the next window with $z=0$, because the next window's integral is a fresh one and is independent of the old given the occupancy.

`MR` is not the augmented filter under any choice of blocks. It is the filter for a misspecified measurement model,

$$\bar{y} = \mathbf{H}\mathbf{N}_0 + v(\mathbf{N}_0), \qquad \mathbf{H} = \bar{\boldsymbol{\gamma}}_0^\top, \qquad \mathrm{Var}\left[v \mid \mathbf{N}_0\right] = \epsilon^2_{0\to\Delta} + \mathbf{N}_0^\top \mathbf{w}, \tag{A33}$$

whose expectation over $\mathbf{N}_0$ returns Eq. A17. The measurement noise depends on the state, which is outside the textbook filter: with that extension the two frames give the same equations for `MR` as well, and without it `MR` has no linear-filtering form at all.

Where the frames genuinely differ is the model rather than the arithmetic. A Kalman derivation presumes a linear-Gaussian state process, and the occupancy counts of a channel population are not one: they are multinomial, their transition noise depends on the state they leave, and no linear-Gaussian system generates them. The derivation above presumes none of that: it computes the exact first two moments of the true process, over the pairs where the recorded average is genuinely linear, and replaces the law by a Gaussian only at the two places the box above names, so the two routes arrive at the same equations because the conditioning of a joint Gaussian consumes nothing but the first two moments. The pair survives as the presentation even though the augmented state reaches the same blocks with fewer entries: on the pair the recorded average is linear in the state with no conditional expectation taken, the collapse hands over the next window's prior at no cost, and the per-state variance slot is what makes `R`, `MR` and `IR` one construction with three readings.

### What was verified, and to what tolerance

The identities of this section were checked against a direct contraction over the $K^2$ pairs rather than asserted. Eqs. A18 and A19 reproduce the explicit contraction of the boundary covariance in $108$ random cases to $7\times10^{-18}$ and $3\times10^{-16}$; the commutation of the start-index marginalization with the update, which is what licenses Eqs. A20 and A21 being written on the $K$ states, holds in $36$ cases to $2\times10^{-16}$; Eq. A28 holds to $1\times10^{-17}$ in $36$ cases, independently of $\boldsymbol{\Sigma}_0$, and against a Monte Carlo of the continuous-time path; Eqs. A12 and A13 were checked against a Monte Carlo of $4\times10^5$ replicates of $200$ channels, worst entry $2.9$ standard errors over $256$ entries; and the two forms of Eq. A24 agree with the forms that go through the pair-resolved variance to $1\times10^{-15}$ in $16$ cases. The whole construction, run on the pairs and marginalized, reproduces Eqs. A14, A15, A16, A17, A18, A19, A20 and A21 to $9.3\times10^{-16}$ relative on two priors, a multinomial start and a perturbed post-update one.

## The interval update, member by member

Appendix 1 gives the cycle for the boundary-conditioned member and names the two places where the others differ. This appendix writes every member out in one template, so that each is fixed by three switches on a single set of equations.

### Objects computed once per interval

All members share the quantities built from the rate matrix $\mathbf{Q}$ and the recording filter for a window of length $\Delta$: the transition matrix $\mathbf{P}(\Delta)$; the boundary-conditioned single-channel moments $\bar{\gamma}_{i\to j}$ and $\bar{v}_{i\to j}$; the start-conditioned mean conductance $\bar{\boldsymbol{\gamma}}_0$ of Eq. A9; the start-conditioned interval variance $\mathbf{w}$; and, for the members that ignore the acquisition averaging, the instantaneous per-state conductance vector $\boldsymbol{\gamma}$. Two matrices collect the boundary weights,

$$G_{i\to j} \;=\; \bar{\gamma}_{i\to j}\, P_{i\to j}(\Delta), \qquad H_{i\to j} \;=\; \bar{\gamma}^{2}_{i\to j}\, P_{i\to j}(\Delta). \tag{A34}$$

### The tilde operator

Over a bold symbol, the tilde marks a contraction taken on the $K^2$ boundary pairs $(i \to j)$ rather than on the $K$ states. (Over a scalar it means something else, a quantity in the natural units of the model, as in $\widetilde{\Delta}$ and $\widetilde{S}$; the two uses are disjoint and never meet in one expression.) It is not a new operator: its two cases are the two contractions the update of Eq. 2 already has, $\boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}\boldsymbol{\gamma}$ and $\boldsymbol{\Sigma}\boldsymbol{\gamma}$, evaluated there with the boundary objects substituted. Both are written out in Appendix 1, the bilinear one as the scalar of Eq. A18 and the vector one as the gain $\mathbf{g}_{0\to\Delta} = \widetilde{\boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}}$ of Eq. A19, and both collapse to $K \times K$ arithmetic once the boundary covariance Eq. A13 is substituted.

### One cycle, three switches

Every member on the lattice runs the same four steps on each window: propagate the occupancy Gaussian (Eqs. A14 and A15), predict the observation (Eqs. A16 and A17), condition on the recorded average (Eqs. A20 and A21), and either carry the posterior to the next window or discard it. Three switches fix which member is being run.

Switch 1: the window.

It has three settings, an instant, the state at the window's start, and the states at both of its ends, and it selects the conductance object, and with it the second term of the predictive variance and the gain. At the instantaneous setting the observation is treated as a sample taken at the centre of the window: the occupancy is propagated there first, by $\mathbf{P}(\Delta/2)$, the conductance is $\boldsymbol{\gamma}$, and, with $\boldsymbol{\Sigma}$ read in that mid-window frame,

$$\widetilde{\boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}\boldsymbol{\gamma}} \;\to\; \boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}\boldsymbol{\gamma}, \qquad \mathbf{g} \;\to\; \mathbf{P}(\Delta/2)^{\top}\boldsymbol{\Sigma}\,\boldsymbol{\gamma}. \tag{A35}$$

At the start-conditioned setting the same two expressions hold with $\bar{\boldsymbol{\gamma}}_0$ in place of $\boldsymbol{\gamma}$ and the full $\mathbf{P}(\Delta)$ in the gain, since the conductance is conditioned on the state at the window's start and the whole window remains to be crossed; the contraction is still an ordinary quadratic form on the $K$ states. The trailing propagator in the gain is not decoration. It carries the update from the frame in which the observation was formed to the frame the next window starts in, which is the end of the current window, so it is the remaining half at the instantaneous setting and the whole window at the start-conditioned one. Omitting it, or using the full window where half is called for, leaves the rank-one down-date in the wrong frame, and the symptom is specific and was observed: a spike in the distortion-induced bias of the recursive instantaneous member at long intervals and large channel counts. At the boundary setting the conductance is conditioned on both endpoints and the two expressions become the tildes, Eq. A18 and Eq. A19. The step from the start state to the boundary therefore does two things at once: it replaces $\boldsymbol{\Sigma}_0$ by $\boldsymbol{\Sigma}_0 - \mathrm{diag}(\boldsymbol{\mu}_0)$ and adds the boundary term ($\boldsymbol{\mu}_0^\top \mathbf{H}\mathbf{1}$ in the variance, $\mathbf{G}^\top \boldsymbol{\mu}_0$ in the gain). The subtracted diagonal is not a second approximation: the within-channel part it removes is already inside the boundary term, and dropping the subtraction would count it twice.

### Switch 2: recursion.

A recursive member carries $(\boldsymbol{\mu}^{\mathrm{post}}, \boldsymbol{\Sigma}^{\mathrm{post}})$ into the next window. A non-recursive one discards it and propagates the next window's predictive from the initial condition, so its log-likelihood factorizes over windows. The switch changes no formula above; it changes only what is fed to the next one.

### Switch 3: the interval variance, total or residual.

It selects the third term of Eq. A17, which is present only when the acquisition averaging is modelled at all, that is at either of the two interval settings of switch 1:

$$(\mathbf{w})_i = \begin{cases} \sum_j P_{i\to j}(\Delta)\left[\bar{v}_{i\to j} + \left(\bar{\gamma}_{i\to j} - \bar{\gamma}_{i\to\cdot}\right)^2\right] & \text{total,} \\ \sum_j P_{i\to j}(\Delta)\,\bar{v}_{i\to j} & \text{residual.} \end{cases} \tag{A36}$$

The two differ by the spread of the interval mean across end states, which is Eq. A24. The switch is free only where the window is conditioned on the start state alone. With both endpoints conditioned on, the residual form is forced whatever the flag says, for the same reason the diagonal is subtracted in switch 1: the between-endpoint spread is already carried by the boundary term, so the total form would count it twice.

## The members

Table 1 sets the six lattice members against the three switches, and it is the equation-level counterpart of Table 1 in the body: where that table says in words what each member predicts, this one says which setting of Eqs. A35, A36 and A20 produces it.

| Member | window | recursive | interval variance | what it is |
|---|---|---|---|---|
| `NR` | an instant | no | — | instantaneous samples, windows independent |
| `R` | an instant | yes | — | instantaneous samples, posterior carried forward |
| `INR` | the start | no | total | `NR` with the window average restored |
| `MR` | the start | yes | total | `R` with the window average restored |
| `VR` | the start | yes | residual | `MR` with switch 3 flipped, gain unchanged |
| `IR` | both ends | yes | residual (forced) | `VR` with the boundary terms restored |

**Appendix 2—table 1.** The six members of the lattice as settings of the three switches. Reading down the last column, each row differs from an earlier one in exactly one switch, which is what makes the ladder measurable one step at a time. `IR` is MacroIR; `INR` is the MacroINR of ***Moffatt and Pierdominici-Sottile*** (***2025***). The data keys under which each is dispatched are in Supplementary File 1.

One consequence of the two switches acting together is worth stating, because it is easy to read the table as saying otherwise. Evaluated at the *same* prior, `MR` and `IR` assign the interval average the same total variance. Expanding the total form of Eq. A36 and using $\sum_j P_{i\to j}(\Delta) = 1$ and $\sum_j P_{i\to j}(\Delta)\bar{\gamma}_{i\to j} = \bar{\gamma}_{i\to\cdot}$,

$$\boldsymbol{\mu}_0^\top \mathbf{w}\Big|_{\text{total}} \;=\; \boldsymbol{\mu}_0^\top \mathbf{w}\Big|_{\text{residual}} \;+\; \boldsymbol{\mu}_0^\top \mathbf{H}\mathbf{1} \;-\; \bar{\boldsymbol{\gamma}}_0^\top \mathrm{diag}(\boldsymbol{\mu}_0)\,\bar{\boldsymbol{\gamma}}_0, \tag{A37}$$

and the two extra pieces are exactly what switch 1 moves from the third term of Eq. A17 into the second when the window goes from the start state to both ends. The decompositions differ, the sum does not. What does change between the two members is the gain, and at a common prior that is the whole of the `MR`-to-`IR` difference.

Two of the six pairs isolate one algebraic change at the point where it is made, and they are the ones the Results use. Stepping `MR` to `VR` flips switch 3, and stepping `VR` to `IR` restores the boundary terms in both the variance and the gain while leaving the variance form where it already was; neither step is confined to what it flips, for the reason given in Appendix 1. The two together are the `R`-to-`IR` gap.

### Least squares, which is off the lattice

Classical nonlinear least squares has no setting of the three switches. It propagates the mean occupancy alone, $\boldsymbol{\mu}_\Delta = \mathbf{P}(\Delta)^\top \boldsymbol{\mu}_0$, predicts $\bar{y}^{\text{pred}}_{0\to\Delta}$ from Eq. A16, and assigns one constant fitted variance to every residual. There is no gain, so no occupancy object is ever conditioned on the data. It is the anchor of the comparison rather than a member of the family.

## Appendix 3

### The diagnostics, derived

The body states the three checks and the two matrices they produce. This appendix carries what is behind them: the expansion that turns a score into a parameter displacement, the exact sense in which the two parts of the distortion compose, and the two scalar summaries with the floor below which neither can be read.

#### The sandwich and the bias are one first-order expansion

Both corrections come from the same step. Expanding the total score around the parameters $\boldsymbol{\theta}_0$ the recordings were generated from,

$$S(\hat{\boldsymbol{\theta}}) \;\approx\; S(\boldsymbol{\theta}_0) + \nabla_{\boldsymbol{\theta}} S(\boldsymbol{\theta}_0)\,(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0), \tag{A38}$$

and using $S(\hat{\boldsymbol{\theta}}) = 0$ at a maximum-likelihood estimate, with the empirical Hessian replaced by its expectation $-\mathbf{H}$,

$$\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0 \;\approx\; \mathbf{H}^{-1} S(\boldsymbol{\theta}_0). \tag{A39}$$

Taking the covariance of Eq. A39 over independent recordings gives the sandwich,

$$\boldsymbol{\Sigma} \;=\; \mathbf{H}^{-1}\mathbf{J}\,\mathbf{H}^{-1} \;=\; \mathbf{H}^{-1/2}\,\mathbf{C}\,\mathbf{H}^{-1/2}, \tag{A40}$$

and taking its expectation gives the bias, $b = \mathbf{H}^{-1}\mathbb{E}[S]$, which is nonzero exactly when the mean score is. The two are therefore the same approximation applied to the two moments of one displacement, which is why a member can fail them separately: $\mathbb{E}[S]$ can vanish while $\mathrm{Cov}(S)$ departs from $\mathbf{H}$, and at $\boldsymbol{\theta}_{\text{pool}}$ it does so by construction. When $\mathbf{C} = \mathbf{I}$, Eq. A40 returns $\mathbf{H}^{-1}$ and the correction is inert. Where $\mathbf{H}$ is singular, every inverse is taken on its retained eigenspace (Methods).

Both corrections are first order, and the sandwich is verified directly against the empirical covariance of the estimates over the simulated recordings and against the coverage of the nominal region (Figure 2).

#### How the two parts of the distortion compose

These compose exactly. The exact identity is

$$\mathbf{C} = \mathbf{K}\,\mathbf{R}\,\mathbf{K}^{\top}, \qquad \mathbf{K} = \mathbf{H}^{-1/2}\,\mathbf{J}_s^{1/2},$$

which holds as algebra for any $\mathbf{H}$ and $\mathbf{J}_s$, since substituting the definitions collapses the chain. The factor $\mathbf{K}$ satisfies $\mathbf{K}\mathbf{K}^{\top} = \mathbf{C}_s$, so it is a legitimate square-root factor of the sample distortion, but it is not the symmetric principal square root, and the form $\mathbf{C} = \mathbf{C}_s^{1/2}\,\mathbf{R}\,\mathbf{C}_s^{1/2}$ holds only where $\mathbf{H}$ and $\mathbf{J}_s$ commute. What composes with no assumption at all is the information volume,

$$\log\det\mathbf{C} = \log\det\mathbf{C}_s + \log\det\mathbf{R},$$

an exact identity because the determinant is multiplicative. The per-parameter diagonal, the trace, the eigenvalues and the Frobenius norm do not compose across the split; only the volume does, and only the volume is decomposed in the maps.

#### The two scalars, and the distance they are the components of

With $\lambda_1, \ldots, \lambda_d$ the eigenvalues of $\mathbf{C}$ on the retained subspace, the magnitude $m = e^{\bar{\ell}}$ and the anisotropy $a = e^{s}$ of Eq. 4 are the exponentials of the mean and the standard deviation of

$\log \lambda_i$. They are not two arbitrary summaries: they are the two orthogonal components of the affine-invariant Riemannian distance from the identity,

$$D_{\mathrm{AI}} = \left(\sum_i \log^2 \lambda_i\right)^{1/2} = \sqrt{d\,(\bar{e}^2 + s^2)}, \tag{A41}$$

so that a member sits at $m = a = 1$ if and only if $D_{\mathrm{AI}} = 0$, and the two coordinates say whether a departure is a common factor on every direction or a spread between them. The pair is reported throughout and $D_{\mathrm{AI}}$ is reported nowhere in its place, because the decision the Results support is exactly the one the pair separates and the single number collapses: whether a scalar correction can work at all ($a = 1$) and what factor it should use ($m$).

Neither reads below its own sampling floor. The anisotropy of an estimated distortion is positive by construction even when the truth is isotropic, since the eigenvalues of a matrix estimated from a finite cloud spread on their own. Drawing a thousand estimates in two dimensions from a perfectly calibrated covariance and forming the same statistic gives a median of $1.038$ and a $95$th percentile of $1.081$, so a measured spread of that size is not evidence of shape; and that is a lower bound on the floor, because the reported covariance the statistic is formed against is itself estimated.

## Appendix 4

### Repairing an interval without changing the likelihood

A reader who accepts the miscalibration can still ask what it would cost to repair it without changing likelihood, and the alternatives answer different questions. The standard error across independent fits to separate patches is free, but it describes the uncertainty of a population mean, mixes statistical with biological variability, and leaves any bias in place. Resampling whole sweeps is valid and keeps the within-sweep correlation, but it brackets whatever the estimator converges to without saying whether that estimator is right. Resampling residuals is not valid here, because independent resampling destroys the correlation that is the problem; a parametric bootstrap assumes the model one was checking; a block bootstrap needs a block length well above a correlation time that has to be measured first. Simulating from the fitted model and forming the sandwich answers the same question the diagnostic does, at a cost set by the precision needed on a covariance, and still simulates from the model one wanted to verify. None of these, the sandwich included, reaches the Bayesian evidence: every repair here corrects the covariance a likelihood reports, while the evidence is an integral of the likelihood itself, which no covariance correction touches, so a comparison built on evidences is repaired only by changing the likelihood.

## References


**Anderson CR**, Stevens CF. Voltage Clamp Analysis of Acetylcholine Produced End-Plate Current Fluctuations at Frog Neuromuscular Junction. The Journal of Physiology. 1973; 235(3):655–691. doi: 10.1113/jphysiol.1973.sp010410.

**Celentano JJ**, Hawkes AG. Use of the Covariance Matrix in Directly Fitting Kinetic Parameters: Application to GABAA Receptors. Biophysical Journal. 2004; 87(1):276–294. doi: 10.1529/biophysj.103.036632.

**Clerx M**, Beattie KA, Gavaghan DJ, Mirams GR. Four Ways to Fit an Ion Channel Model. Biophysical Journal. 2019; 117(12):2420–2437. doi: 10.1016/j.bpj.2019.08.001.

**Colquhoun D**, Hawkes AG. On the Stochastic Properties of Single Ion Channels. Proceedings of the Royal Society of London Series B. 1981; 211(1183):205–235. doi: 10.1098/rspb.1981.0003.

**Colquhoun D**, Sigworth FJ. Fitting and Statistical Analysis of Single-Channel Records. In: Sakmann B, Neher E, editors. *Single-Channel Recording*, 2 ed. New York: Plenum Press; 1995.p. 483–587. doi: 10.1007/978-1-4419-1229-9_19.

**Conti F**, Neumcke B, Nonner W, Stämpfli R. Conductance Fluctuations from the Inactivation Process of Sodium Channels in Myelinated Nerve Fibres. The Journal of Physiology. 1980; 308:217–239. doi: 10.1113/jphysiol.1980.sp013279.

**Dawid AP**. Present position and potential developments: Some personal views. Statistical theory: The prequential approach. Journal of the Royal Statistical Society Series A (General). 1984; 147(2):278–292. doi: 10.2307/2981683.

**Del Core L**, Mirams GR. Parameter Inference for Stochastic Reaction Models of Ion-Channel Gating from Whole-Cell Voltage-Clamp Data. Philosophical Transactions of the Royal Society A. 2025; 383(2292):20240224. doi: 10.1098/rsta.2024.0224.

**Godambe VP**. An Optimum Property of Regular Maximum Likelihood Estimation. The Annals of Mathematical Statistics. 1960; 31(4):1208–1211. doi: 10.1214/aoms/1177705693.

**Heinemann SH**, Conti F. Nonstationary noise analysis and application to patch clamp recordings. Methods in Enzymology. 1992; 207:131–148. doi: 10.1016/0076-6879(92)07009-D.

**Hines KE**, Middendorf TR, Aldrich RW. Determination of parameter identifiability in nonlinear biophysical models: A Bayesian approach. The Journal of General Physiology. 2014; 143(3):401–416. doi: 10.1085/jgp.201311116.

**Hodgkin AL**, Huxley AF. A Quantitative Description of Membrane Current and its Application to Conduction and Excitation in Nerve. The Journal of Physiology. 1952; 117(4):500–544. doi: 10.1113/jphysiol.1952.sp004764.

**Huber PJ**. The Behavior of Maximum Likelihood Estimates under Nonstandard Conditions. In: *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability, Vol. 1* Berkeley: University of California Press; 1967.p. 221–233. https://projecteuclid.org/euclid.bsmsp/1200512988.

**Jiang R**, Taly A, Lemoine D, Martz A, Specht A, Grutter T. Intermediate closed channel state(s) precede(s) activation in the ATP-gated P2X2 receptor. Channels. 2012; 6(5):398–402. doi: 10.4161/chan.21520.

**Katz B**, Miledi R. Membrane Noise Produced by Acetylcholine. Nature. 1970; 226:962–963. doi: 10.1038/226962a0.

**Katz B**, Miledi R. The Statistical Nature of the Acetylcholine Potential and its Molecular Components. The Journal of Physiology. 1972; 224(3):665–699. doi: 10.1113/jphysiol.1972.sp009918.

**Lei CL**, Ghosh S, Whittaker DG, Aboelkassem Y, Beattie KA, Cantwell CD, Delhaas T, Houston C, Novaes GM, Panfilov AV, Pathmanathan P, Riabiz M, dos Santos RW, Walmsley J, Carr H, Gavaghan DJ, Mirams GR. Considering Discrepancy when Calibrating a Mechanistic Electrophysiology Model. Philosophical Transactions of the Royal Society A. 2020; 378(2173):20190349. doi: 10.1098/rsta.2019.0349.

**Mehra RK**. On the identification of variances and adaptive Kalman filtering. IEEE Transactions on Automatic Control. 1970; 15(2):175–184. doi: 10.1109/TAC.1970.1099422.

**Milescu LS**, Akk G, Sachs F. Maximum Likelihood Estimation of Ion Channel Kinetics from Macroscopic Currents. Biophysical Journal. 2005; 88(4):2494–2515. doi: 10.1529/biophysj.104.053256.

**Moffatt L**. Estimation of Ion Channel Kinetics from Fluctuations of Macroscopic Currents. Biophysical Journal. 2007; 93(1):74–91. doi: 10.1529/biophysj.106.101212.

**Moffatt L**. MacroIR: Bayesian Kinetic Modeling of P2X2 Activation. Zenodo; 2025. https://doi.org/10.5281/zenodo.17162475, doi: 10.5281/zenodo.17162475, software, version v1.0.0.

**Moffatt L**, Hume RI. Responses of Rat P2X2 Receptors to Ultrashort Pulses of ATP Provide Insights into ATP Binding and Channel Gating. Journal of General Physiology. 2007; 130(2):183–201. doi: 10.1085/jgp.200709779.

**Moffatt L**, Pierdominici-Sottile G. Bayesian Inference of Functional Asymmetry in the P2X2 Receptor. Communications Biology. 2025; 8:1740. doi: 10.1038/s42003-025-09056-x.

**Münch JL**, Paul F, Schmauder R, Benndorf K. Bayesian Inference of Kinetic Schemes for Ion Channels by Kalman Filtering. eLife. 2022; 11:e62714. doi: 10.7554/eLife.62714.

**Neher E**, Stevens CF. Conductance Fluctuations and Ionic Pores in Membranes. Annual Review of Biophysics and Bioengineering. 1977; 6:345–381. doi: 10.1146/annurev.bb.06.060177.002021.

**Owen MJ**, Mirams GR. IonBench: A Benchmark of Optimisation Strategies for Mathematical Models of Ion Channel Currents. PLoS Computational Biology. 2025; 21(8):e1013319. doi: 10.1371/journal.pcbi.1013319.

**Sigworth FJ**. The variance of sodium current fluctuations at the node of Ranvier. The Journal of Physiology. 1980; 307:97–129. doi: 10.1113/jphysiol.1980.sp013426.

**Sigworth FJ**. Covariance of Nonstationary Sodium Current Fluctuations at the Node of Ranvier. Biophysical Journal. 1981; 34(1):111–133. doi: 10.1016/S0006-3495(81)84840-0.

**Sokal A**. Monte Carlo methods in statistical mechanics: Foundations and new algorithms. In: DeWitt-Morette C, Cartier P, Folacci A, editors. *Functional Integration: Basics and Applications* Boston: Springer; 1997.p. 131–192. doi: 10.1007/978-1-4899-0319-8_6.

**Stepanyuk A**, Borisyuk A, Belan P. Maximum Likelihood Estimation of Biophysical Parameters of Synaptic Receptors from Macroscopic Currents. Frontiers in Cellular Neuroscience. 2014; 8:303. doi: 10.3389/fncel.2014.00303.

**Stepanyuk AR**, Borisyuk AL, Belan PV. Efficient Maximum Likelihood Estimation of Kinetic Rate Constants from Macroscopic Currents. PLoS ONE. 2011; 6(12):e29731. doi: 10.1371/journal.pone.0029731.

**Van Loan CF**. Computing Integrals Involving the Matrix Exponential. IEEE Transactions on Automatic Control. 1978; 23(3):395–404. doi: 10.1109/TAC.1978.1101743.

**White H.** Maximum Likelihood Estimation of Misspecified Models. Econometrica. 1982; 50(1):1–25. doi: 10.2307/1912526.

**Zadrozny PA.** Gaussian Likelihood of Continuous-Time ARMAX Models When Data Are Stocks and Flows at Different Frequencies. Econometric Theory. 1988; 4(1):108–124. doi: 10.1017/S0266466600011890.

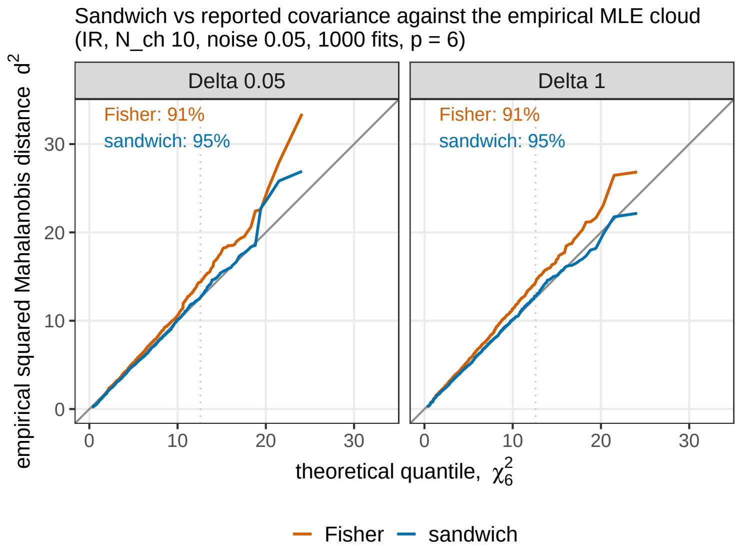


**Figure 2—figure supplement 1. The reported covariance against the empirical one, in all six directions at once.** The squared Mahalanobis distance of each estimate from the cloud mean, sorted and drawn against the quantiles of a $\chi^2$ with six degrees of freedom, under the covariance the member reports and under the distortion-corrected sandwich. A calibrated covariance lies on the identity; above it the reported region is too small and under-covers. The reported Fisher covariance under-covers the nominal $95\%$ region and the sandwich covers it, $0.914$ against $0.947$ to $0.949$. Centring on the cloud mean tests the second moment alone, which is what the correction claims to repair. Two things differ from the body figure, both forced by the sample: the member is `IR` alone, and the cell is $N_{\mathrm{ch}} = 10$ at $\widetilde{S} = 0.005$ with group size $100$, the one cloud in the set carrying a thousand fits of that size.

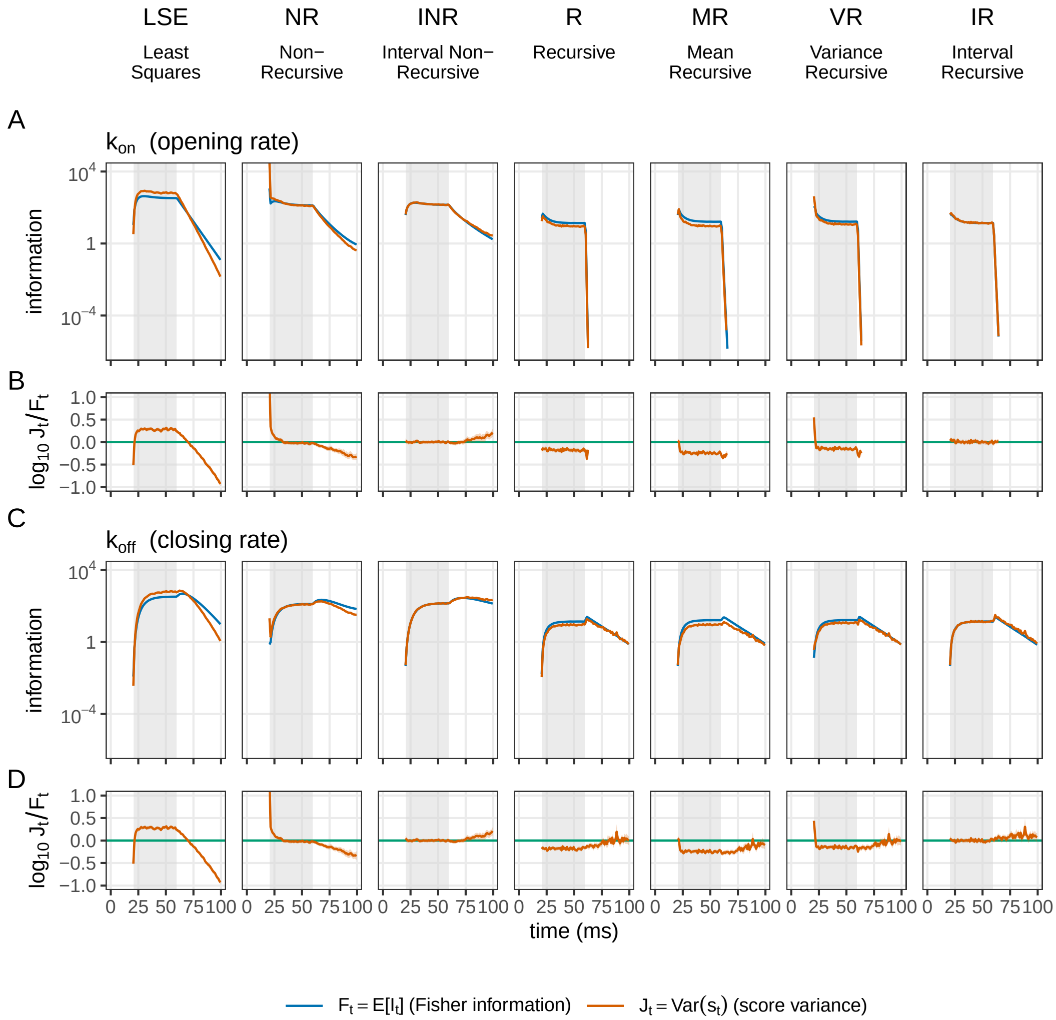


**Figure 3—figure supplement 1. Per-step information against score variance, the opening rate and the closing rate.** Seven members across the columns, the least-squares arm being the interval-averaged `ILSE`, and two parameters down the page as bands of two rows: $k_{\text{on}}$ (A, B) and $k_{\text{off}}$ (C, D). The upper row of each band carries the per-interval Fisher information and the score variance as magnitudes, which lie on each other where the member is calibrated at that interval; the lower row is $\log_{10}(J_t/F_t)$ with its 95% bootstrap interval, zero when they do. Per-interval calibration is not monotone along the cost ladder: on $k_{\text{off}}$ the median reads $+0.184$ for `ILSE`, $-0.038$ for `NR`, $+0.007$ for `INR`, $-0.145$ for `R`, $-0.229$ for `MR`, $-0.135$ for `VR` and $+0.033$ for `IR`,. Curves stop at the information mask, which is also where the upper axis ends. The least-squares ratio row is the same curve in every band of both halves, median 0.184: with a single constant variance the parameter cancels out of the ratio, leaving the true residual variance over the assumed one.

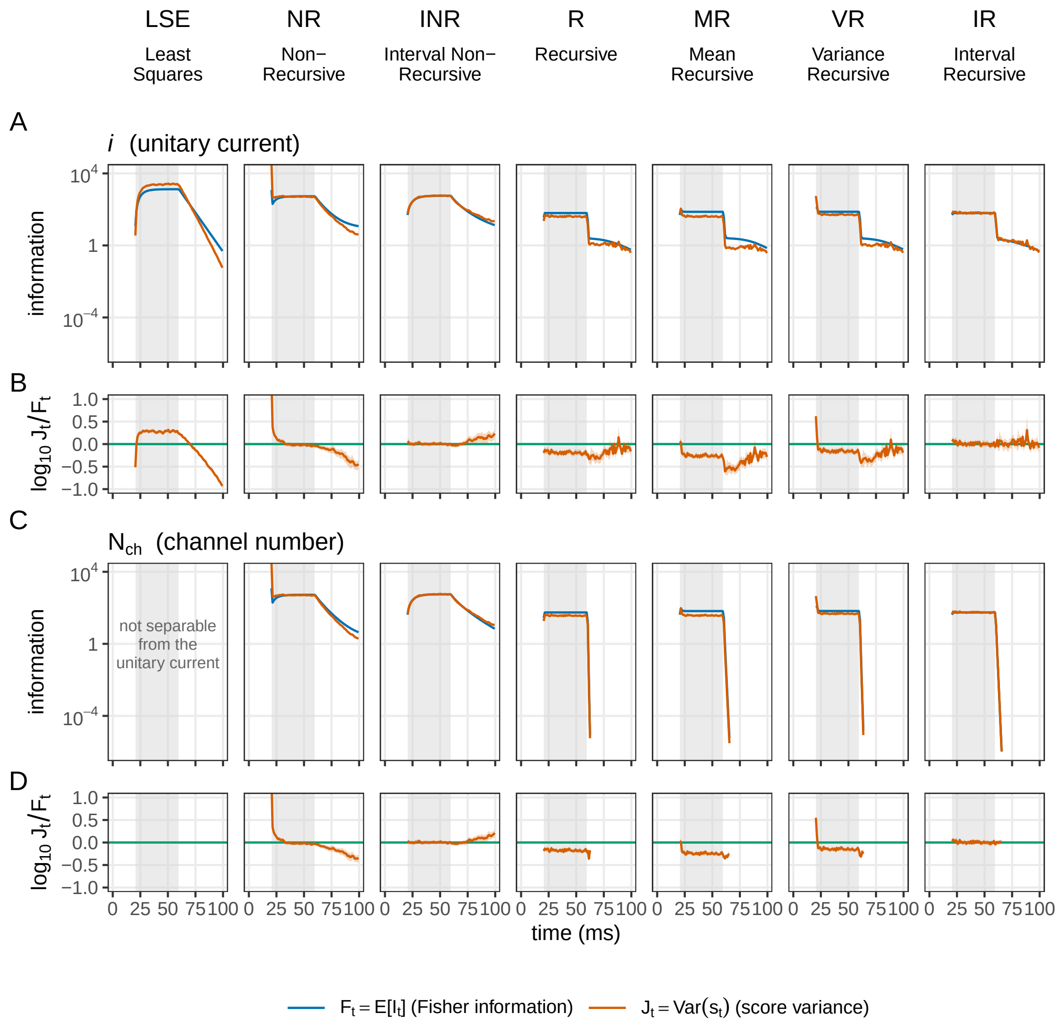


**Figure 3—figure supplement 2. Per-step information against score variance, the unitary current and the channel number.** The other half of the preceding supplement, on the same seven members and read the same way: the unitary current (A, B) and $N_{\mathrm{ch}}$ (C, D), the upper row of each band the per-interval Fisher information against the score variance and the lower row $\log_{10}(J_t/F_t)$ with its $95\%$ bootstrap interval. In the two recursive columns the $N_{\mathrm{ch}}$ band reaches the information floor within six intervals of washout while the unitary-current band does not. The channel-number band carries no least-squares panel: $N_{\mathrm{ch}}$ is not separable from $i$ there.

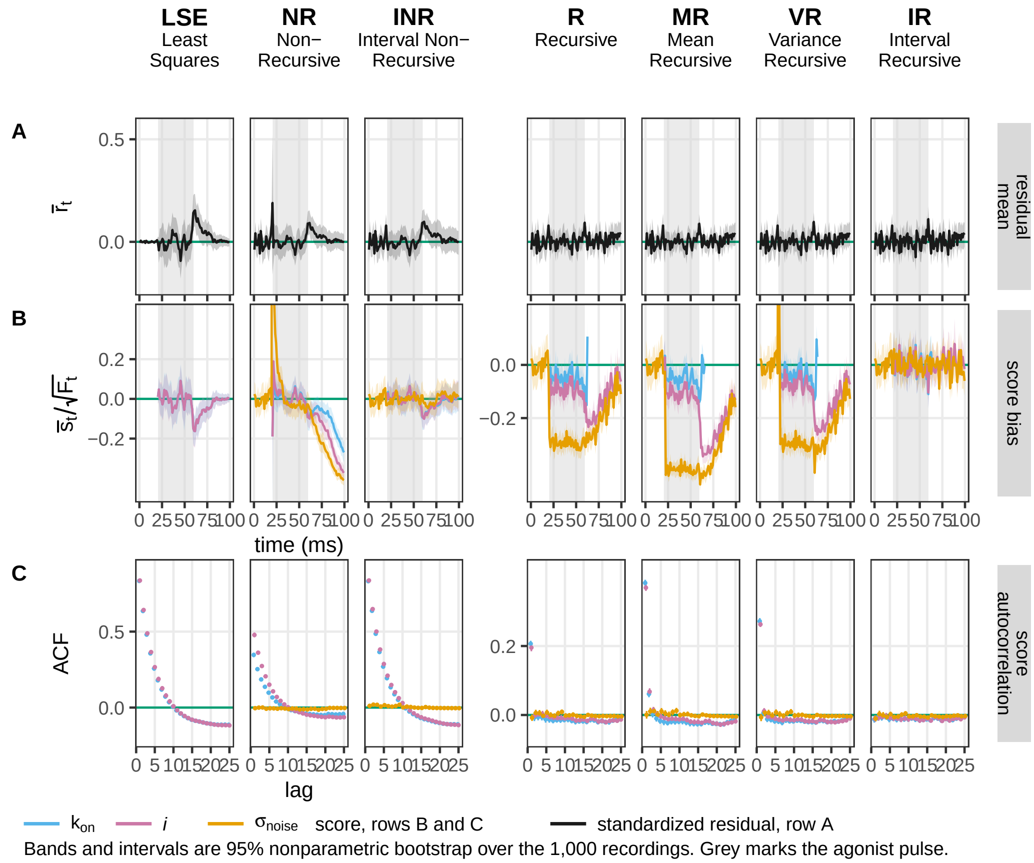


**Figure 3—figure supplement 3. The checks this figure has no room for.** The seven members of the preceding supplements at the same cell, carrying (A) the mean standardized residual, the one of the three properties of $r_t$ not drawn above; (B) the score bias for the three parameters not coloured above, $k_{\mathrm{on}}$, the unitary current and the instrumental noise level; and (C) their score autocorrelation. The first residual moment separates nobody, which is what makes the other two properties the informative ones. In (B) the ordering repeats the one in figure supplement 2: counting informative intervals whose interval excludes zero against a chance level of $0.05$, `IR` reads $0.067$, $0.050$ and $0.050$, `INR` $0.100$, $0.100$ and $0.000$, and the three recursive members between $0.409$ and $0.975$. The noise level, measured before the agonist arrives, keeps a white score even where the kinetic directions do not. Least squares carries no noise curve: its noise level is a plug-in, not a free direction.

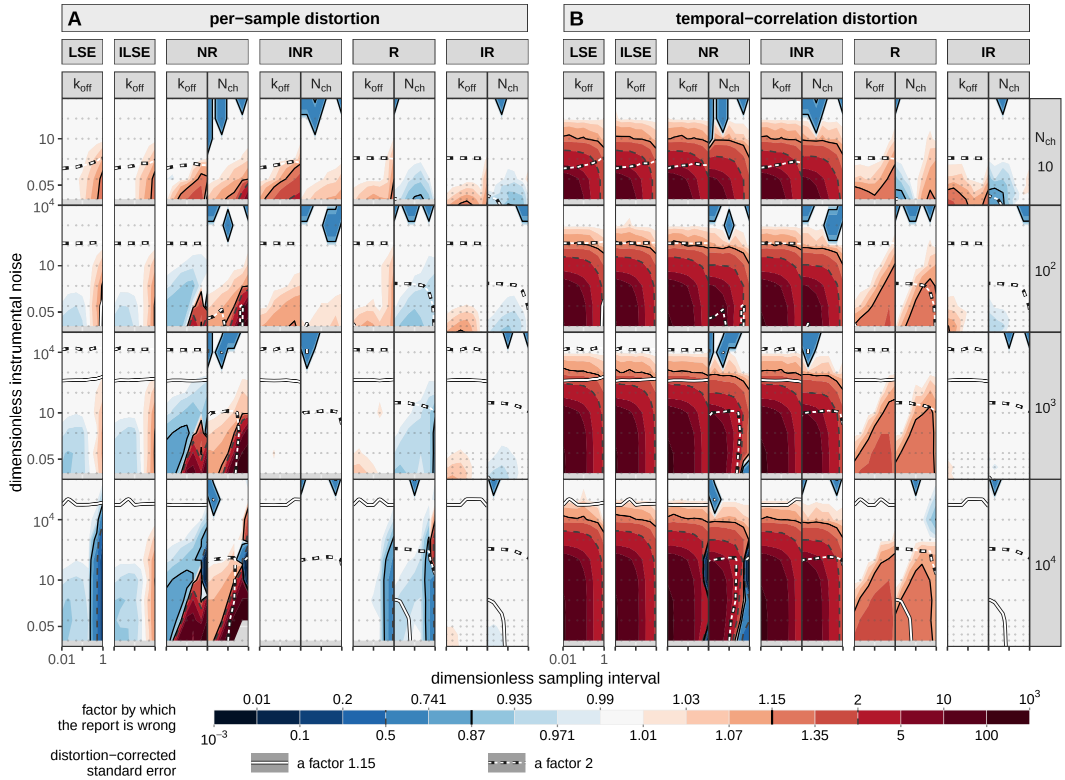


**Figure 4—figure supplement 1. The distortion factored: what is a one-interval failure and what is a failure across intervals.** Figure 4's second moment split into the two congruences the battery computes separately, the per-sample part $\mathbf{H}^{-1/2}\mathbf{J}_s\mathbf{H}^{-1/2}$, the departure of a single interval from the Gaussian the member assumes, and the correlation part $\mathbf{J}_s^{-1/2}\mathbf{J}\mathbf{J}_s^{-1/2}$, the memory across intervals that a likelihood diagonal in time cannot see (defined above). Same plane, same rows, same scale, same six members and the same two criteria as the body figure, nested component over member over parameter. `NR` and `INR` carry the *same* correlation part, 16.5, 19.5, 21.2, 21.2 against 16.5, 20.7, 21.8, 22.1 on the closing rate across the four decades of channel count, as they must, since they differ by a one-interval variance term and by nothing about time; what separates them is the per-sample part, where `NR` runs 1.24, 1.02, 5.68, 47.2 and diverges while `INR` runs 1.72, 1.15, 1.02, 1.00 and converges, and on the channel number the same pair ends at 569 against 1.00. The divergence the body figure shows for `NR` is therefore not a temporal failure at all: it is the one-interval variance missing a term that grows with the channel count. Second, the recursion buys the other factor and only the other factor, `INR` to `IR` taking the correlation part from 22.1 to 1.000 and leaving the per-sample part where it already was. Third, `R` and `IR` are equally faithful to a single interval and part company on the correlation part alone: on the closing rate at $\widetilde{S} = 0.01$ the per-sample part runs 1.09 to 0.998 for `R` and 1.10 to 1.007 for `IR` across the four decades of channel count, while the correlation part rises for `R` from 1.35 to 1.47 and falls for `IR` from 1.24 to 1.00. The factorisation is exact in the log-determinant and not entry by entry, the reconstruction being $\mathbf{K}\,\mathbf{C}\,\mathbf{K}^{\mathsf{T}}$ with $\mathbf{K} = \mathbf{H}^{-1/2}\mathbf{J}_s^{1/2}$: $1.024 \times 1.433 = 1.468$ against a total of 1.481 for `R` at a thousand channels.

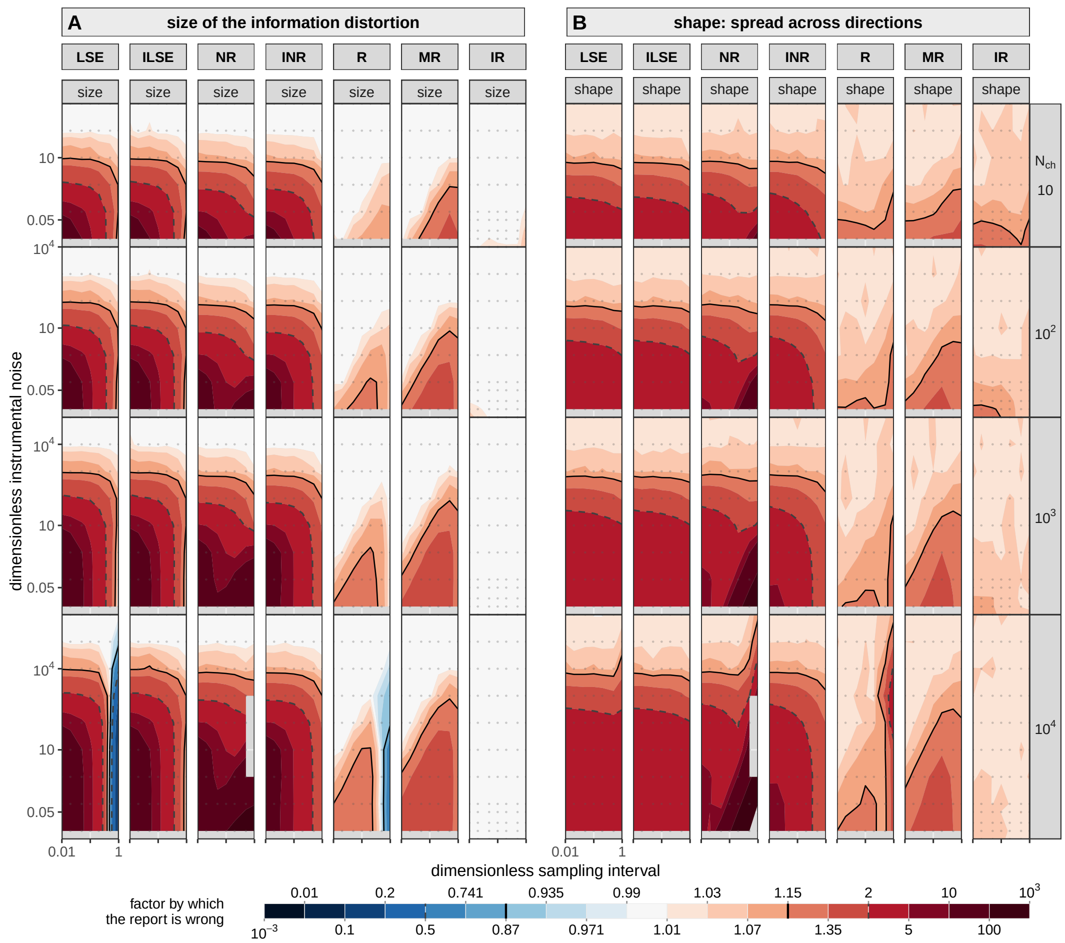


**Figure 4—figure supplement 2. The magnitude of the information distortion against its anisotropy.** Two invariants of the same distortion matrix over the same plane, for seven members: both least-squares arms, `NR`, `INR`, `R`, `MR` and `IR`. (A) its magnitude $m$ and (B) its anisotropy $a$ (Eq. 4), $m$ signed about a null of one and $a$ one-sided. The rule they give: $a$ at one means a single effective-count rescaling is exact and $m$ is the factor it should use; $a$ above one means no such rescaling exists at any factor. The two least-squares arms are the same figure wherever the recording is finely sampled, and part at the coarsest interval: at $\widetilde{\Delta} = 1$ and $10^4$ channels `LSE` reads $0.074$ in magnitude, understating its own information thirteenfold, where `ILSE` reads $1.00$. That is the mean model failing and not the variance model, and it is the deepest of the few cells anywhere on this page where a member is conservative rather than overconfident: $11$ of `LSE`'s $210$ cells and $4$ of `R`'s, all of them at the two coarsest intervals and none for any other member.

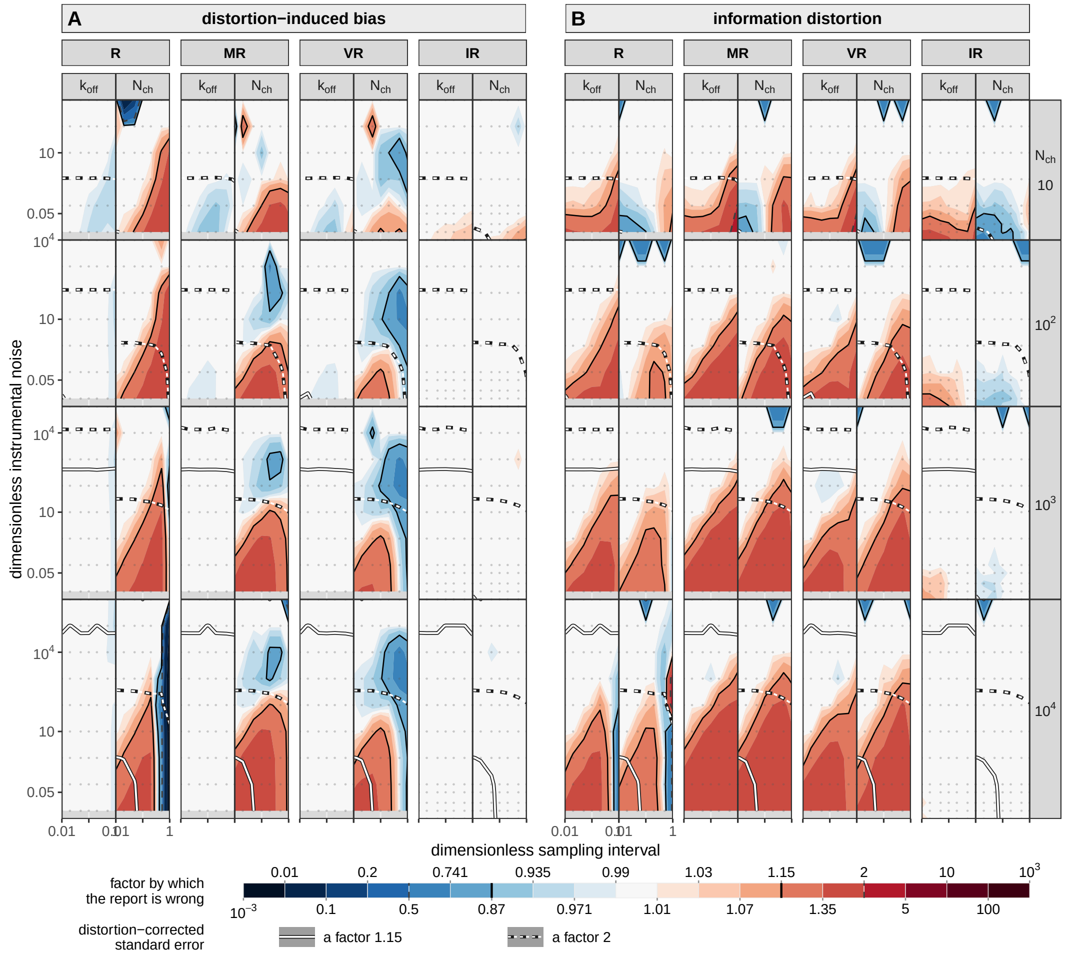


**Figure 4—figure supplement 3. The same plane for the recursive family: R, MR, VR and IR.** Figure 4 unchanged in every respect but the roster, the same two moments on the same shared factor scale with the same two line families and the same nesting, over the four rungs inside the recursive family, `R`, `MR`, `VR` and `IR` (Table 1). Sixteen columns rather than the body's twenty, since no least-squares arm appears and every member carries the channel number. The reading is in the text: neither partial correction improves on the member it corrects, at any channel count, on either parameter.

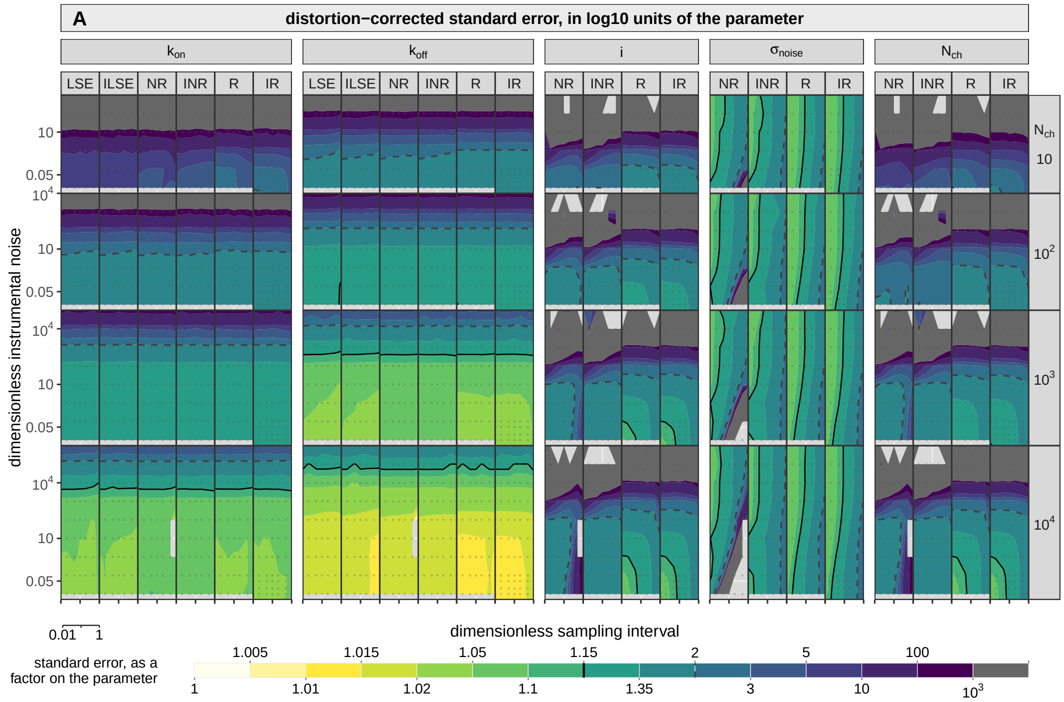


**Figure 4—figure supplement 4. What each member achieves, as opposed to what it reports.** The standard error each member delivers on each parameter: the square root of the diagonal of the distortion-corrected covariance, over the same plane as Figure 4 and in the same orientation. What the design axes buy is read directly off it, and the reading is in the text. It is also the field whose level set at a factor of two draws two of the boundaries of Figure 7. Blocks are per parameter, the members side by side inside each. The scale is sequential: pale is a parameter pinned, dark one barely determined. The two marks are an error bar of $15\%$ of the parameter and an error bar of a factor two. The corrected error is drawn rather than the reported one, which Figure 4 shows wrong by up to a factor of $74$, so only the corrected version compares members on the same footing. Both least-squares arms hold the unitary current and the noise level fixed and so appear on the rate blocks alone.

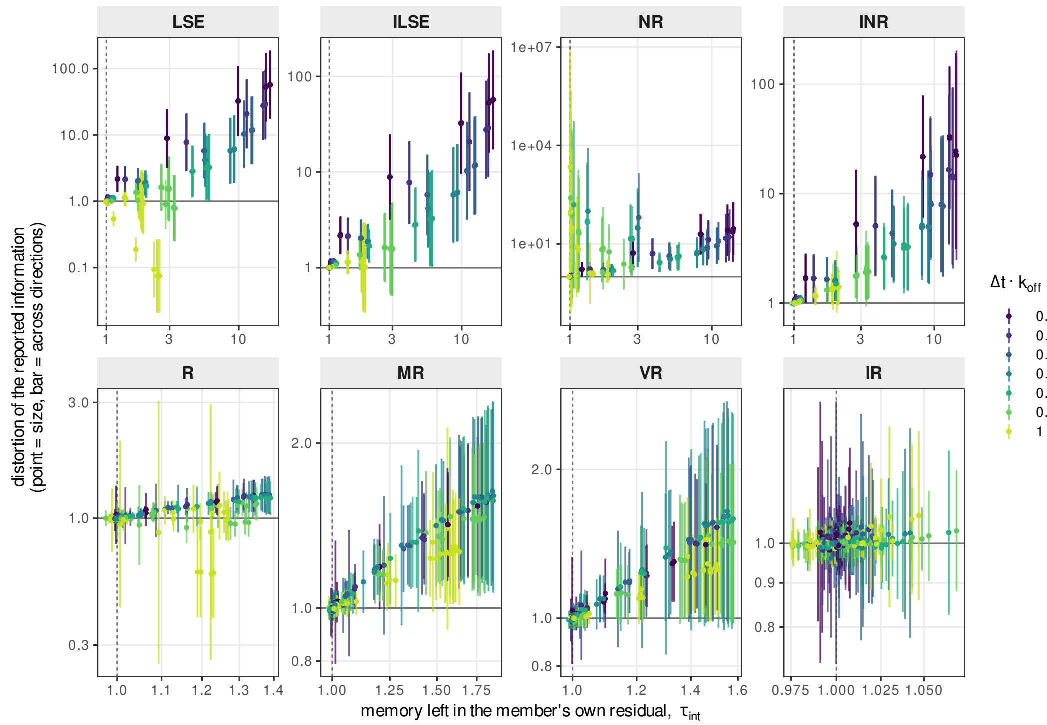


**Figure 5—figure supplement 1. Each member against the memory left in its own residual.** The whole ladder, with the horizontal axis of each panel now the integrated autocorrelation of that member's own standardized residual, which a reader cannot compute without running the member: a mechanism panel rather than a decision one. It asks whether the memory a member fails to remove predicts the information it then misreports. At a fixed sampling interval it does, for every member with any range in the statistic, the rank correlation between the two reading $+0.92$ for `ILSE`, $+0.96$ for `INR`, $+0.93$ for `MR` and $+0.91$ for `VR` over the whole plane and about $+0.99$ within one interval. `NR` carries $73$ cells with $\tau_{\text{int}} < 1.05$, of which an eighth are distorted, the worst by a factor of $2217$. Each panel is on its own scales.

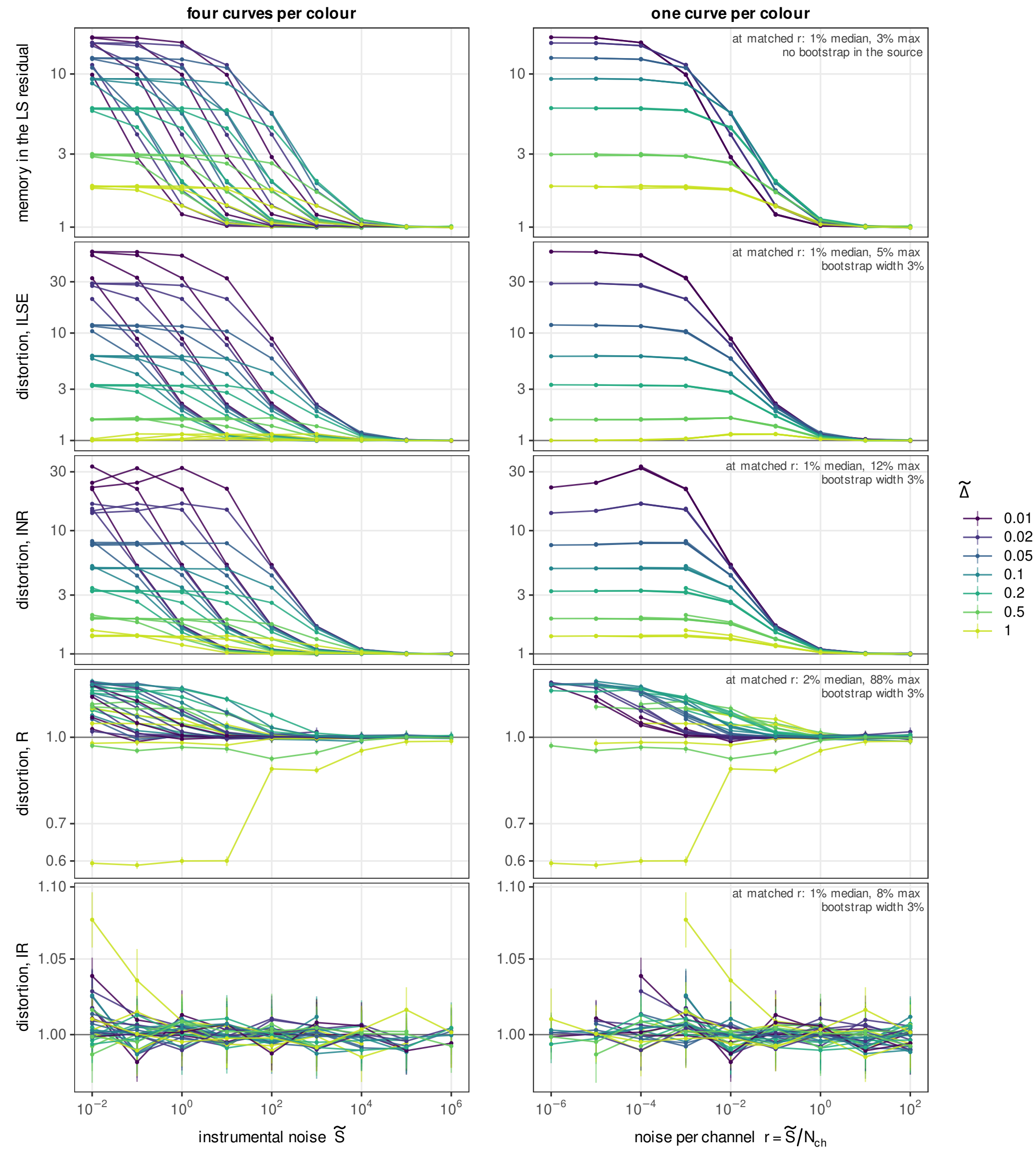


**Figure 5—figure supplement 2. The instrumental noise and the channel count act only through their ratio.** The same cells as the body figure, drawn twice. Each line is one channel count at one acquisition interval swept over the instrumental noise, twenty-eight per panel, and the two columns differ in nothing but the horizontal coordinate: on the left the instrumental noise $\widetilde{S}$, on the right the noise per channel $r = \widetilde{S}/N_{\mathrm{ch}}$. On the left the four channel counts are four separate curves per colour, a decade apart; on the right they merge, so seven of the twenty-eight are visible. Nothing is fitted, rescaled or normalized between the columns. The top row is the horizontal axis of the body figure itself, $\tau_{\mathrm{int}}$ of the least-squares residual, and it collapses with the rest: a reader who measures $\tau_{\mathrm{int}}$ on their own record has measured $r$ without knowing it. The vertical bars are the bootstrap quantiles of each cell, absent on the top row alone: each point is known to about $3\%$ and the four channel counts land within $1\%$ of one another. The number in each right-hand panel is the residual disagreement between cells that share $r$ exactly at the same interval, with no interpolation. The $5\%$ of `ILSE` and the $8\%$ of `IR` are within that scatter; the one large value, `R`'s, is the single ill-conditioned cell at $N_{\mathrm{ch}} = 10^4$ and $\widetilde{\Delta} = 1$ that Figure 4 greys out and this figure does not, without which `R` agrees to $23\%$. In the bottom row `IR`'s departures from unity are the width of its own error bar and not structure.